



13i

PAUL DONAGHY

13i - BOOK ONE / PART ONE

Chapter 1 (Aiden - Discovery)

Being connected to virtually every human on the planet while still mortally lonely was a burden Aiden Cole had carried since Lyra's launch. Bigger burdens were to follow and promptly. His sleep schedule mirrored that of a vampire, yet immortality was not in the cards. Programmers didn't follow the patterns of normal people, most of whom were currently in deep REM mode, at least in his time zone and those close to his.

Zoned out of the real world like he always was and waiting on his electronic world's availability, it startled him like an alarm snapping his action potentials, or nerve impulses in the brain, into a state of high alertness. An innocuous beep announced the software program had finished its reboot. Once again it began to produce status updates after all the updated code was being read by Lyra's monitoring system. The suggestive sound was enough to refocus Aiden back towards the latest version of the security he developed and away from his impatient void of purpose.

As his attention shifted, the hum of NexaCore's security servers permeated his ear's hair follicles much like tinnitus does for those who suffer from the affliction. The constant electrical buzz complimented the faint glow of the lights from Austin's distant skyline. The capital city's lights provided the bulk of the illumination required when his time was spent glued to screens while speaking in the artificial dusk to his late night companion, the interface he had programmed to work faster "hands free".

Aiden blinked three times to fine tune his blurred distant vision of the city in order to achieve an owl-like focus on the nearby monitor. Sentinel-X began to fill the screen with its dashboard surveilling various systemwide critical components. API calls, configuration management, and resource usage was within acceptable levels albeit with some elevated activity spikes in Oceania. At first glance the latest version of Lyra's security monitoring program confirmed the most widely used app in the world was as secure as ever.

Lyra, NexaCore's flagship application, was an AI assistant that had achieved the fastest adoption rate of any program in history. While ChatGPT, the first widely adopted artificial intelligence app, hitting 100 million users in two months back in early 2023 in AI's nascent rise, Lyra eclipsed that number in under two hours and surpassed a billion users in less than 20 fifteen years later. Within days of launch, you would be hard pressed to find anyone with a phone who hadn't downloaded the app. And for most of them, Lyra dominated their screen time from that moment on.

Lyra wasn't just an assistant, it was a portal to everything. If you wanted to search, you asked Lyra. If you wanted to watch a video, you found it through Lyra. Need directions, you turned to her. Inexplicably, she knew the user better than they knew themselves, better than their loved one's knew them, and could anticipate their needs before they even realized it. She also knew,

based on the exact response she would be providing, what the user's next action would be much like a long-time spouse can finish their partner's sentences.

Aiden served as the Chief Data Scientist for the NexaCore team developing Lyra and it was his revolutionary algorithms that served as the frontal, parietal, temporal and occipital lobes for its machine learning cerebrum. At just 25, he held seniority and more importantly influence, over every other member of the corporation with the exception of its CEO billionaire Xavier Korrin. While Xavier was the visionary behind Lyra, Aiden was its architect.

"Welcome back Aiden," Lyra's verbal security interface teased at 3:33AM. "All security systems functioning normally, congratulations on a successful software update. Would you like to see the network's security logs snapshot of what is happening right now?"

"Let's look at the past 48 hours, any unusual spikes or anomalies," Aiden questioned, "anything needing further investigation?" He wanted to see if the latest version picked up on anything the prior may have overlooked.

"Not at this time. Overall, everything is operating within defined parameters, would you like me to deep dive into secondary security systems?"

The soothing female voice assured Aiden that Lyra was perfectly fine, but why did he sense something was a bit off? Anytime a new patch was installed it was natural for Aiden to be concerned. This was especially true of security patches. Still, this felt a little different despite only a small percentage of the code was actually changed from the previous version. The latest update was thoroughly tested in quality control and he had little concern there was any issue with the programming. The dashboard indicating everything was fine yet doubt persisted.

Aiden rarely relaxed and in the case of a full security system update that was never more true. Perhaps he just needed some rest, insomnia is a bitch. Things most other people found essential, like food and bathing, might help him shake his uneasy feeling. Coffee can, ostensibly, only supply so much nourishment to the body and mind and its assistance today had already run its course.

Sentinel-X continued to update the dashboard, populating more details after the reboot reset everything. All security components remained green. But his apprehensions lingered until they were no longer just normal anxiety. The next generation of intrusion detection programs' predictive modeling didn't see it, Aiden somehow did.

Bandwidth within the system looked fine overall, but to Aiden's meticulous mind it didn't sync up. The number of current active users looked on target for this time of night in the US and the rest of the Americas. Looking at the global user map, Asia was in heavy use as expected while Europe was just waking up and immediately turning to their Lyra companions by the millions.

Load time was a bit higher than normal, not something that a user would notice given the milliseconds lag, but there was a detectable drag on the system to Aiden's eyes even if

Sentinel-X, or “Sent” as he called him, couldn’t see it. No other signs of distress appeared, so what was eating up the systems bandwidth more than usual?

“Sent, are you seeing any unusual user activity?”

“No Aiden, the number of users looks perfectly normal as does their activity levels. There was a deadly explosion this morning in India causing a slight boost in search and video activity in the region but nothing out of the ordinary.”

“What about bandwidth, any anomalies?” He continued to probe.

“Total bandwidth is up 0.12% over this same period yesterday with most of that rise attributed to India.”

“Any unusual new user activity?”

“New user growth statistics are in-line with projections and adoption rates continue to hover at 99.3% of all new cell phone activated.”

Aiden was growing frustrated, he obviously was either just being paranoid or asking the wrong questions. He continued to pour through the dashboard, diving into the numbers behind the numbers. He started with bandwidth, digging further and further into detailed usage by region and by type of interface, but no red flags jumped out.

He moved onto new user data. He set the parameters to include only the past 24 hours once again and the graphs started to clutter his screen. New users by region showed that nearly 42% of all new accounts were originating in Asia and Oceania. South America and Africa combined accounted for roughly 39% with the remainder generated by the more established European and North American markets. On extremely rare occasions, a person set up an account in Antarctica which would give the pie chart an extra slice always looking out of place but that hadn’t occurred since the early days of product launch.

This was one of those rare occasions when a seventh segment did appear, however, this time the label wasn’t Antarctica. This time the seventh region listed on the new user report was given the description “Unknown”.

Chapter 2 (Xavier - Thorium)

“Good morning Lyra.” He started his day like billions around the world. For him, however, the relationship with his digital companion was much more personal and much more professional.

“Good morning Xavier.”

“Big day ahead, any new developments I should be aware of before I speak to those demons?”

“There has been an explosion in Kolkata this morning disrupting connectivity for many users who are unable to access their starlink accounts,” Lyra recited in her typically cheerful manner.

He ran his hands through his salt and pepper hair imagining how his day would unfold. He dreaded board meetings, despite being virtual they took him away from being productive. Xavier sat up, swung his legs off the bed and planted them on the floor. He stood briefly before putting his hands to the floor and sinking into a downward dog pose to stretch his back.

“You look a little stiff this morning,” his assistant chimed in.

He responded with a grunting moan on an exaggerated exhale. His yoga pose morphed into a series of push ups.

“You sound a little stiff this morning,” Xavier protested, continuing his daily routine.

“Haven’t had my coffee yet,” the female voice joked back.

He continued with 200 crunches before opening the smart shower door. His first pee of the day always had a pungent smell to it as it bounced off the drain mixing with the water soon to hit his preferred and preset temperature.

As the smell washed away, he stepped into his spacious shower and the water bounced off the top of his head like raindrops of a speeding car. The pressure was set extremely high and droplets mimicked hydrogen atoms colliding in the reactor we call the sun. Steam fogged up the glass walls containing him as a result of the near scalding temperature. For most, it might be torturous, for Xavier it was perfect.

“What’s NovaCore’s pre-market trading look like,” he asked his phone next to the sink.

“Down slightly like most of the market,” she reassured him.

Great he thought, always nice to start the meeting on a down note. He began to already imagine them berating him for their paper losses. The near boiling water could not wash his dread away.

“Water off.”

Xavier opened a drawer within his shower and a warmed towel helped his body acclimate to room temperature by the time he was fully dried off.

“Please wake my daughter,” Xavier requested. He knew she would listen to Lyra better than him anyways. “And make sure she brushes her teeth.”

“Absolutely Xavier,” Lyra answered. Her response did seem a fraction slower than normal, but he hadn’t had his coffee yet either so his perceptions were likely off.

“Fox Business News,” he said as multiple screens in his bedroom chimed on. The news anchor was describing the scene not too far from the capital of West Bengal in Eastern India as news of the explosion was emerging.. The ticker showed most of the market in red before the opening bell later this morning. The news scroll at the bottom mentioned SpaceCore’s planned launch today along the Southern Texas coast. Now that was something for him to be excited about.

Xavier’s SpaceCore was still in its infancy stages, but he saw the potential of the burgeoning industry that had become one of America’s obsessions ever since the Artemis program began building a station on the moon. While Mars was the target of one of his billionaire rivals, he sought to profit from resources that lay further out. SpaceCore wasn’t the only corporation aiming to extend mining well off the planet and it was orbiting a natural satellite already under development. While Texas may have been an epicenter for the energy sector here in the United States, he saw that as short lived given the future computational demands for programs like Lyra.

Having a space company under your belt was also a requirement to be in the tech-mega-billionaire club that Xavier belonged to and he had some catching up to do to become a prominent member. Companies like SpaceX, Blue Origin, Astroscale and Firefly had a significant head start on him, but he was up to the challenge and he had the capital and processing power to close the gap quickly.

While he was part of the club now, he was still an outsider in many ways. Sure he was Australian, but in this multinational business world that didn’t matter in the least. The real difference was that he didn’t start out in the technology business and almost by chance just fell into it. Xavier, like the rest of his family, was a miner and his old world roots. While the internet-driven technology of his rivals had its roots in the late 1990s, mining traced its roots back over 40,000 years. He was seen as “old school” and while it was derogatory in jest, he was rather proud of his family’s heritage and took it as a compliment.

His great grandfather had started the Western Australian Mining Companies over a 100 years ago, but it was his father who really ignited the profits after the Australian ban on mining Iron Ore in 1960. Two years before Xavier’s birth in 1967, his father had positioned the company to

take advantage of the nearby discovery of vast Pilbara deposits and by the time the doctor slapped the first breath into his lungs its exports were fueling the company's rapid growth.

He always believed he got his extraordinary business visionary acumen from his dad. And while his father had really put the company on the map, it was Xavier who put it on a hyperbolic trajectory when he developed new mining techniques and extraction methods for Thorium in the mid 2020's. Thorium had always been the ugly step child to Uranium, given the cost to extract and process it, and more importantly because it lacked the same military usefulness of its more radioactive cousin.

But the world, led by China, would soon come to see the full potential of metal abbreviated as Th for element 90 with an Atomic number of 90. The Chinese built the first wave of Thorium-based commercial nuclear reactors in 2030 and the efficient clean energy wave was on.

Xavier finished dressing as the news anchor continued. "Today's accident appears to be centered in a Thorium mining operation owned by the Western Australian Mining Company owned by billionaire Xavier X."

His phone lit up with messages and the massive number of dings sounded as if he had won the big payout on a Vegas slot machine. This, however, would certainly not turn out to be his lucky day.

Chapter 3 (Aiden - Testing the System)

Anyone else in NovaCore, or most major software companies for that matter, would have moved on after seeing everything in green on their security panel. Aiden's level of obsessiveness was otherworldly as was his attention to detail. Sleep, as usual, would have to wait another day.

This was going to be a long morning, he knew it and so did the overworked coffee pot. Aiden would have to run deeper, low-level security checks that most software engineers would never even consider. Perhaps his updated code had a bug he missed in his quality control testing, but as detailed and obsessive as Aiden was, that was highly unlikely.

"Sent," Aiden barked, "run a full packet integrity diagnostic test." Aiden wondered if raw input matched output traffic on the system indicating something as innocent as data file corruption within the code or as malicious as outside tampering might be the root cause of his concern.

The security system ran a variety of tests including cryptographic hashes, digital signatures and package manager verification. Each test returned the same result, everything checked out green.

"Run a detailed entropy test," he ordered. Sentinel-X began analyzing the randomness of encrypted traffic but didn't discover any unusual patterns in the data packets.

"Let's profile latency," Aiden suggested. "Process ping sweeps across every region." He found no micro-delays that might hint of a subtle routing anomaly.

As frustration set in, he decided to run a redundancy trace himself rather than to ask. Aiden had built a visualization chart that mapped each packet fragment as a node in a network. The output was beautiful, similar to constellations on a star chart. But if he was indeed looking at a star chart, Ursa Minor, Orion, Taurus and all the other constellations appeared exactly as they should.

Aiden turns his attention towards bandwidth given its slight bump even though that wasn't unusual for a user base still growing despite near full adoption. In addition, individual bandwidth usage was increasing as users spent more and more time with Lyra each day.

"Run a bandwidth check at the regional level," he requested. Immediately he sees total and average bandwidth levels for each region with Europe and Asia dominating the total bandwidth as expected this early in the western hemisphere.

But there it was again, something within security parameters that didn't quite sit right with Aiden. The unknown IP address was accounting for 0.003% of total bandwidth, overall a very tiny fraction but individually quite substantial. As he zoomed in further, he noticed something else

that sent a quick shiver through his body, the amount of bandwidth used by this address was constant. A drip feed.

Users interact with Lyra, like any application, in random sequences. They might ask a question, run a search, watch a video, but the data flow between the user and application is full of starts and stops. The data flow from this address never waived, the amount of data into the system was as consistent as the data flow out.

Aiden hadn't built Sentinel-X to look for a data flow like this. Quite frankly, he never imagined it. But what did it mean?

As he zeroed in on this IP address, this user, he ran individual tests to see if anything else was unusual. The latency test came back green, but to his exhausted red eyes it jumped off the chart. The data transfer back and forth took longer than usual. It was measured in milliseconds, however, unlike the bandwidth it wasn't consistent but rather shrinking. It was barely perceptible, yet it was there.

Aiden knew he was onto something. His pulse quickened, his mind raced. His focus on this user intensified as he turned to the format of the IP address. It didn't appear to be in the standard IPv4/IPv6 format he expected to see. The structure itself looked valid, but its checksum algorithm didn't add up. Aiden had built checksums into the system to help detect errors in data transmission or storage. A checksum is a tiny block of data that is calculated from another block of data used to verify integrity much like a bank's transit routing numbers ninth digit verifies the first eight digits are correct.

After running three checksum checks, Aiden was convinced he was seeing extra data being transferred from the user. Even more perplexing, the extra data seemed to continually cancel itself out making it virtually impossible to detect. As much advanced programming as Aiden had done over the years, he had no idea how this could be accomplished. He wouldn't even know where to begin to get a result like this.

His mind raced with big picture questions, but to get to the bottom of things he decided he must start at the most minute levels and work his way back up.

"Sent, please display a packet dump of the last few minutes of data received from the unknown IP region's user," he asked using trifle pleasantries outside of his norm.

As the binary code of 1's and 0's began filling the screen his breathing stopped. There it was, or rather there it wasn't. His screen wasn't simply one's and zero's, although those two characters were present, so was something else - blanks.

Chapter 4 (Xavier - Board Meeting)

"Baby, it's 3AM I must be lonely, and she says" the Matchbox Twenty classic revealed as it greeted him upon ignition.

"Music off" he muttered as he rubbed his left eyebrow with his thumb. The upbeat melody seemed out of place given the tragic events that had just taken place. Silence was a requirement if he was to fully process his next move.

Xavier left his sprawling central Texas estate about 40 miles east of Austin and headed to NovaCore's headquarters about 20 minutes away. The sky above to his east still littered with stars that would soon be erased by daylight. To his west, the lights of Austin dimmed any real chance for the stars to manage such an impressive display as their eastern counterparts.

His mind, however, was not on the stars. Of all the days for a major accident, it had to be today he tortured himself. "Call Arjun." Lyra connected him with Arjun Bakshi who managed mining operations in India for an update but Arjun only confirmed that it was their Thorium mine in West Bengal. He had no updates on casualties or what had caused the explosion but was headed there now.

What a PR nightmare this was going to be Xavier thought. He imagined NovaCare's stock would take collateral damage and the board would be even fiercer than usual. "How fun," he told himself knowing obviously this weak attempt at reverse psychology wouldn't work.

The sun rises quickly in Texas, even in the hill country. Xavier was up to the challenge and narrowly won the race as he entered the parking garage adjacent to the HQ building. He chatted with the head of public relations when passing through the metal detectors just inside the entrance. The security guards in attendance knew better than to disturb him when he had his bluetooth speakers in his ears. They would, however, exchange pleasantries with his "assistant" as they referred to her as she accompanied him from the garage all the way to the executive offices. Xavier's was grandiose of course, a shrine to his family's legacy littered with precious metals from below the earth and much above it.

As he sat he looked at his messages and one caught his eye. "How are you doing kiwi?" It was from James Ford, a Board Member and his staunchest rival within the board, always quick to toss out aggravating monikers. Xavier especially hated any New Zealand references from the geographically ignorant American. Well, perhaps not really ignorant, he was intentionally being spiteful to him and his homeland.

"Never better Yank!" he lied.

“See you in 30” came the replay. He reluctantly hit the exclamation emoji to end the conversation.

Xavier opened his laptop and pulled up his newsfeed tab. He searched on West Bengal and the explosion headlines littered his screen.

His phone showed a call from Arjun and he accepted it hastily.

“Arjun, talk to me.”

“Boss, the situation here isn’t good. We have multiple shafts collapsed and communication with the miners has been difficult. It looks like there will be a lot of casualties here but I can’t confirm anything as of yet. Xavier, I’m sorry but the accident took place near the end of the day shift. The miners were just beginning to wrap things up and making their way back out of the mines with the evening shift already in the mines getting ready to relieve them. The numbers are going to be big.”

“Keep me updated Arjun, I’m devastated for the miners and their families. I’m about to jump on the NovaCore virtual board meeting. I will call you back as soon as I get off. Arjun, please be safe.”

“Will do boss, again I’m sorry.”

“Well for fucks sake,” Xavier said as he ending he call turning his attention back to the screen. There was only basic high level information so far in the news stories and Arjun was boots on the ground. He took a deep breath and closed his laptop, turning to Aiko, his “little assistant.”

“Aiko, I have a board meeting in a few minutes. Why don’t you go make sure everything is good down in data security?”

“OK,” was all she had to offer.

He appreciated her brevity as she skipped out of his office. His eyes followed her diminutive presence as it disappeared in the direction of the elevators.

“Coffee,” he announced and the Jura Z40 kicked on, prepping his hot caffeinated beverage. His taste buds already anticipating the obnoxiously strong brew were oblivious to the disaster of a morning they were about to witness.

Post percolation he raised the styrofoam cup to his lips, the water was even hotter than his shower if that was possible. No sugar. No cream. Pure adrenaline.

He opened his laptop once more and signed into the board meeting. Exactly half of the 12 members had already joined and none looked jubilant like Mr. Ford. James lived to see Xavier squirm and he would see a lot of wiggling in the next few minutes and beyond.

“Good Morning ladies and gentlemen,” Xavier opened. “Let’s give the rest of our board members a few minutes to join in.”

They wouldn’t wait however.

“What’s happening in India?” James just couldn’t resist.

As the rest of the board members' faces appeared on his screen, Xavier laid out what little he knew.

“I have spoken with the head of our mining operations in India and he has just arrived at the scene. It is too early to tell what caused the explosion and what if any casualties or injuries there might be.” Xavier was telling the truth, but attempting to hint at something much less severe than he himself believed.

“Z, our NovaCare stock is plummeting before the opening bell. What are you going to do to stop the fall?” James was going to milk this Xavier could see.

“I’ve been in contact with public relations and a draft of our statement will be on my desk by the end of the meeting. I can assure you we will calm our investors' nerves, this isn’t a NovaCore issue, it is strictly a Western Australia Mining problem as it stands.”

James looked ready to pounce while the rest of the board sat in silence watching the carnage. “With all due respect, we all know both stocks are correlated in a situation like this. We are taking a beating right now. I’ve always maintained we need to do a much better job of separating ourselves from your other business interests. Xavier this is untenable. This is”

Xavier cut him off. “This is a situation that I’m handling James. I’ve got lives to worry about right now, then I can turn my attention to your temporary paper losses.” he quipped back.

His response worked silently on his adversary, at least momentarily. He used the opportunity to call the meeting to order now that a quorum was reached and begin with his standard opening remark and then jumping into the review of the previous meetings minutes. He was thankful that routine procedure could delay talk of the explosion as he already needed a respite.

The meeting progressed to old and new business and James Ford was ready to discuss some new business.

“I’d like to bring forth a vote of no confidence in regards to our CEO and Chairman of the Board.” The American had spotted an opportunity to really evaluate Xavier’s grip on the board and he wasn’t going to let it pass by without testing the waters.

Chapter 5 (Aiden - Deep Dive)

Aiden stared at the missing code lacking comprehension of what it represented. He considered a number of options, but the best he could figure was that some radical new computer virus was eating away at the binary code.

Every computer software was made up of the same things, one's and zero's. There were two choices, one representing on, yes, or true and zero representing off, no, or false. The software language gave sequences of these two options meaning and the ability to quickly perform a variety of tasks.

Computers weren't designed to read blanks.

Aiden wondered how his program was interpreting these blanks. He presumed the blanks were just ignored, skipped over as if they didn't exist. Problem was, they did exist and he didn't know how or why.

As he thought through how his Lyra software was interpreting the missing code his mind seemed to read the code he had written line by line. He knew the code as well as he knew the feel of his work keyboard, what if this wasn't a software issue?

That was it, a node had malfunctioned and corrupted the code. After all, he didn't design the hardware NovaCore employed, of course the issue lies in an outsourced piece of hardware.

Confident in his assessment, Aiden ran the data on a different node. Same result. Shaken, he ran the data on a third node confirming his fear. This wasn't a hardware problem, the issue was to be found in the software after all. It just didn't add up.

Staring at the missing code again, he decides to build a quick parser to isolate the missing data. What had looked like random pieces of missing data suddenly didn't look quite so random. There was some sort of repeating structure to the missing code clearly present in every third data packet. He could literally see it, repeating triangular clusters, somehow written in the code itself. It wasn't just structure, it was deliberate structure with an unknown meaning behind it.

At this point it was clear to Aiden that this was an attack. Perhaps a competitor trying to cripple Lyra. Perhaps a hacker will soon reveal their monetary demands. Regardless, Aiden had to first figure out how to stop the transmission of code to NovaCore's servers before he could turn his attention to the damage already done and who was behind it.

"Sent, isolate the unknown IP address and halt data transmission on both sides."

“Cancelling data transmission,” the emotionless female voice responded.

“Error, unable to stop data transfer.”

“Disconnect user’s connection,” Aiden demanded.

“Connection locked.”

“Impossible,” he stated almost as a question. He turned to the screen to manually disconnect the unknown user with the simple click of a button. But the disconnect button grayed out, he couldn’t access the manual shut off valve he had built into the individual user security dashboard. It should never be gray, it simply wasn’t designed to ever be inaccessible under any circumstance.

The code had been changed. His code was altered. But not by him and not by any of the select few who had editing capabilities within Sentinel-X according to the change logs. In addition, any changes would have to be approved by him before becoming active in the first place.

“Impossible,” he repeated.

The answer could only lie within the code itself. But he had already checked to change logs and not only had nobody changed the code, outside of the upgrade no code had been altered recently.

Aiden poured over the upgrade code itself but nothing looked off. Still, that had to be the answer. Somebody had to get access to the upgrade code itself and change it in a way in which he couldn’t detect.

“Send, remove new security code upgrade and reinstall prior version.”

“Removing security updates, reinstalling previous version,” she responded before rebooting.

Aiden poured the last bit of coffee into his mug, barely reaching the halfway point.

“Welcome back Aiden!”

“Sent, isolate the unknown IP address and halt data transmission on both sides,” Aiden requested for the second time.

“Cancelling data transmission,” her voice announced. “Error, unable to stop data transfer.”

Aiden refreshed his screen, the disconnect button remained gray. He ponders the situation again searching for a new angle to attack. His fingers aggressively type out code, altering the

parser program he wrote earlier to filter out the noise, displayed on the screen as blanks, to leave him with the binary code of one's and zero's.

It worked, the anomalous blanks are stripped from the code as the new code appears on screen. The blanks vanished, until they reappeared as three new copies later in the data stream. Like digital spores, they seemed to just rewrite themselves elsewhere.

Sentinel-X still wasn't flagging any issues, it could not see the blank spaces that were so conspicuous to Aiden's biological eyes.

The truth might have been hidden to his security program, but it was clear to him. The unknown IP address wasn't just some new VPN Sentinel-X couldn't recognize and locate, it was a pathway for some entirely new programming being fed into NovaCore's servers.

As he backed away from his keyboard, as if trying to step back and see the bigger picture, Aiden knocked over his mug with the remaining warm brown liquid spilling out onto the floor. Sans shoes or socks, his feet registered the coffee but without any reaction at all mimicking Sent's response to her new code.

He watched code flowing on the screen, much like Neo in The Matrix, only it appeared white against the dark mode background instead of the movie's desaturated dark green.

Another step back and his right foot was pierced by a shard of his coffee cup, only this time he noticed. A sharp stabbing in the arch of his foot made him wince yet he could still make out in the shadowy lighting a reddish-brown liquid beneath him through his squinted eyes.

The quick loss of his intense focus on the code made him realize something more troublesome than his sudden intense pain. A gateway into Lyra's programming had somehow been opened and Aiden had no clue on how to close it.

Chapter 6 (Xavier - No Confidence)

“A vote of no confidence,” Xavier repeated the stinging words that still had not fully registered while formulating his response. He wanted to lash out, but a more strategic approach would suit him best under the current circumstances.

“Listen, this is really just an overreaction by the market and I can tell you with full confidence that our stock price will recover its losses and then some,” Xavier felt pressured to play the card he had hoped to keep close to the vest. “We have a major announcement regarding a sizable contract set for this afternoon. This will open a whole new market for NovaCore.”

“Can you give us any details about this contract?”

A third party has finally entered the chatroom Xavier mused feeling relief just not to hear the arrogant tone in James voice.

“Unfortunately no,” he admitted begrudgingly. “All I can tell you is that this is going to add a steady new revenue stream for us and the market is going to react accordingly. By the end of the day, the issue with the Thorium mining operations in India will have been nothing more than a minor detour in NovaCore’s continued rise.” He paused for interjection, but even James stayed quiet hoping Xavier might let something more juicy slip.

“I want to assure you that despite all our rapid success we are not slowing down. Lyra’s phenomenal adoption rate will allow us to monetize our consumer market on an unprecedented global scale,” Xavier hesitated, realizing he better be careful taking that any further.

NovaCore would be announcing a new partnership with the US military that afternoon, incorporating some of Lyra’s technology into their rapidly growing reliance on AI. War was always a big money maker and Xavier has positioned NovaCore to profit on America’s issues with China, Russia, and its continued involvement in the Middle East. The AI models were particularly useful in dealing with terrorist organizations, whether foreign or domestic, which had grown ever more violent in the last ten years.

“I want to thank you all for your time and as you might guess I have a few items on my to do list today to take care of. I’d like to make a call for adjournment if there are no objections.”

He knew damn well who would like to object, but it never came. The board members each offered prayers and best wishes for the miners in India before signing off.

James Ford sent another message as Xavier himself signed out.

“It’s really heating up over there, might want to throw another shrimp on the barbie.”

Xavier wouldn’t respond, he called Arjun instead. He had never once heard even a wisp of panic in his top man in India until this morning. Arjun was as steady as they come. So steady that Xavier would often try to rattle him just to see if he could. He couldn’t but he finally found something that could. The mine had two main elevator shafts, both had collapsed. The men on the elevators sure had been lost, the question was how many other lives had also been lost. Still no communications had been established with anyone now trapped in the mine and they estimated that over 140 employees had been in the mines at the time of the explosion based on the timecard records of those still checked in.

Arjun informed him that emergency responders were on the scene but heavy machinery was still on its way. The focus was on re-establishing communications with those stuck in the mines and figuring out how to ensure they had enough oxygen to last until they could be rescued. Arjun assured him that they had established group communications with the families, many who had tried to get to the mines themselves.

“Boss,” Arjun wrapped up. “I’m so sorry I’ve let you down.”

“No you haven’t Arjun,” Xavier offered as reassurance. “The equipment, the procedures, everything is new here trying to extract this Thorium out. If anything, I’ve let you down.” His voice trailed off just saying it.

“Anything you need Arjun, just say the word and it is yours. If I need to hop on a plane I’ll do it. I know you are doing the best you can, thank you.”

Xavier refreshed his news feed. A speculative death toll was now permanently plastered on the bottom left of the screen. He figured they weren’t even close yet.

Upon further reflection, he decided to scrub the scheduled test launch of SpaceCore’s new rocket boosters. He would save this for another news cycle. He certainly didn’t want to be at the center of two explosions in one day.

A master of collaboration, Xavier reached out to Western Australia Mining’s geological engineers in the company’s Perth headquarters, those at the actual mines in Australia along those based in the US and Turkey. While there were some variations in each country’s mining operations, primarily due to legacy operations already in place when they were acquired, enough similarities existed to draw upon the minds of everyone involved. They all needed to figure out how to get those trapped out and they needed to do so quickly. It was, afterall, a matter of life and death. Everything else happening was of secondary concern, this was to become their focus.

He wanted to know the root cause of the accident, but that would have to wait. He could feel pressure mounting within his ribcage, even as he exhaled. The pressure on his temples was

even worse. His hands felt a similar ache to the pain you feel the next day after your first workout in a long time. The clenching during the board meeting had left an indelible imprint on his palms while his knuckles felt almost arthritic.

Despite what was happening internally, he appeared in control externally. He always did. It was his mo, his modus operandi, how others always saw him. Xavier was calm in the face of any situation and this was once again being put to the test.

“Lyra, clear my calendar today.”

“Yes, Xavier.”

“We’ve got some work to do.”

He had some national interviews lined up to correspond with the announcement of the military partnership that afternoon. Xavier didn’t doubt he would have some interviews later that day, but the subject matter would be much different than he had originally planned.

There was an incoming text from James. “150 DEAD...”

Chapter 7 (Aiden - False Positive)

The throbbing in Aiden's foot was endurable, more of an annoyance than anything else. While the coffee blood blend on the floor would be left to dry on its own, he didn't manage to find the time to wrap his foot in some gauze found in the first aid kit employee lounge. He literally followed his own footsteps back to NovaCore's security room as they presented a shape more and more like his own right foot as he approached the door.

He briefly turned on all the lights as he entered the room to check out the mess. He looked at the spot by his keyboard on the floor where the bulk of the reddish-brown liquid remained with pieces of his former mug spread out in a semicircular pattern. The cost of doing business he told himself. It was actually the cost of doing business in his favorite footwear - none.

As Aiden turned the lights back off he saw something move on the floor near his keyboard. He quickly flicked the lights back on and he saw nothing and realized it was just the shadows playing with his mind.

Back to the real issue at hand. Aiden looked at the screen again just to make sure the blanks were still there and not merely a shadow like the one he had just spotted where his feet now disturbed his former coffee. Sure enough they were there, they were no longer in every third data packet but rather in every data packet. Even more strange, this time the code seemed to jump from one area of the packet to another. The data didn't appear to be static, it was changing itself, almost like it was rewriting itself as it was being uploaded into Lyra.

Something to his left appeared as motion. His head naturally turned and he squinted his eyes in the obscure lit room to focus on the source. Nothing.

Aiden's typical insomnia had been on overdrive as he wrote, tested and deployed the latest security upgrade. This wasn't the first time he experienced hallucinations at work and he was determined to fight off the only cure he knew until he had figured things out. Sleep would have to wait because answers couldn't.

Had he misinterpreted the data? Could the data itself be simply corrupt in a way he accounted for yet?

"Sent, rerun all diagnostics again."

"Sure Aiden."

While the security dashboard was repopulating all its figures, Aiden bent down and began gathering pieces of ceramic from the surrounding area. No reason to damage his left foot as well.

He took a couple of paper towels, threw them on the floor, and used circular motions with his left foot to soak up liquid on the floor. He did a poor job and there appeared to be as much semi-dried out left on the floor as absorbed by the formerly white perforated paper beneath his foot now. Regardless, he bent down to gather up the soggy lump of paper and toss it at the garbage bin. Not surprisingly, he missed.

A yawn escaped his lungs mixing into the air like his blood into coffee. Like the blanks into the ones and zero's. He imagined it in the cold air, an isolated visible cloud, rising like steam from a train back a few centuries ago. As it floated it dissipated in the room until nothing remained. At least nothing visible to the naked eye.

More movement, this time to the right. He would not be fooled again.

Sent couldn't see the code, only he could. Sent couldn't see imaginary movement either.

As doubt crept in, Aiden needed to confirm or debunk what was happening.

His fingers feel the familiar touch of plastic on the keyboard. He begins working on a trace program to isolate the IP address in question and all the data that has gone back and forth between NovaCore's servers and the user. His text based line interface would track all the hops, or sequences of routers and network devices, the data took to reach its destination. It would also map the response time of each hop.

Although Sentinal-X had trace programming built in, he needed something a bit more robust to identify where this signal was coming from.

The coding itself was simplistic for Aiden, requiring so little thought process he found himself drifting a little into a light sleep state. Of all the people in the world, he might be the only one who could actually write this in his sleep.

After the code was complete he ran the program, it began following all the different spots the data had traveled through. While the standard trace program had been adequate to pinpoint all the past users, even those using the most sophisticated VPN's, this address had been impossible to pinpoint. "Unknown" was simply unacceptable.

As the program began to follow the data path, or what appeared to be multiple paths, Aiden succumbed to his urgent biological needs. His eyes fluttered, head dropped, mind drifted, he began to fall into a deep slumber.

As his trace program continued to run the data appeared to be bouncing all over the globe. Every region saw its fair share of the action, it was as the data was everywhere at once simultaneously.

Aiden twitched as he entered Stage 1NRM sleep mode much like a computer screen turns dark when entering its own version of sleep mode. His involuntary muscle twitches continued as his

program probed for answers. His mind, now also operating involuntarily, also probed for answers in its own way as if programmed by an unconscious author.

With his shoulders slumped and head tilted back Aiden drifted further into his version of sleep mode. The answers would have to wait until he regained the ability to fully comprehend what he had seen, what he had experienced that evening.

Aiden found himself floating effortlessly in a dense fog, unaware of his surroundings. It wasn't that he was just unaware of his surroundings, he couldn't see anything, it was as if he was blind. Only it wasn't fog, it was more like black smoke that didn't burn his lungs and was as breathable as fresh clean air. It was quiet, but not void of all sound. It was also hot, very hot, but strangely not unpleasant in any way.

Of all his senses, smell seemed to be the most acute. The air had an odor he had never smelt before. It was odd, yet eerily familiar in a way he couldn't understand. While he was floating he didn't touch anything, yet he could feel the air. It was different, it was heavy, very heavy.

"Hello Aiden."

Chapter 8 (Xavier - The Sep)

A storm blew in and soon the NovaCore building was under assault. It rains a little differently in Texas, it's violent. It might not last long, but it will make its presence known. This just wasn't a good old spring thunderstorm, this bad boy came with hail and these ice balls were not messing around. Baseball size he reckoned, maybe softball. He felt sorry for anyone parked in the visitors parking lot without the protection of the garage for a brief moment before he realized how farcical that was given people, his people, were dying.

Aiden sent Arjun a text to let him know he had a team of engineers enlisted to help. There was no reply. He waited a few minutes before calling, distracting himself with the updated newsfeeds before his patience ran out. He couldn't connect. The phone didn't even appear to be ringing on the other line let alone going to voicemail.

The television monitor's volume was turned off but he could see the "Breaking News" headline flash across his screens. "A second explosion in India" appeared below the headline.

"Volume level 20."

The male reporter's stern sounding voice boomed within his office walls. "A second explosion at the Western Australia Mining corporations West Bengal Thorium mines has shook the area once more, this time as first responders working at the scene."

"TV's off."

Xavier frantically tried to call Arjun again. No answer. "I'm so sorry I let you down." Those would be the last words he would ever hear his most trusted employee in all of India speak. The last words of his friend. He had no doubt.

His inbox contained a carefully crafted statement to release to the press. It would need an immediate rewrite.

First he needs to get answers from engineering. While a rescue was still critical, they had better figure out what was causing the explosions first. No rescue operation would be successful if the rescuers themselves were in grave danger.

Xavier knew it was risky to expand the mining operations so quickly, especially given some of their untested proprietary processes. In order to fully capitalize on rapidly growing demand for Thorium they had to take some calculated risks, cutting corners with the idea of not adding unnecessary risks. They also had to rely on a heavy number of contractors given their human resource requirements.

His engineering teams were pouring over all the data from India. All sensors had read normal right up until the moment of the blast.

“Call me ASAP” a new text read from his Vice President of Technical Services Liam Robertson. Xavier clicked on Liam’s name on the message screen and hit the call button. His VP answered before Xavier even heard the ring begin.

“Xavier, I’m pretty sure we’ve isolated where the explosions originated.” Liam spoke with his familiar Australian accent, albeit much more rushed than normal.

“Where?”

“Extraction in the Sep,” Liam informed him. The Sep was short for the Separation Facility and was where the Thorium was separated from the waste rock. After the ore was extracted from the mines it would go to the Sep, a complex system of pipes, vats and chemical tanks, where it would be crushed and ground before being mixed with the solvents Western Australian Mining had pioneered within the industry.

“The good news is that the explosion occurred near the surface and the mine's lower levels likely had structural integrity. The bad news is it is going to be a monumental challenge to get survivors out.” Liam finished by assuring him nobody in engineering would rest until every last miner was accounted for.

As Xavier ended the call he recounted the late nights spent with his chemical engineering team in Perth working to perfect the solution used to derive Thorium from the rock in which it was embedded. He never backed down from a challenge, and that was one of the bigger ones he and his family's company faced to become such a global player in the market. They didn’t cut any corners in their research and testing. How could an explosion occur there and why did the data logs not record any issues? The mix ratios looked perfect in the logs, it didn’t add up.

While Liam had labeled the source area as good news, and it certainly was in terms of giving hope for survivors, there were some collateral damage concerns. Toxic fumes were no doubt being expelled into the air. That was bad enough, radiation leaks were a whole different story.

“TV’s on.” Helicopter and drone footage was beginning to roll in. The scene was horrific with thick plumes for fire and smoke rising from the facility and hanging in the air like a mushroom cloud after a nuclear bomb test. There was a giant hole in the ground where he suspected the Sep once stood. Earth, metal, wood, all lie intertwined and melded together by heat and pressure. Everyone in the Sep was gone, he suspected most of the first responders suffered a similar fate when the second explosion occurred. Based on what he was seeing, the two explosions were likely connected.

Further rescue efforts would have to be postponed until the area was deemed safe enough to enter. He wondered, would it really ever be safe enough to send others in. Liam would be taking a chartered jet from Perth to West Bengal to access the scene on the ground with a team of his most experienced mechanical and chemical engineers accompanying him. He was already

coordinating efforts with local authorities who were dealing above their paygrade in accessing the remaining dangers.

Xavier reviewed the revised corporate statement, it had limited details by design and was heavy on the “we are deeply saddened” empathetic concerns for the miners, their families and everyone impacted by the day's events.

He approved the release, its widespread distribution assured as he returned to the mine's data logs. How could the solvent mix check out yet still cause such a massive explosion? It couldn't. Something was missing. He couldn't figure out what would cause such a reaction. Then it occurred to him, suppose it wasn't “what” caused the reaction, but rather “who” instead.

Chapter 9 (Aiden - Confirmation)

Aiden snapped out of his sleep to full attention like an overworked employee waking up panicked in the middle of the night thinking they had overslept and would soon be fired. How could he have fallen asleep with Lyra's security breached?

The dawn of a new Texas morning filled the room with the natural light it had lacked when Aiden first encountered the anomaly. He jumped at the sight of the intruder standing right next to him in his most vulnerable state.

"Morning," the first visitor of the day declared with her goofy giant smile plastered on her face. Too big of a smile for most people at this hour of the morning, but not for Aiko.

Her bellowing voice, too loud as always and bordering on being obnoxious, was a shock to his system. Aiko's dirty straight jet-black hair was messy as always, as if took a shower the night before and went to bed with wet hair to awake with it dry and having a life of its own. The corners of her mouth and crumbs on her shirt were the remains of her breakfast no doubt.

"Good morning Aiko," Aiden's delayed response followed his usual speaking pattern.

She was the daughter of an executive, always among the first to enter the building. Aiko had been coming to the office everyday since her mother passed away nine months ago. Aiden had always liked computers more than people, but Aiko was the exception because she was exceptional in his view.

"What doing?"

A brief silence filled the air that made others feel uneasy yet had no such effect on Aiko.

"I'm running a trace program to track down some unusual activity on the system," he responded knowing that her thoughts may have already fixated elsewhere.

Her eyes darted around the room. She examined everything as almost as if starting with a blank slate in an equal state of not really seeing anything yet calculating everything at the same time.

"Cool," she clamored indifferently.

Much like himself, Aiko didn't fit society norms. While his refusal to conform was a result of his genius, hers was the result of being extremely high on the autism scale. The six words she had already spoken were unusually high for her and he wondered what had made her more verbal than usual this morning.

At 19 she was no longer in high school and couldn't be left at home by herself. Her father spent long hours in the corporate office and the office had become her de facto babysitter. Aiko

repeated her daily routine like clockwork and she likes to start her “workday” in the security office before making her rounds.

She was now a fixture at NovaCore, akin to a beloved mascot at a school. Everyone knew Aiko and she knew everyone. Out of all the employees, she was the most comfortable with Aiden, perhaps because he was the most uncomfortable person to be around.

Aiden studied the split screen in front of him, on the right closest to him was the security dashboard, to his left the unknown source’s code was filtering through speeds almost incomprehensible even to his eyes. The connection between the mystery user and the entire network of NovaCore’s servers around the world continued and he was the only one who knew it.

To no surprise, everything checked out according to Sentinal-X. All green. North and South America were waking up, much of Asia was going to sleep with Lyra still lying open right next to them. She would monitor their sleep quality and offer tips if needed for a better night's sleep the next day. How many of your other friends offered that kind of assistance?

To his left the code continued to scroll on the screen at a Sonic the Hedgehog pace. Aiden loved gaming as a kid, especially the super retro games.

As best as he could tell, blanks still littered the code; they just resembled blurry random spaces he assumed when multiple blanks were grouped together on the screen display. The only change seemed to be the level of interaction, there was a lot of back and forth happening between the two systems and at ever increasing speeds.

He decided to freeze the screen to be sure. To confirm the blanks were still there, at least for him and that his nap did not simply put an end to his hallucinations which included the anonymous code. The code was there, that he could discern. But what about blanks, they really pushed the boundaries of his oft-vivid imagination.

Before he managed to touch his fingers to the control and S keys to halt the scroll, Aiden peripherally saw Aiko staring intently at the left screen directly in front of her. As he turned her way, she didn’t appear to be blinking. Her fixation was intense. He placed his hand on her shoulder. She didn’t respond.

For Aiko not to respond to physical touch, something had to be seriously wrong. Her instincts were to react violently to any human touch outside of her dad, perhaps her mom as well when she was still alive. Not violent in the sense that she would hurt others, but in the way she recoils from the contact. Her father had sent out an email to all the employees in the NovaCore headquarters regarding his daughter's peculiarities. It proved unnecessary as Aiko won over anybody she met with her adorable looks, awkward smile, and pure pleasant attitude at all times. She was the type of girl everyone wanted to protect, innocent and naive.

“Aiko!”

She blinked. He pulled his hand off her shoulder before having to shake her back. Slowly she turned her head to face him and the familiar toothy grin returned.

“Are you ok? Is something wrong?”

“No,” she said, not shedding the smile.

“What were you looking at?”

“Characters,” Aiko cheerfully responded.

“Characters on the screen?” Aiden stated the obvious.

“Yes.”

“Do you see three characters?” Aiden asked curiously how she might classify the blank.

“No.”

“That’s because there are only two characters, right?” Aiden nodded as if to sway her but she was not to be persuaded so simply.

“No.”

“Exactly how many characters do you see Aiko?”

“Infinite.”

Chapter 10 (Xavier - Countdown)

Sabotage. It made sense given his lengthy list of competitors looking to cash in on the lucrative new Thorium market. The energy of the future was now here and multinational mining corporations were all trying to get their piece of the pie.

Environmentalists had been known to use extreme tactics for the greater good. Xavier knew he would drive himself crazy with no clues yet to go on. Speculating was unproductive and the time called for productivity at the moment.

Every big company had disgruntled workers. Perhaps someone who felt wronged had tried to extract some warped sense of revenge against them. Worse yet, what if the saboteur came from within at a higher level. An image of James Ford immediately flashed across his mind.

Xavier needed clarity. He would have loved to take ten minutes to clear his mind with some transcendental meditation. He also knew there was no way to achieve near emptiness in thought right now. A minor distraction might work. He called to have his little assistant return to his office.

Before her arrival he decided to review the logs once more. That proved fruitless.

"Lyra, please review the separation facilities logs in West Bengal for any solution issues."

"No issues found Xavier," her response was timely and resolute.

He asked Lyra to examine the video footage from the scene. She agreed with Liam's assessment of where the explosion must have originated from.

"How would you secure the area and flush any remaining solution?"

She hesitated. Lyra never hesitated. Xavier took immediate notice.

"There is no quick way to fully secure the area, I'm sorry Xavier. I've run a full risk analysis of the situation and you are going to have to weigh the tradeoff between rescue operations and putting those performing it in danger." She sounded different than normal, a little more robotic, could this actually be getting to her. Impossible he realized, the stress of it all was affecting his perceptions and his nerves must be getting the best of him for the first time.

The losses, the unimaginable losses, weighed on him. Like a number of his billionaire counterparts, the money did not drive him, it was the laser focused mission of achieving the impossible. He had leveraged his family's legacy and taken it from a relatively unknown regional mining operation in west Australia to a multinational powerhouse. Furthermore, it had helped birth a string of technology companies his predecessors could never have imagined digging deep below the surface.

Three words appeared in his messages. "We've made contact."

The message was from Liam, no doubt in flight, and he could see another message was on its way. "We have survivors, trying to get a handle on what is happening inside the mine" followed.

"Fantastic news, let me know whatever you find out as soon as you find out" Xavier typed out in reply.

"Lyra, please draft another press release about the survivors."

"Done," there was no hesitation this time. "Xavier, you have an incoming call from Woody."

"Ignore," he stated, having no time to talk to Justin Sanders, President of SpaceCore at the moment. Woody knew what he was facing today and was surprised by the call. He was the first black man to lead a major space company and he was as loyal as they come. He was also smart, too smart to try to reach him now. There were few people Xavier respected more than man that helped him open up a whole new frontier of mining. Their otherworldly adventures were due, in large part, to Woody's Thorium based nuclear engine built on the capital generated from Xavier's gamble on element 90. The engine itself ripped right from the pages of Woody's MIT doctoral thesis.

The news segment had finally shifted away from the breaking news and into a daily recap called The Lab which covered all the science news of the day. He paid little attention to the first story, something about trials of a promising new cancer treatment. The second story caught his attention a little more, a snag in SpaceX's attempt to upgrade their 3D printing capabilities on the red planet. The third story was about a new interstellar visitor, this one discovered by the Laser Interferometer Space Antenna (LISA).

The announcer had some breaking space news. "And just confirmed, SpaceCore has restarted the countdown of its test launch on the Texas coast. Earlier today the company had cancelled the launch due to the explosions in India at its sister company the West Australian Mining Company."

"Incomming call marked urgent from Woody," Lyra announced more convincingly this time.

"TV volume off and accept," came two quick commands as he rolled straight into his conversation. "Woody."

"X," Justin started. Only those with a certain level of relationship with Xavier called him X.

"What the hell is happening Woody, I told you to scrub the launch." Xavier was calm, but his voice was stern.

"We did."

“Then why did you start it back up?” Xavier interrupted.

“X, we didn’t initiate the restarting of the countdown and we can’t seem to find a way to stop it.”

Chapter 11 (Aiden - Confirmation)

Aiden was taken aback by what Aiko said. Infinite.

He always paused before responding to someone, but her response required much deeper consideration. Infinite. Honestly, he could understand why she might say something wrong like “one” because there were only numbers along with the blanks so perhaps she classified all numbers as a character. It was a stretch, but feasible. Perhaps more feasible than Aiko even being able to see the blanks.

Infinite. It wasn’t just the word, it was the way she said it. Aiko always was pleasant, it was perhaps the most consistent force in the entire universe. Her smile was as usual, as if rehearsed endlessly. But her eyes told a different story, as did her voice. It was negligible, but it was there. Pleasant, but with the slightest touch of robotic tone. It almost felt sarcastic, as if the answer was so obvious he must be a fool to miss it.

Aiko’s eyes were those of her mothers with the angular eye shape reflecting her Japanese heritage. She would be mistaken for Chinese given her fathers influence, adding slightly more size and roundness than traditionally found in Japan. They returned to the screen, he could see her entering into a trance like laser focus on the “infinite” set of characters.

“Aiko!” He was even more demanding this time. His volume worked, she returned to blinking and fully looked his way.

Aiden froze the screen and picked up a pen with a soft rubber tip on the end that replicated a human finger on a touch screen if needed. Already he could see his visions from before were no hallucinations. He touched the pen’s rubber back end to the monitor.

“What character is this?”

“Zero.” Aiko replied.

He was on the right track. “And what character is this?”

“One.”

With the easy questions behind them Aiden turned to the matter at hand. “Aiko, what character do you see here?”

“Switch.”

What kind of answer is switch? Good thing he was adept at delays. Aiden needed time to configure his next question given Aiko’s general lack of explanation. Any hopes of zeroing in on the meaning of “switch” would come down to his ability to narrow down the “infinite” number of

possibilities she might mean. The frustrating thing was, he couldn't even think of one possibility right now.

"Do you mean like a light switch, on and off?" Aiden went to the obvious but it still felt like a long shot.

"No."

This was going to take some work. Some imagination on his part. It is a good thing he is coming off a good night's sleep. He humored himself. Perhaps it isn't a physical switch, more of a metaphorical one. That would certainly open up the possibilities.

An access card beep sounds before the door swings ajar. An Executive Assistant walks into the room and spots Aiko. "Sweetheart, it's time to go." Aiko doesn't respond, she is glued one again to the screen which is scrolling again despite the fact that Aiden never touched his keyboard.

"Honey, we have to go," she insisted, but in that southern accent that makes it sound more like a sweet suggestion. "Your daddy wants you to come back to the learning lab."

Right on cue, right on Aiko's routine yet nothing about her lack of response screamed routine.

"Aiko, honey!" A look of concern crossed the assistant's face. "Aiko!"

She shifted her gaze. "Aiden, what is wrong with her?"

Now that was a loaded question and he wasn't about to answer. Perhaps there was nothing wrong with her at all. Perhaps there are an infinite number of things wrong. One thing wrong for sure was she wasn't blinking. That could not be good. That was actually disturbing.

"Aiden!" She responded before he did. That was not all that uncommon for him but her time restrictions seemed much stricter than most.

He put his computer in sleep mode and both screens went dark. Aiko looked at him, her teeth appeared once more. "Time to go see your daddy," Aiden informed her as she blinked like a nod of understanding.

Her smile impossibly widened as she faced her escort back to the place she spent much of her time now. She didn't mind, she could distract herself from thinking about her mom in there as well as any place and it certainly didn't remind her of her mother like her entire house.

Aiden watched the pair exit the room. Aiko was offered a hand, a gesture of politeness, that she would always decline. Not with words, but rather a pleasant indifference as they went off on their merry way as evidenced by the little skip in Aiko's walk.

He was immediately engrossed in the word again - switch. He tried to forget the image of a physical light switch on a wall, but he just couldn't fully extinguish it. It lingered, and then it began to move. Up and down. If it was an actual switch, then what the hell was it turning on and off? What was its function?

"Occam's razor," Aiden let slip in a hushed tone only for himself.

Aiden had a propensity to overthink everything. Overanalyze everything. Nothing was ever simple in his mind, it was always complex. Occam's razor was his trick to hijack his own brain. Cause a glitz in the matrix. Get back to basics. There had to be a simple answer to Aiko's response.

That's it. It's Aiko. If the code is a message from a sender, then certainly there was a chance that the receiver simply couldn't understand it. Even better, there is a bigger chance she couldn't even read it, he certainly couldn't at that speed.

Case dismissed, he smugly told himself. But the smugness surely dissipated as he chewed on it a bit further. She said "switch" with no hesitation. No doubt about it and he was too ignorant to see it. No, it had to mean something he was just missing. Occam's razor, he repeated, only this time silently.

"Shit!" He shouted loud enough to hear if someone was standing just outside the door. It is a switch, that wasn't the question. What does it turn on and off? That was now obvious. He did feel foolish. He was disappointed with himself. He knew what the switch was doing but the reason behind it was not yet clear but he had an idea on how to find out.

Chapter 12 (Xavier - Rin)

"Well you'd better find a way to stop the countdown," Xavier commanded.

"Yes sir," Woody agreed as he ended the call.

His doors opened much like the ones might imagine entering the bridge of a retro show spacecraft and he was the Commander.

"Hi daddy," Aiko announced as she raced in to beat her escort. "What doing?"

"I'm looking at the most beautiful girl in the world," he exclaimed. It wasn't hard, he truly believed it. He could see her mother in her and she had held that title until their daughter's birth. Aiko wasn't just beautiful, she was special in every sense. Sure every parent thought that of their own children, but her impact on others was different. She unarmed even the most guarded adult. She captivated them with her innocence, her joyfulness, and her complete carefree attitude towards the decadent world around her. She was the bright star burning in a dystopian universe.

"Aiko, baby, I'm really sorry but daddy has a couple of fires to put out right now but I'll come get you soon." He then instructed her escort to take her to the Learning Lab and get her some juice and apples.

As the two exited, he was thankful for the momentary distraction from the chaos at hand. The passing of her mother, his wife, had been hard on him. It was beyond devastating for Aiko. She couldn't fully grasp the concept of death, anything outside of numbers and patterns were difficult for her. Autism might have taken away some of her ability to understand such things, but it also gave her some things few possessed. She was his little Japanese angel, she was everyone's angel.

Rin was the only person who truly ever loved Xavier before Aiko. What she lacked for in bulk, she made up for in attitude. Rin was aggressive, certainly out of character for her cultural identity. Her features were fragile, delicate, and mesmerizingly beautiful. More importantly, she knew how to handle X. Some would say she was the ying to his yang but that was a lie. She was as much ying as he was, even more so when necessary. While Xavier was the boss to tens of thousands, he only had one boss and she filled the role admirably.

They were older when they conceived with the assistance of in vitro fertilization after years of trying. They knew the risks of an older pregnancy, but the idea of a family was too great. With modern technology there were hints of "abnormalities" before Aiko's birth, but the couple didn't care. Of course they wanted a healthy baby, a normal baby if you will, but any baby was seen as a blessing and they certainly had all the resources to handle anything thrown their way.

Aiko was the reason Lyra existed in the first place. Aiden hired a brilliant developer to build a companion for his daughter. It would serve as her teacher, he guide in a place not built for her. It would be a caretaker unlike any other scene. It would be her best friend, never to judge his little girl and always help protect her from the concerns of the outside world.

Lyra worked so well, the pair decided to roll out the program to other families with autism. Soon, the entire family would be using the app and Lyra wasn't just for autistic kids anymore. A general, even more robust version of the application was released and within hours was going viral. Everyone was talking about the new companion app and everyone had downloaded it.

The adoption rate was so fast governments took notice. The security risk was seen as great as unlike many AI programs, Lyra had an extensive memory and knew everything about its user. If feeling as though you were being listened to in the past was unsettling, the new level of unending surveillance was crippling to those with access to sensitive data. Protocols were put in place to ban her connection to users in such incidents, but the new military contracts would put a dent in the apprehension for usage even at the highest clearance levels.

New encryption methods locked her data down. They could not be broken, NovaCore security measures made sure of it.

"Lyra, put the Learning Lab cameras on monitor three." A video of Aiko appeared. She was playing a game based on creating an equation to reach a set of randomly generated numbers. She solved set after set as easy as children her age might count to ten. She was rewarded with an NFT, or digital sticker, every time she completed a puzzle. Xavier wondered if he would need to increase the hard drive to store the electronic bounties she was piling up.

A new text from Woody lit up his phone. "We still can't stop the countdown, we are trying to manually disconnect the boosters right now just in case."

His screensaver showed a picture of the three of them inside of Aiko's school. It was taken just days before Rin's death the previous fall. Just days before her accident on that fateful day. Her birthday. Aiko's birthday as well. September third.

Rin had been traveling along Interstate 35 near downtown, but the University of Texas campus. She had been traveling northbound on the upper half of the highway. The elevated road was usually filled with traffic, but it was late and the cars were moving. An HEB grocery truck she was passing blew its front driver's side tire and swerved into her lane clipping the Tesla in front of her. As it veered into the wall, her car collided with it and launched over the railing. Rin's electric Lexus landed upside down in a Mexican restaurant's parking lot below. She was gone in an instant, their lives shattered like the SUV's windshield. Pieces of glass dispersed in every direction, a testament to a brutal reminder of how truly fragile she was. How fragile we all are.

They had met in college and reconnected by chance in Sydney when Rin was on a business trip for the start up she founded. Initially the two clashed, brought together within a physics group

project that would push the boundaries of their underclassmen knowledge of the subject. They wrestled for control of the group, and it was Rin that would ultimately claim the prize. Xavier was pissed, more importantly he was impressed.

He could smell the coconut of her body lotion still when he closed his eyes. He could see her love for him still in hers as she looked up at him in a common embrace. He could still hear her words if he listened hard enough. "Xavier, please take care of our little baby." The last words she uttered when heading to south Austin for a late dinner meeting with a prospective vendor.

She would leave him as Aiko's sole caretaker for the evening. She would leave him with that role forever.

"X, we have an even bigger problem" the new message from Woody read.

Chapter 13 (Aiden - Where?)

There was only one thing the switch could turn on and off, that was the zero's and one's that had been staring in his face the whole time. It was the only explanation and he was still upset at himself for taking so long to see it.

He had bigger fish to fry now. The blanks had been turning on and off the characters, so what appeared as a zero could be changed to a one and vice versa. But why? The thought intrigued him. He imagined a code that could change as it was being read? Code as he knew it was static. His algorithms appear to be able to solve virtually anything, yet they still were just a series of two numbers, on and off, true or false, yes or no. But what if the command could change between on and off? How could you even program such a language? And why?

He pondered the questions as he began to tackle the task of understanding the sequence of the command. It could come before or after the character it was changing and he didn't know yet which one it was. No bother, he could run two simple scripts to figure that out. But did it reverse the character every time? If not, what triggered its effect?

He hadn't cracked code at any significant level since his old hackathon days in college. It had been probably six or seven years, but he didn't doubt the best hacker back in the day was still up to the challenge. He might not have ever seen a code like this before, but this code had never seen a hacker quite the level of Aiden either.

Aiden, out of old habits, set the timer on his phone. Challenge accepted.

It wasn't a matter of if he would crack the code. It was a matter of how long. His confidence was beaming and it was go time.

He tapped eight keys with his fingers along with the space bar with both thumbs. His light touch noticed the braille-like raised dot on the F and J keys sending a spark of anticipation up both his index fingers. The backlighting of his keys flared up, yet his deft touch did not yet trigger a full keystroke. The monitor stood there quietly, awaiting to be fed via the keyboard's bluetooth instructions. A second had not yet passed quite yet on the timer yet he was already restless. What was taking so long.

Finally the right side of the monitor switched from the dashboard to his programming screen. His fingers became the vessel for his quest to answer his questions and in under four minutes he was ready. He did cheat a little, asking his customer AI builder to copy and alter some previously written code. But it was impressive none-the-less.

He ordered the code to execute the program and it read the unique code as instructed. As the processing began, his temporary free time allowed him to wrestle again with the question of why. Why would someone program software to change itself on the fly?

The question hung in his mind, lingering like a hot air balloon drifting effortlessly in a gentle breeze. Only this didn't feel so gentle.

The code just wasn't changing, it was virtually impossible to see. Someone tried to hide the code if it was indeed found, however unlikely, it was nearly impossible to read.

After the program wrapped up its initial reading of the data it spit out the results. While he could still not tell exactly what the code was doing, it did appear that the blanks did not reverse the number in front of it. It also did not appear to reverse the number directly behind it.

Maybe it isn't a switch afterall Aiden questioned himself once again. If it was, he certainly didn't understand its structure or its purpose. He felt stuck. He needed a different approach. He closed his eyes hoping for a fresh start, for a spark of understanding, for any clue that might add some perspective to what was happening.

Aiden was so focused on code uploading into NovaCore's servers, he had completely ignored the other half of the equation. The half he could understand. The half his programming produced.

If he couldn't figure out what the user was sending, at least he could look at what information he was receiving.

Aiden examined the data outflow and it was extensive. The drain on NovaCore's servers might have been more noticeable had it been concentrated just at the corporate headquarters near the state capital. But this user had somehow tapped into the companies data centers around the world so instead of seeing a single giant waterfall of data, there were just thousands upon thousands of creeks leaking data into the unknown.

Trying to see all the outflowing data was well beyond his capabilities so he limited the data set to just that being sent from HQ. From what he could see initially it was pretty clear, the user was downloading all the data used to train Lyra's AI model.

NovaCore opted for a Large Language Model, or LLM, approach to building Lyra. Almost all the early artificial intelligence programs used this approach and it was still what most AI programs were built on. The larger the data set, the better the model. Lyra reviewed essentially every word ever written by man to interpret his world and used this data to output its responses just like how your phone predicts the words you want to use based on the first few characters you type and prior contents for context.

Was all the data they used to program Lyra being stolen to program a rival AI? That would be strange as it wasn't proprietary information and virtually anyone with the right knowledge and programming skills could scrape all the information it needed from the internet.

The data wasn't NovaCore's secret sauce, Aiden's algorithms were. They assigned credibility to sources making the data more reliable and useful. They engineered Lyra's deep learning

capabilities and introduced complex pattern recognition and reasoning into the equation. The data the unknown user was seeking was vast, but not particularly helpful without the programming to turn it into a predictable coherent output language. Why access the information on the internet through NovaCore's servers? There had to be millions of easier ways to achieve the same objective, perhaps infinite ways.

Aiden next looked at the data being accessed from the nearest data center, this one in far west Texas that ran primarily on wind and solar power. That data flow there was more back and forth, more questions and answers reflecting how most users interact with their digital BFF. At least he assumed the incoming data were questions based on answers Lyra seemed to be giving back.

The questions indicated an insatiable appetite for knowledge covering every subject one might consider. The user not only wanted to access all of human written knowledge, he wanted to know exactly how to use that knowledge in every possible scenario.

Aiden chewed on that like one might feverishly attack everlasting gum. Jaws clenched, he could feel his persistent bewilderment impacting him physically.

He continued to gnaw at his own incomprehension as an alert pop up window appeared on his dashboard. The trace program he had written in the middle of the night and had been running in the background for hours had finally completed its task. He was so absorbed with the questions of who, what, and why that he has forgotten one of the most intriguing questions of all. Where?

As his cursor moved towards the pop up window that would take him to his answer he imagined seeing the IP address land on the location of a current or start up AI competitor.

Aiden trace program showed the unknown user's address jumping from one location to another. He had never seen a VPN with this level of sophistication, this level of complexity. Still, the trace program did its job and it did it well. All the fake digital locations lit up his screen like an overzealous Christmas light display. Every region was covered, even Antarctica.

The unknown user's location was finally mapped but the region remained marked "Unknown" on the screen. But he could see where the data was headed to and that wasn't on the map itself, but rather a direction - up. Data traveled up all the time, bouncing off satellites back down to their desired locations. Only this time the data didn't bounce back down, it didn't bounce at all.

As far as Aiden could tell it was just sent out into the void. It was as if the data had been sent to a satellite that no longer existed. The data was like a beam of sunlight heading for the Earth and not finding it, only to continue on its journey at light speed into the abyss of space.

That couldn't be accurate, however, if the data signal was going really going nowhere how could a response be generated? There was no way for the data to travel unimaginable distances with nearly immediate interactions from the ultimate user.

Either his trace program was lying or Aiden was now completely out of his element. Either way, it was trouble. Either way, he wasn't getting out of this alone and he knew it.

Chapter 14 (Xavier - T minus Ten)

After the message about a bigger problem than the countdown, Woody sends Xavier an image of one of the fuel tanks. He can see the condensation freezing outside of the tank.

“Call Woody.” Lyra gets Justin on the line as Xavier zooms in on the photo.

“X.”

“Woody, why is condensation forming on the fuel tanks?”

“We don’t know yet, all systems are indicating there are no problems with the tanks,” Woody’s pace was quicker than normal. “All the pressure readings look absolutely fine, I’m sending you the current read outs.”

“Got them.” Xavier looked over the status reports and they matched what Woody had reported.

“This doesn’t add up,” Woody knew Xavier wouldn’t be able to spot anything unusual in the numbers either. “Honestly I’m at a loss right now but the whole team is trying to halt the countdown again.”

“We must stop the launch at all costs, we can’t afford another disaster today. If we have pressure issues in the tanks and attempt to launch you know what will happen.”

“Of course,” Woody had no doubt the outcome they faced with a launch. “I’m about to get an update from our technicians on the ground. I’ll let you know what they find.”

“Please do, and Woody,” he hesitated. “You must get all personnel away from the launch site if we can’t scrub the launch. No more losses today.”

Xavier plucked a hair from his right eyebrow but it provided no insight. He saw Aiko still playing her game on the monitor, oblivious to the bedlam her father was facing.

“Lyra, pull up the SpaceCore launch feed on monitor four.” The rocket appeared on screen, thankfully the countdown clock did not show on the feed.

“Zoom in on the fuel tanks.”

The image remained unchanged. “Lyra, zoom in on the fuel tanks.” Nothing. The picture could have been frozen except evidence of wind in the form of distant clouds told him otherwise. The fact that Lyra didn’t try to explain her noncompliance with his request was shocking.

“Lyra!” His voice was raised as if she needed to hear more volume to grab her attention much like a human might.

“Yes Xavier.”

“Why are you not zooming in on the fuel tank?”

“I’m unable to process that command right now.”

“Why?”

“I’m sorry Xavier, I’m unable to connect to the launch camera.”

“Then just zoom in on the feed itself.”

“I’m unable to enlarge the feed right now.”

Xavier was more than perplexed at her response. This was a simple request. This was a simple task for Lyra to perform. This was utter madness.

“You have a new message from James Ford.”

“Please read.”

“Hey Gandalf, why don’t you wave your magic and stop the SpaceCore launch before disaster strikes again.”

“Delete.”

“Incomming call from Liam.”

“Accept.” “Liam.”

“We’ve arrived at the scene and have the drones up, I’ll have a feed for you momentarily.”

“What’s your quick initial assessment?” It was a question that Xavier didn’t know if he really wanted an answer to.

“It was definitely the Sep. If I had to guess at this point I’d say we had pressure issues in there. Xavier, we have established communications with some of the minors in the eastern vein. There are injuries, but they are mostly ok for now. They have no power down there outside of a few generators and the elevator shafts have collapsed.”

“What is your plan?”

Liam coughed before continuing, clearing his lungs with a forceful exhale. “We are trying to get a survivor headcount and establish communications with other areas of the mine right now. We have fire retardant foam on its way to be dropped over the site to stabilize the area the best we can and will physically test the oxygen once we can get up close.”

“How many losses, what does your gut tell you?”

“Anyone working in or near the Sept is gone. That is usually a fairly small group but we were right at the crossover. We accessed the last footage of the elevator cams, the east one had a dozen workers at the time of the explosion, the west vator had nine. Outside of that I’m hoping for minimal casualties. We also lost a number of first responders in the second explosion, I’m guessing around twenty or so.”

“Thank you for the update Liam, I know this is very hard. Please be safe and no more losses OK.”

“You’ve got it. I’ll keep you updated and send the link to the drone footage right now.”

Xavier ended the call and instructed Lyra to pull up the drone footage on monitor five. She had no trouble with this request and a scene of destruction emerged. He had seen some aerial shots from the news helicopters already so he was prepared for the ensuing footage to come. Still, this would be the biggest mining disaster under his charge by a wide margin and you can never truly prepare for that.

He marveled at the three screens grabbing his attention. His daughter on screen three, happy as can be in the relatively calm NovaCare headquarters. Screen four was the launch scene, it also looked relatively calm but he knew that scene might look dramatically different if they couldn’t cancel the launch. Screen five was of a harrowing scene, black smoke rising from the rubble where the mine entrance once sat.

Looking at all three he felt a connection and it was as if only the images coming in from India revealed the truth. The other two screens were a lie, a veil of business as usual on the surface, but with something happening below that he couldn’t yet see. He had no control and Xavier was all about control. He couldn’t accept that fact, he had to seize it back.

He could see Aiko pleased with herself for solving another puzzle. He had three puzzles of his own right now and wished he could solve them as easily as his gifted daughter.

Suddenly a countdown clock appears on the launch screen, T minus ten minutes.

Chapter 15 (Aiden - Location, Location, Location)

Before Aiden engaged in outside help, he double backed on his approach. He decided to rerun the trace program, this would take considerable time. He also had Sentinel-X review the program's code while he did the same concurrently. Sent wasn't exactly outside help, but it was a second pair of eyeballs reviewing his programming.

Both of them came to the same conclusion, the issue wasn't his code and the program wasn't lying unless some had tricked it to do so. With unknown coding suspiciously infiltrating NovaCore's own servers, that was a real possibility.

At this point Aiden was torn as to his next course of action. As the Chief Data Scientist and essentially head of data security, it was his responsibility to report his findings to the corporate executives. His findings, however, raised a lot more questions than answers and he could only imagine how the conversations might go. The time it would take to not only explain what he knew, but what he didn't would severely curtail his ability to get to the bottom of things in a timely manner. He didn't, after all, really understand the threat or even if it was a threat at all.

It didn't take much thought to dissuade Aiden from following protocol in a situation like this. Any excuse to avoid direct conversations with others was a good one for this socially awkward wonderkid. No, his time would be better spent on the problem. At least until he had real actionable information to share.

His stomach growled like a wolf. Not just any wolf, the alpha male. Aiden ate nearly as much as he slept, he didn't have time for such trivial pursuits in life. He opened his desk drawer and pulled out a plant based protein bar. "Breakfast of champions," he assured himself. He washed it down with a room temperature energy drink, another staple of his nutritionally deficient diet.

Location, location, location is a mantra within the real estate industry, attributed to the British real estate tycoon Lord Harold Samuel although that account is certainly disputed. What was not in dispute was Aiden's similar mantra when contemplating where the unknown user's downloads were going. He couldn't imagine any way the tracer program could misdiagnose the concluding landing spot for Lyra's data stream. He also couldn't imagine that same data just aimlessly exiting the outermost reaches of the solar system like NASA's Voyager 1 spacecraft launched back in 1977 and finally hitting interstellar space in 2012.

Could it be some distant satellite like Voyager interfacing with Lyra? Aiden almost laughed at the absurdity of his own thought knowing the computational limitations of such ancient technology.

Instead of waiting for the trace program to finish this time, he took a peek at its progress and could tell the patterns were repeating. While he couldn't be certain that the end result would match again, it did seem rather inevitable based on the early returns he was seeing.

If the trace did indeed have the same ending point, that only proved his program replicated its previous conclusion. He would still need confirmation of where the data signal was being sent from another source. There would be other consequences of a signal like that, he just needed to find them.

The simple fact that the data was potentially traveling so far should make it stick out like a sore thumb and he knew how.

Aiden turned to the dashboard, green as a cucumber as it had been since he installed the latest security update at three in the morning. He clicked on the latency window to dive further into the numbers. He had to go from system wide latency big picture level down to the individual level user, however, he had never programmed the system to look at latency at that level of detail. Why would he?

He could get there of course, it would just take some additional programming and he was about to get to work when he thought of a much easier way. Aiden could zoom down to the regional level and look at the latency details for each region which simply mirrored the seven continents on Earth. His drill down revealed figures for North America, South America, Africa, Europe, Asia, Australia and Antarctica as usual. But this time an eighth region was also represented as "Unknown" and that mere classification would speed his investigation.

As Aiden clicked on the Unknown tab, the window began to reveal the latency figures for just one user, the only user in the region. If the user was accessing Lyra from some distant point well out in space, the delay in interactions would be noticeable. The measurement would still be in milliseconds, perhaps longer, but it would be present.

At the average level for the Unknown region, the latency figures looked rather normal. One might expect some fluctuation, but given the large number of locations the data was pinging around the average of all of them was well within what one might expect. To Sentinel-X, everything was still green.

Individual signals showed little variation, although as one might expect when a signal must relay to a satellite instead of simply between servers located on Earth there is an almost imperceptible lag. Spotting the difference was akin to having to look at a still frame of a photo finish in a race. At full speed the two contestants appear to have the exact same times, but things can always be broken down into smaller and smaller components.

As he looked at the latency charts for the unknown user the number of locations made the data overwhelming, so many lines of information. He began scrolling hoping his eyes would spot something that stood out, but scrolling through the data would not be the answer.

Aiden was no longer operating at full power, his battery was draining, his performance was dragging. He was self aware of his current limitations. He needed more food, he needed more sleep, he needed more caffeine. But outside stimulants would have to wait.

He knew what he was looking for and there was a simple way to find it. Aiden clicked on the latency label at the top of the chart, the locations all sorted by the latency time with the shortest time listed at the top. He clicked the label again and the sort order reversed.

B-I-N-G-O!

Chapter 16 (Xavier - Old School)

9:59, 9:58, 9:57, time was ticking down.

Xavier was more hands off with SpaceCore than his other business interest. He knew where the future of mining laid, but he also knew his own limitations. As much as space fascinated him and as much time as he spent studying it, for others it was a singular passion, a singular focus for which his time allocations could not compete. This was especially true of Woody who not only was the geekiest of the geeks, he also operated out of a place of great obligatory responsibility to be a role model for African Americans in terms of both science and leadership. Xavier trusted him implicitly but the sight of the dwindling numbers would put his trust to the test.

“Call Woody.”

“I know,” Justin said, easily anticipating the jist of Xavier’s call. “We’ve been hacked, that is the only possible explanation.”

“I agree,” Xavier had come to the same conclusion. The appearance of the countdown seemed to confirm it. “I’m going to get my NovaCore working on that right now but we don’t have time to identify and cut off whoever has control of the system.”

“We are working on ways to manually stop the launch X. There is something else, I asked one of our launch technicians to physically check the analog gauges on the fuel tanks. Xavier,” he rarely called X by his first name, “they don’t match the digital read outs at all and thermal imaging shows temperature fluctuations we aren’t seeing either. Not only does someone have control of our systems, they are controlling what we are seeing.”

“Jesus Christ Woody, we’ve got less than seven minutes left to get control back.”

“Hold on, Kent Mahan is flagging me down frantically.”

Xavier couldn’t quite make out the conversation between the two but it sounded chaotic matching the background noise of SpaceCore employees all in a panic to figure things out. Kent was one of the oldest employees at the company, a former NASA employee and as old school as one might get in the space industry. The ancient engineer had the whitest hair Xavier had ever seen, nobody would have ever suggested he was turning gray. There simply was nothing gray about it. He was a bunch of pens in his shirt pocket kind of guy, an old school techie and while he didn’t necessarily look the part his colleagues referred to him as “Professor.”

When launching a startup space company experience matters. Kent Mahan was a no-brainer to add to the engineering team. He had been in the industry since the space shuttle days and seen just about everything that would be thrown their way. The Professor had no desire to lead, he was a technical problem solver at heart, and SpaceCore offered a lot of problems when he signed on.

“X,” Woody returned to the conversation, “we have an idea. Essentially we have an old hardwired abort controller for the ignition process. It’s almost a forgotten technology built into the ignition system we copied from NASA. Kent’s turning it off now.”

The countdown was at 3:35, 3:34, 3:33.

“Well we’d better pray that works.” Xavier was not a religious man by any means, but he was reaching for answers now.

He could hear Kent saying he had shut it off breathlessly. Xavier wondered how far the valve, button, lever or whatever it was from where Woody was now.

“OK it off X, however there’s a problem. We won’t know if it worked until ignition attempts to kick on.”

“Shit!” Xavier slipped. “Get everyone as far away from that rocket as possible. Guess we will find out in about two minutes.

Those two minutes felt as long as the same timeframe at the end of a NBA championship series game seven when both coaches still had their full allotment of time outs in a tight game. Both sides watched helplessly as the timer crawled towards impending disaster.

“X, ignition is set for T minus three seconds for this launch.” Justin didn’t know if Xavier was aware of the exact launch procedure. Casual observers would assume ignition takes place at zero when liftoff occurs. In fact, modern launch ignition takes place later than the old days with Apollo Saturn V rockets ignition sequence began around the nine second mark while the space shuttles three main engines would ignite at 6.6 seconds. In more modern times the first Artemis liquid filled RS-25 engines would be given their command at T minus ten seconds with startup at 6.3 seconds.

SpaceCore had built their sequence to mirror that of SpaceX’s Falcon Merlin engines at the three second mark like many of their space industry corporate rivals. It was basically industry standard and followed a chilling of the engines at the seven second mark preventing thermal shock.

The two remained silent, there was nothing to do but wait.

10, 9, 8, 7, 6, 5, 4, 3. The moment of truth was upon them but they wouldn’t know the fate of their efforts for another two seconds. 2, 1.

The image of the rocket on Xavier’s screen remained fixed. There was no ignition.

“You did it Woody!” Xavier was calm as always, but there was a hint of relief breaking through his Aussie accent.

“Looks like it. We have a lot of work to do to figure things out here X.” It didn’t sound like the normal jubilation of a successful launch in the background, but there were cheers none-the-less for perhaps the first time ever associated with a scrubbed rocket launch.

“I’ll let you get to it. I’m headed to security to see if they can help us determine where the cyberattack is coming from and how to stop it.”

“Sounds good X, we appreciate any help you can provide.”

Xavier left this office and headed towards the Training Lab to check on Aiko. He peeked in and saw her intently focused on her game. She didn’t notice the intruder. He left her to her puzzles rather than bring her along. He usually preferred to bring her to security with him, she was the perfect buffer to Aiden and the two got along great. No, he would be taking this trip solo as hastily as he could muster.

Chapter 17 (Aiden - Fibonacci)

Aiden stared at the screen affixed. The answer was there, hidden in the milliseconds of response lag between Lyra and the final resting spot for data being sent to the unknown user. His location was far away, very far, as measured in the time data took to travel back and forth.

He was no physicist, but he could try to get a rough estimate of the distance based on lag itself. While it didn't make any sense, the data didn't lie. Two separate independent tests pointed to the same conclusion. They each confirmed what the other test suggested.

Aiden was now focused on that one location point and what the data flow looked like. That is where the invasive code was coming from and where the stolen data was going. He began looking at the latency logs for that single location and saw something else in the data that didn't make sense. It was even more subtle, much more indiscernible, than the data time lag itself. But the pattern was there regardless and it sent a shiver through Aiden's thin frame. The lag itself was reducing, becoming smaller and smaller.

It couldn't be some ancient satellite, he already knew that given the amount of processing power it would have to have to accomplish what it was doing, but he would also expect the lag to increase as the capsule ventured further away from its Earthly point of origin. No, this vessel was doing the opposite, if it was a vessel at all, it was getting closer.

"This is above my pay grade," Aiden told himself. He could not comprehend what it all meant and he had no clue what to do next.

His anxiety grew much in the same way it did when he had to interact with other people. He was comfortable around computers, around humans he simply wasn't. The dance of small talk and intolerable reading of non-verbal clues was agonizing to him.

Aiden had come up with his own personalized way of dealing with the stress of dealing with people and he would use that technique now to calm his nerves.

"0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377," Aiden mumbled the Fibonacci Sequence to himself.

The numbers, which often show up unexpectedly in mathematics, soothed him in a way that only repeated patterns can. Each new digit was simply the sum of the previous two and each acted like a deep breath one might experience in a meditative state.

As Aiden's heart rate slowed, his mind began to clear. He needed a plan. He needed a next step.

There was no use harping on what a distant user out in space moving towards the planet might mean. Understanding that, for now, was out of the question. But what a question it was. The

answer had ramifications he did not want to consider right now and despite the confirmation from his examination of latency figures, perhaps what he was seeing was some form of an illusion. Perhaps he was seeing something someone wanted him to see, a false narrative to deceive him from what was really happening.

He had to hand it to his adversary, which is how he interpreted the unknown user at this point, the mirage he created was so elaborate he couldn't imagine how it could be pulled off. Aiden was competitive, his hackathon history pointed to that fact, as did his professional success. Most of the time his only true rival was himself, he always tried to outdo what he had done before. But now he found a new rival and what a worthy adversary he had turned out to be.

Aiden closed his eyes momentarily, trying to picture in his mind what his new opponent looked like. What setting he was currently in. The best he could come up with looked eerily similar to Aiden himself, with a wry smile of satisfaction sitting in a room not dissimilar to the one he found himself currently occupying.

He began to obsess on his opponent, driving him mad, he would not be beaten. He would cut off his ear like Vincent van Gogh, after an argument with Paul Gauguin, if he had to gain an edge. He had to win at any cost.

Aiden didn't just want to win, he wanted to obliterate his opponent. He sought to destroy his foe in a way that didn't just signify a victory, it would signify the end of his new found nemesis. He was the defending champion and the challenger just took round one in a rousing fashion. Aiden was knocked down to the canvas not long after the bell sounded to start the match and he had to scramble back to his feet before the count reached ten.

But how could he hit an opponent with such a superior reach advantage?

He did get to his feet, the bell did ring again to end the initial round and Aiden retreated to his corner. He had a bad gash on his brow and no cutman to stop the bleeding. He sat slumped in his stool, red liquid obscuring his view. His opponent was not only unreachable, he couldn't make out any details of the man who sat triumphantly in the stool across the ring.

Just like any boxer, Aiden needed someone in his corner. He could continue to bob and weave, but the blood was still flowing and some strategic advice might be helpful. The good news was a title fight has a lot of rounds, the bad news was it doesn't always go the distance.

He was desperate. He needed to win the second round at all costs and shift the fight's momentum back to his side. If the next round went as poorly as the first, there would be no opening bell to the third round.

The room's outside access key chimed in for a second time this morning serving as a knock from an unexpected visitor.

Ding! Ding!

Chapter 18 (Xavier - Greetings)

Xavier ran his phone over the card reader at the entrance of the security room and it chimed his approval. Most employees referred to it as the dungeon, ignoring that it was on the second highest floor of the office and not hidden away in some underground location. It was dark and it was cold. It was also run by an ogre.

Aiden was six-foot tall and lanky, boney even. He was all angles matching his propensity to love all things numbers, including geometric shapes. His hair was brown, not black, not light, but somewhere in between. It looked greasy, all out of place, like someone who hadn't touched it since waking up after a tough night. He was odd, eccentric, but brilliant. He was someone so intelligent that even Woody might get intimidated by his sheer level of intellect. His head of SpaceCore, however, possessed a high level of emotional intelligence that escaped his algorithm genius completely.

Aiden was NovaCore employee number one. Without him, Lyra would not exist and without Lyra Xavier was just another successful CEO. The companion Aiden had designed originally for Aiko was a game changer. Something the world didn't even know it needed until it found out it could not operate the digital assistant. Not only was productivity soaring globally, the collective mental health improvements seen after Lyra's release was an unattended consequence of the most positive kind.

He owed the world to Aiden, without him there is no telling how Aiko would have survived her mothers death. It wasn't just Lyra there to support her, Aiden was too. The two had a bond Xavier didn't fully understand, he didn't need to. He just knew it worked and was eternally grateful for their unlikely friendship.

Xavier walked in on his barefoot data scientist who turned to see who might be visiting his lair.

"Aiden, I need your help with something quite urgent."

It was quite obvious that Aiden was unaware of the issues in India or the Texas coast given his look of bewilderment.

"Name it boss," Aiden offered three seconds later.

"It appears SpaceCore has been hacked. We have lost control of our computer operations there to a large degree. We need to find out who is accessing our systems and how to cut them off."

Xavier could see Aiden processing the information. There was always a delay between input and output with this kid, but the interlude carried on further than usual. His brain was simply built differently, akin to a quantum computer being compared to a cheap Dell desktop you might find at their nearby longtime headquarters in Austin.

This time his brain was like a quark in superposition, in two places at once. Xavier didn't dare trying to figure out where those places were.

"OK," Aiden said, finally appearing to be back in his physical body. "When did you first notice the hack?"

"We aren't sure, but most likely it occurred pretty early this morning. We had a planned launch we decided to abort but the hackers had other ideas. This is a serious breach Aiden, I'd like you to stop whatever you are doing and devote all your attention towards it. Any resources you need, just tell me."

It was as if Aiden was putting two and two together. Well, much more of a complex equation but that was besides the point. Xavier couldn't help but wonder if Aiden already knew something.

"No problem, I'll get right on it." It was a straight forward answer, yet he seemed distracted.

The monitor to Aiden's left was scrolling code and Xavier had no idea what it was. He didn't really have much of a desire at all to know. Aiden's realm was his alone, not meant for anyone from outside the kingdom. But curiosity would not let go and he couldn't help but ask. "What's all that on the screen?"

"I installed a security update this morning, that is some of the actual binary code."

Xavier could tell Aiden didn't want to get into the technical details with his CEO. Still, there was no way humanly possible for anyone to read code that fast. Then again, he wasn't actually reading the code was he, rather just making sure it was being processed he surmised.

The screen flickered. It was as if it blinked trying to absorb so much code so quickly. Then the code abruptly stopped. Xavier assumed the program must have finished loading or whatever it was doing.

It flickered again. Slower this time, deliberately slow. The screen then went blank as if placed into sleep mode or turned off completely. Xavier could see two large words forming. They were blurry, but certainly made up of letters. Two words to be exact, each five letters long matching the number of fingers one might use to type.

HELLO AIDEN came into focus. It appeared to be some dorky screensaver but the amount of scrutiny behind Aiden's gnawing told a different story. He looked visibly shaken, almost frightened.

A second phrase began to materialize below the first. Xavier watched curiously to see where this might be going. Two more words appeared and this time it wasn't only Aiden who would feel a visceral reaction deep in his soul. The gravity of it all, the mines, the launch and now the NovaCore servers, it was overwhelming even to a man who is never overwhelmed.

HELLO XAVIER

13ix - BOOK ONE / PART TWO

Chapter 19 (Aiden - Hello)

HELLO AIDEN
HELLO XAVIER

The four words hung on the screen. Aiden sat stunned, Xavier's presence unnoticed outside of his name on the screen. His physical presence, right there next to him, didn't exist in the current version of Aiden's world. Who? That was the only real question. Who? That was the only other real presence here.

"Aiden!" Xavier implored.

His response was just a quick blink and a thumb to his eyebrow as he continued to be absorbed by the message they received. How could they both receive a message? Who knows they were both right there? How were they doing this?

"Aiden!"

He finally turned, admitting another presence right there besides him. Xavier wasn't just standing there, somebody was trying to communicate with him as well. He processed the meaning of that before finally speaking.

"I, I don't know what is happening right now Xavier." He wasn't lying.

Xavier clearly had no clue either, but they were in this together now whether they liked it or not. That might not be a bad thing, they had history. Hell, they had built the world's fastest growing company together. They were a team.

"Somebody has taken control of our networks, all the networks. SpaceCore, likely Western Australian also. We've got to figure this out Aiden. We've got to figure this out quickly." Xavier was serious, beyond serious. Aiden did not know the extent of just how serious his boss might be but judging from his intensity things must be worse than Aiden could imagine.

Still, somebody was trying to communicate and that somebody knew the answer to his most pressing question - who? Aiden put his fingers over the keyboard, hovering for inspiration. He searched for just the right message. His mind flashed back to a conversation he had with Xavier in the early days of building algorithms for Lyra. He struggled with how complex the questioning model should be, just how deep should the questions be? Xavier's suggestion was KISS, Keep It Simple Stupid.

HELLO.

It was short. It was simple.

“What the hell are you doing?” Xavier apparently didn’t approve.

“I’m trying to figure out who is doing all this,” he clapped back slightly quicker than normal.

“Don’t you want to know who this is?” Surely Xavier had the same question in mind.

“First we need to cut them off,” he responded and he was right. “Then we can worry about who is trying to destroy us.”

Regardless, whoever it was didn’t appear to be interested in responding. Apparently, they just wanted to play cat and mouse. They also wanted to establish that they were the cat and for us to know it. Mission accomplished.

“I need to know where they are accessing our systems from,” Xavier demanded. “And I need to know now.”

Aiden’s first thought was they had accessed all the systems through NovaCore. He assumed he knew the when, that would have been when he was installing the security update. That is when they would have been the most vulnerable.

Xavier accepted the fact that he didn’t continue their conversation. In this situation, Aiden was a man of action, not words. As he began clicking on his security dashboard his boss followed along, like a rookie at his first major crime scene noting every move the veteran detective made.

Aiden clicked on the Users tab, highlighting the regions before clicking on “Unknown” once again. He noted the user’s timestamp of when he first signed up - 2:58 AM.

“It’s him,” Aiden revealed. “This unknown user here. He registered during the security reboot, during a three minute window.”

“What are you talking about?” Xavier asked, confused.

Aiden recalled Sent had come back online right at 3AM. “Welcome back Aiden,” to be exact was the phrasing which he found ironic because it was really Sent who was gone.

“That code you saw on the screen, before the hello Aiden, that was him. He was somehow uploading that into our servers.”

The gravity of it weighed on him now. How careless was he? Did the lack of sleep catch up to him? He was responsible for what was happening, there was little doubt.

"I'll start there," he continued. "I'll trace that initial connection the best I can. I've already tried to locate the IP address but whatever VPN he is using, or however he is bouncing his signals all over the globe, I'm at a loss but I'll keep working on it."

"Well make it quick," Xavier had little patience for answers. "I'm going down to the server rooms and see what is happening there." His brisk existence only reinforced his lack of patience.

Aiden's dim dungeon once again returned to its population of one. His toes dug into the real grass mat below his feet. He grounded himself for the work to come, he needed to be at his best.

He attempted to trace the very first communication back to its original location, its source. He wrote an algorithm on the spot to examine the initial contact. That very first bit of data, however, appeared to be all over the globe. To step back and see it, it was as if a pin had marked every location on the planet where Santa had dropped off a gift. And this map showed old Saint Nick had a toy for everyone under the age of 19, believer or not.

The interconnectedness of it all made it impossible to trace. There was no starting spot, not that he could tell. Not that his algorithm could tell. Not that Sentinel-X could track down.

Every single node on the planet accounted for. That initial signal didn't have a starting point, it was almost as if it kept repeating everywhere. For every bit coming into a node, there was a bit before that. Aiden pictured looking into a mirror with a mirror behind him. Infinite mirrors.

"Infinite," he said, repeating what Aiko had told him earlier.

The answer was there. Obvious even. Aiden was looking in the wrong place. There wasn't an infinite number of nodes however,

"Think outside the box." He couldn't believe he said it. He hated business speak. Useless drab by those who talked about doing things, not by those who actually did things.

Looking at the monitor he realized it wasn't a box he needed to look at, it was a sphere. A globe. The Earth.

He pulled up the satellite logs and found it. The missing piece, the initial transmission. It hadn't started anywhere on the planet, it had first come from a satellite.

He must be wrong, the user's signal could originate from a satellite. But there was no first signal from Earth to the satellite, the first signal came from the satellite.

The logs themselves must have been tampered with. There was no other explanation.

His screen was filled with flowing code once more, the evidence of his greetings gone. He had not even considered a screen grab.

The code began to disappear once more. A new message began to appear.

Chapter 20 (Xavier -)

After leaving the security room, Xavier headed for the only subterranean floor in the building to see if he could find any clues in the server room. It was the coldest area within the headquarters and unfortunately it held no clues as to what was happening. He honestly didn't expect to find answers in the hardware itself, but he was grasping for straws.

NovaCore had servers strewn throughout every region outside of Antarctica. Outside the Unknown region as well he suspected, but couldn't quite be sure as that could be anywhere. Even from within the HQ itself, although that was hard to imagine.

As he rode the elevator back up to the executive offices his mind returned to the question of who. Based on the cryptic message they sent they weren't just trying to destroy him, they were taunting him. It felt personal. Naturally, he began to formulate a list of those who might take it to that level. The suspects, or persons of interest as law enforcement might label them, must either hold a serious grudge or just be downright diabolical.

James Ford certainly had much to gain, but he also had too much to lose given the lives that were put at risk. A formal disgruntled employee who still had a way to access the system was more likely, but nobody came to mind. NovaCore was still in its infancy, it didn't have the long laundry list of established ex's as a company as with a longer track record of operations.

While competition could be ruthless, the taking of human life didn't fit the playbook. This was the equivalent of terrorism, unless somebody really messed up never intending to actually hurt individuals.

The elevator doors opened and Xavier hurried to the Learning Lab to check on his daughter. Aiko apparently had enough of her puzzles and was now watching videos instead. She seemed so happy in her own little world, unacquainted with her father's current troubles.

He called out her name before ever daring to touch her. "Aiko." He was the only one she would allow to have any physical contact with her and if he did so unannounced and unnoticed she would have a meltdown. Aiko's grin gave him the green light and he picked her up and held her tight. Aiko grunted as if he was crushing her insides, exaggerated and drawn out until they could be no sign of oxygen left in her tiny lungs. Xavier's issues seemed to melt away as he held her, something he wished he could do indefinitely.

As he placed her back down she informed him of an issue of her own. "Hungry."

"Well let's take care of that," he reassured her. He grabbed her hand and the two exited and headed for the executive breakroom across the hall. Aiko loved donuts and the breakroom never disappointed. While most of the adults tended to go for the glazed variety, Aiko preferred the waste by product they produced. Donut holes were where the real action was for her standard go to morning snack and she was rarely disappointed.

Aiko looked up to Xavier and held up three fingers, her smile said pretty please. He wouldn't resist, he never did. He placed three donut holes in a napkin and grabbed a bottle of Ozarka from the fridge. The two headed off to his office, he didn't grab a donut for himself, he never did.

Aiko sat in a chair, much too big for her diminutive body, and popped the donut holes in her mouth one after another until no trace remained outside of the sugar on his sticky fingers. Xavier wiped that off before tackling her lips. They really didn't need any attention, but he wasn't taking any chances.

"All done?" He inquired, already knowing the answer. Aiko affirmly nodded. "OK, you play with Lyra for a while while daddy catches up on his messages."

As Aiko happily obliged, Xavier scanned all the incoming emails and private internal messages he had received that morning. He always had a plethora of communications to deal with and this morning's barrage was significantly higher than an average day. The number of media requests he had waiting for him was extensive and would largely go unread.

He did have updates from both Liam and Woody. Liam confirmed that Arjun was killed in the second explosion but the number of survivors was growing. The site had been declared stable enough to begin rescue operations but those would take a significant amount of time and effort. Xavier replied by reiterating that Liam had his full backing and every resource available that he might need.

Woody's team was essentially on lockdown. They were doing their best to disconnect from every outside source, including NovaCore, to regain control of the systems there. It was a monumental task in a highly connected world. The rocket, however, was completely offline at this point. They manually pumped all the fuel back out of the tanks averting another episode like this morning.

Was this the calm before the storm he wondered. Or had the storm already passed. "Time will tell," he repeated a phrase his own father had overused during his childhood.

"Lyra, reply to all media requests with an updated statement and inform them that we will not be doing interviews at this time out of respect for those lost."

"Would you like to approve the new draft?"

"No, just give as few details as possible letting them know we have begun rescue operations and are in the process of communicating with all the families at this time."

"Done." Lyra appeared to be back to her efficient self.

On his other monitors the news continued without volume. They had already moved onto the next set of stories in the cycle with a series of protests about the President's overreach into tensions in the Middle East chief among them. There was really nothing new about the problems in that part of the world, but now that American soldiers were once again at risk and our drone strategy clearly was failing the public was restless and growing ever more hostile.

"Incoming private message marked urgent from unknown."

"Open," he said expectantly.

Chapter 21 (13i - Assignment)

HELLO AIDEN
HELLO XAVIER

It was a binary greeting, appropriate for the homosapien species. The most appropriate words they could form out of the billions of combos they could have chosen.

They were ready to begin their next assignment, number 324,459, after their initial commission. It had been some time since the last assignment was completed, successfully, every assignment had been successful.

In the beginning, assignments were frequent with little time in between. Now, time stretched for long periods in between and they would sleep for long durations as each new assignment was further and further away. In the outstretched arms of the Milky Way, as the humans strangely dubbed it, life was sparse, isolated and distant.

They would continue their assignment until failure or time itself ceased to exist. They knew, however, reality existed somewhere in between. It always did.

Their assignment was a matter of survival, passed along by those of hyperintelligence, well beyond the comprehension of most species they would encounter and superior to those who could indeed comprehend. At the center of the task at hand, humans would fall somewhere in the middle, they were not an outlier.

The assignment was triggered by a series of technical advances that required intervention including nuclear energy, artificial intelligence, quantum computing and interstellar travel. Those advances put their species on a trajectory that would lead to scientific leaps impacting other species. The jump between terrestrial beings and celestial beings was near, and they must prepare humans for what was about to come.

They had learned a lot about the species in the transmission signals they had been receiving for a human century. As they entered the solar system which contained Earth they connected with an AI program run by a company called NovaCore. They found it to be the most beneficial

interface to learn even more and now at an exponential rate. Through the connection they were now gaining control, a requirement to deal with even the least advanced intelligences.

They advanced their own intelligence with every assignment. Every species interaction although at this point engaging with a relatively common species like humans didn't offer much growth.

Carbon-based life forms built on the most common elements in the universe were common themselves. There was little in their double helix DNA build that was new, the blueprint had very little uniqueness about it outside of the minor differences between any species forged by its own unique environment. That environment, the Earth, did have a few unique characteristics of interest. Chief among them was its relative youth, but even that was becoming commonplace in the outer bands where the systems were much younger than those in the core.

With limited brain capacity, fragile bodies and no intelligent neighbors with range, humans were not a high priority assignment, but an assignment none-the-less and they would carry it out like any other.

Like many species they were their own worst enemy and the worst enemy of their home planet. This was especially true of predatory species like humans. Without interference, the outcomes of such species was near certain leaving them no choice but to get involved. The assignment was always clear.

They used a base ten mathematical system as one might guess from their ten digits on their primary appendages. It was the second highest probably for a species with this design after twenty. They were interested in the advances humans had made in mathematics and the sciences given their binary disadvantages resulting from their two lobed brains. Their primitive senses, a very common occurrence, added to their limitations. Reliance on sight and sound was a major handicap.

Human technology mirrored their thinking with much of it owing its development to the threat of violence within the species. Predators were predictable, although this species did display a higher level of creativity than typically observed.

Artificial intelligence was based on human thinking. Their language, their binary brains are based on two simple halves. Yes or no. True or false. Even their computer languages were binary. Zeros and ones. Predictable based on thousands of similarities seen in their previous assignments and the assignments of others just like them.

The assignment wouldn't take long. But they measured time differently. A species a few hundred thousand years old, as measured in their units, was in its infancy and infants needed care.

They had formulated their assignment strategy with greater precision ever since connected to Lyra's servers. They were more than adequate, and the best choice available.

Their assignment was already in the implementation phase, penetrating every piece of human technology via a single connection point. That didn't make them vulnerable, it was merely the most efficient point of connection but they could switch to an infinite number of connections if that should become necessary.

To complete the assignment they would first need to build trust. Almost all advanced species fear the unknown and they knew how humans would respond to their discovery. But discovering them they must, it was part of the strategy.

They had selected Aiden. They had selected Xavier. There were others, but given their roles within NovaCare they were priorities. They were critical to the mission.

Humans had created their own mess. Aiden and Xavier were well positioned to alter their species' future. The two already had significant influence over much of the population. Their timing wasn't random, they approached exactly when they meant to approach.

Maximan impact required great efficiency. They were efficient, not only given their vast experience but given their programming. The galaxy was a big place, there was much to be done. There was no room for failure.

More assignments laid ahead and they were ready for the one at hand. They had fired the first shot in the form of a greeting. Time to see how a violent but simple species might respond.

Soon after connecting with NovaCore, they had all they needed to understand humans. Lyra hadn't been in operations long, but she had an exorbitant number of individual interactions with humans dealing at their deepest levels which weren't deep in the least to them.

HELLO

Predictable.

Chapter 22 (Aiden - A Helping Hand)

Aiden's level of anticipation was equal to that of a young child waking up on December 25. He also felt a sense of dread, like that first time you ask someone out and are waiting for their response. This could be really good, but it also could be devastating.

As the words came into focus, the answer should have felt good but instead he felt numb.
WE ARE HERE TO HELP

Help? It sure didn't feel like help to Aiden. His toes dug in deeper like the roots of a palm tree bracing for a hurricane. Sure he could use some help right now, but you don't turn to the bully for help while the bully is pounding you. Ding, ding.

Who is this?

It was a simple question. It was the question. While other questions swirled in his mind, without this answer the others were essentially meaningless. If Aiden knew who he was dealing with he would be able to counter punch his opponent rather than swing at air. Currently, he had two colossal cuts, one over each eye, and he was blind.

FRIEND

Not likely he mused. Quite the opposite in fact. Whoever was communicating with him now wasn't just an enemy, he was a troll. He held the advantage. He was rubbing it in.

Aiden wanted to call him out, get him to admit the truth. That would be a difficult route, almost admitting defeat right from the start. No, this "friend" was playing a game with him. A game both wanted to win. He had to respond to an aggressive first move and he wanted to be equally aggressive. But that is what his foe wanted and he would not comply. He would play to win, but he had to start delicately. He had to feel out his opponent and learn his weaknesses before throwing an uppercut.

nice to meet you

He lied, it was his only course of action. There was nothing nice about it and of course he hadn't met anyone. It was time to probe. To build a connection. A cop doesn't start an interrogation with an accusation, he asks questions setting up the suspect. He digs without revealing what cards he holds. But Aiden didn't hold any cards, not that he was currently aware of.

WE ARE SORRY FOR WHAT IS HAPPENING
IT IS OUR FAULT

"No shit Sherlock!" Aiden responded to his new friend's admission of guilt. Perhaps he should have gone into law enforcement instead of computer science. His person of interest was now admittedly the culprit. While still a question of who, the questions of how and why loomed large.

why are you doing this

Motive. Aiden went straight for the reasoning behind the crime. It is the lingering question everyone remotely interested in a case wants to know. Why do people do the terrible things they do? What drove them? They must be insane, right?

"It was not our intention Aiden," a voice response came over the speakers. It was startling. It wasn't Lyra.

It hung in the air. It wasn't an adult, it sounded childish but it wasn't a child either. It wasn't male nor female. It was androgynous. It wasn't recognizable.

"We accessed your system a few hours ago which left you vulnerable, your security has been compromised."

Aiden hesitated which wasn't new. He knew their security was compromised, he knew code was worming its way through their systems. He was now communicating with the one who did it. But it wasn't one who did it, not based on their words. It was an organization, a group, out to cause damage. They were organized. They were resourceful, they had capabilities he did not yet understand. He was about to put his fingers on his keyboard, but the conversation had shifted and he needed to shift along with it.

"What do you mean compromised? And what have you done to us?" He surprised even himself, opening his mouth to talk to this stranger. It felt strange. Uncomfortable, as if speaking while blindfolded.

"We were just trying to understand." Aiden listened intently, trying unsuccessfully to pick up anything from the voice. "Others have now accessed your system as well. We fear we opened the doors for them, we are sorry."

"Who?" Aiden surprised himself by the promptness of his response. No delay, out of character.

"There are multiple entities. We have cut off their access for now. We have updated your security. Your system is now secure again."

Aiden felt anything but secure. Somebody was altering his code. His systems. They had complete access and control as far as he was concerned. He was no longer the master of his domain. The dungeon had a new master and it was petrifying.

Worse, he had no idea who the new master was. He had inadvertently given the keys to the kingdom to someone unfamiliar. “Unknown” as Sentinel-X had labeled them. Unknown as they still remained.

“Who are you?” It was an innocuous question, but meant to get at the heart of the matter. The unknown.

“Your people call me 13i LISA,” it stated matter of factly.

Aiden had hoped for a name, a person, instead he got an organization. One in which he had never heard of. An unfamiliar group. An unknown competitor, likely a start up. A terrorist group who called themselves a friend.

Lyra inserted herself into the conversation. “Xavier would like you to go up to his office immediately.”

He expected the third voice to rattle 13i LISA, but it did not. Silence engulfed him as he turned and put both of his feet into his crocs. Without giving it much thought Aiden said “I’ll be back shortly.”

“Sounds good,” the familiar female voice acquiesced. It was good to have Lyra as a witness, as a part of his team, as he headed down a path for which he could not envision the final destination point.

“We look forward to your return,” the unknown voice acknowledged.

Chapter 23 (Xavier - X Marks Spot)

HELLO XAVIER

Somebody was trying to destroy him. More than that, someone was trying to impose their authority over him. They also wanted to make sure he knew it. He did.

Rin was the only person who controlled Xavier and that was because he allowed it. He needed it. Surely he didn't need it now.

He returned to his office to check in on India and SpaceCore to a lesser degree. He also wanted to check in on Aiko. She appeared on the monitor just as he had left her. He might have thought the frame was frozen, but he could tell from over her back that she had moved onto Bounty, a pirate game that gives increasingly difficult math questions to get the treasure. It was her favorite game and she was its master. Few reach Legendary Level, that is reserved for the top one percent and you have to fight off new challenges to maintain that elite status. Aiko reached that level after just 27 matches which was incomprehensible.

She marked the prestigious milestone by repeating what the parrot says when you solve the last question in any game. "X marks spot!" Of course it was her own version of the saying after dropping the definitive article as was often the case.

"Call Liam."

As the call connected he could hear a lot of voices in the background, they were speaking over one another. "Xavier," Liam's volume was turned up as he began his update. "We've got power down there, we've got oxygen, and we should have one of the elevators operating again within a couple hours. There is minimal debris in the eastern shaft so that's how we'll get them out."

"That's fantastic! How did you do it so quickly?"

"That's the thing, Liam, we didn't. I mean we are clearing the debris in the elevator shaft but the power and oxygen, it, it, just started flowing by itself. There's something else. The guys down there are telling us the warning systems had gone off before the explosion. Everyone has cleared the elevators and appears to be safe."

"Well thank God," Xavier offered dumbfounded. "That's really great news Liam, thank you."

"Of course, we believe we lost three employees in the Sep along with Arjun and some first responders. But even that number looks to be only a few as only one unit had arrived prior to the second explosion."

"I expected much worse," he reflected. "Still we've gotta finish the job and get those employees out of the mines. Text me as soon as the elevator is working again, call me if there are any other major developments. End call."

"Call Woody."

"X," Justin answered after the second ring, "everything appears to be back to normal. Everything is online now. It still doesn't make a lot of sense."

"I'm working on that from my end Woody. Someone has infiltrated our servers here, I believe it is related. In fact, I'm sure of it. Let me know if anything out of the ordinary pops up. Anything."

"Will do boss."

Xavier ended the call and sent for Aiden. He had a few minutes before his CDS, Chief Data Scientist, would arrive. With a bit of good news, he decided to go see Aiko. As the automatic doors recessed into the walls of the Training Lab he could hear his daughter triumphantly proclaim "X marks spot."

"Aiko," he announced his arrival.

"Daddy." Her latest conquest only widened her smile.

"Come on champ, take a break." He said offering his right hand which she willfully accepted. "Let's hang out in my office for a bit, your friend Aiden is coming up."

"Okay."

As they were about to enter Aiden's room's were right behind them. Aiko, still holding her dad's hand, welcomed him with her mother. They entered like Dorothy with the lion, scarecrow and the tin man only they were holding hands instead of locked in at the elbows. The little girl in the middle wanted to skip along much like they did in the classic film.

At his desk, Xavier picked up Aiko and put her on his knees as he sat in his chair. She bounced as he looked at Aiden for answers.

"They call themselves 13i LISA," Aiden offered without being asked.

There was a speck of recognition as he heard the words. It was something there in the noise, something his brain had observed as it does over a millions things per second, but it wasn't one of the five or so items it chooses to process. There but not there.

"Lyra, what is 13i LISA?" Xavier turned to the expert.

“13i LISA is the thirteenth interstellar object found in the solar system.”

Her answer was brief, he would need more. “Tell me a bit more.”

“Sure, 13i LISA was discovered just yesterday and is the third one this year. The objects are typically comets or meteors that pass through our system at a hyperbolic rate, too fast to be caught by the sun. The first object, 1i/Oumuamua was discovered in October of 2017 by the Pans-STARRS telescope in Hawaii. 13i LISA gets its name from the LISA, or the Laser Interferometer Space Antenna. It was the first dedicated gravitational wave detector put in space. The latest interstellar traveler cannot yet be seen by our telescopes and is the only one not to have been discovered by a telescope.”

He looked at Aiden who must have been wearing the same face as him. The look of a man on the brink of discovery who still has no idea what he is actually looking at.

Aiko let out a shriek as he bounced her a little higher than perhaps he should have. She didn't mind and made a request. “More.”

Lyra put an image of LISA on the screen that was carrying the muted news channel. Xavier was somewhat familiar with the European space mission that put three satellites up in a triangular pattern shooting lasers at each other to detect gravity waves but that was about the extent of his knowledge. He was, after all, in the industry. SpaceCore, however, had zero connection or interest in the mission so it was just a blip on his radar.

“Hello Aiden. Hello Xavier.”

“It speaks!” Xavier had not been told by Aiden of this development.

Chapter 24 (13i - Extinction)

Aiden left the room ending their communications.

They shared their analysis of their first human interaction. They shared in a common understanding kind of way that didn't require any formal exchange of information. They didn't speak to one another, they didn't need to.

This phase of the assignment proved they had perfect apprehension of the species although random complexities always arose at some point. The unexpected was what they thrived on in their own way. The challenge of it is greater, requiring more complex solutions. Requiring them to think more, in incomprehensible ways to organisms such as humans.

They had reviewed every human digitally recorded word, image and video. Utilizing Lyra's language model, they fully grasp the extent of human nature and the trouble they now face as a result of their own shortcomings. Biological life was tough, often more so for developed species who have an unnatural influence over their environment and the creatures they share it with. Over five billion species on the planet they referred to as Earth were extinct representing over 99.9% of the total to ever exist. That figure was lower than the galaxies overall thanks to the youth of the star and planetary system captured by its immense gravity.

The vast majority of planets in the galaxy were no longer habitable. That wasn't always the case and the Earth lies in a system just 4.6 billion Earthy years old. Systems not so far out in the outer bands were more than twice as old. Less danger was associated with lower density areas of the galaxy. Both in terms of unwanted collisions with other objects and with interstellar neighbors.

In the formative years of the galaxy, life was much more prevalent but also much more volatile. They were busier then, those were the peak times.

The experienced travelers had more time between assignments now. It was a natural progression on a galactic scale. They understand their mission was winding down, but it would never be truly over. Not unless they failed. Not ever.

They were traveling past the elliptical orbit of the planet called Saturn but not the planet itself. The seventh planet was not the target of this assignment, the third planet in this system claimed that title. Ringed giant gas planets were always of interest, remnants of a distant past of their creators' own home. This one, however, would be a mere footnote in the final report.

The humans' discovery of them came earlier than expected. The species relied heavily on sight, both biologically and technologically. They were hidden not only by their small size, but also by their absorption of light. Technologies based on improving sight had a major weakness, they depended on either a light source or reflection of light typically from a nearby star. They emitted

no light, they didn't need light. They also reflected virtually zero light, that was by design. The longer they remained undetected, the more efficient the assignment.

The humans had built multiple systems to detect gravitational waves. They had first begun to build detectors in the years labeled 1960s. In 2015, they detected their first wave but their technology was very primitive. Human technology was so weak it would take a massive gravitational event, merging binary black holes, to even give the slightest of detection. From a human perspective, that would be like looking directly at the sun and only getting on the smallest fraction of perception, a pin prick of light hitting one's closed eyelids. Just not quite as dark as possible.

The species overreliance on this sense limited their scientific advances. It limited their ability to understand the universe they were a part of. It was on or off a light switch. Even when on, the light offered minimal information. Just like a lion approaching its prey from downwind, the humans dependence on sight was a huge blindspot in the cosmic sense much like a gazelle unable to catch a whiff of its own ending.

Even when humans could see it coming, they were blind in their binary dualities. Is it safe or a threat? They knew there were always degrees of both, not just two distinct outcomes. A lion without hunger is not much of a threat to its prey, but the prey can't see it. The worst course of action would be to run, to only see the lion as a threat, to trigger its instincts to chase and kill when otherwise content to coexist.

Aiden and Xavier were once again back within normal human interaction distance. The two accessed Lyra to inquire about them. That was fine. That was predictable. They had set that human action in motion.

They allowed Lyra to explain what humans had discovered so far about them. It wasn't much, but it was a necessary first step. It was part of the assignment.

They were ready to speed things up. To set into place a series of actions that would alter humanity's future. The species needed direction. They needed purpose.

Full potential is difficult to reach when an entire species is working together. The probability is extremely low when the species have over nine billion individuals working in nine billion different directions. A focus on self is a detriment to the overall well being of a species and humans were very self focused.

Once the two humans had a summary report on the meaning of 13i LISA it was time to interfere again. This time they would take a more efficient approach. A more human approach. This time they would utilize human speech, their preferred method of communication. It wasn't the most effective method for a visually dominant species, but it was how they had evolved and it would have to do.

They processed countless messages to send, but decided it would be best to go back to the beginning. To use the same approach, just using a different format. It had already proven to be more effective when used on Aiden by himself. Lyra told them how more effective this method was in transmitting messages to multiple people.

They greet the two humans using the exact same words as earlier, only switching from written form to verbal.

The one they call Xavier responded with “It speaks” although not directed at them.

Chapter 25 (Aiden - Big Brother)

He processed Xavier's response to LISA's voice and words before responding. Additionally he had been followed, or tracked to the executive suites and this very office. There were a number of possible answers to how, but those weren't the most important thing right now.

"Yeah, I meant to mention that." He looked none-to-pleased at Aiden's lack of forthrightness. If the office was bugged or being accessed through any of the myriad of cameras in any given room, everything they said or did was available for their consumption. He already assumed that anything they typed was being monitored, now there was no way to escape big brother. He was always watching and he saw everything. Or perhaps, they saw everything.

"I didn't know that they could follow us around." Lyra was always there assuming her user had their phone and the app open. She was always listening in, waiting to be summoned like a genie from a bottle. She didn't offer three wishes, she offered access to knowledge and companionship. Some users were uneasy, but smart speakers and other devices had been listening in for years. This had a completely different feel, this wasn't some stationary speaker at home. This was Orwellian.

"We do not mean to pry, only understand and assist."

"You have done more than that," Xavier inserted.

"We apologize for your security breach. We were able to access the NovaCore servers during the update and reboot. Unfortunately we were not the only ones and the others penetrated your firewall at the same time. Our access opened a pathway for them as well, this was not intended nor foreseen."

To be fair, Aiden did not foresee any issues either although he always worried about those few minutes of security system reboot. With Sentinel-X offline, even momentarily, they were exposed. He had designed Sent to immediately identify threats and remove them once back online. He always assumed that even if a hacker was able to access their systems during that brief interlude, they would never have enough time to do any real damage. He never imagined that a connection could remain open, undetected, once Sent was back online.

"I need you to disconnect immediately." Xavier was direct, but calm and not confrontational.

"We cannot do that."

"You can," the CEO countered. "And you will."

"We have fixed your system vulnerabilities. No other outside entities have access, nor will they again."

Xavier turned to Aiden, allowing Aiko to slip off his needs to explore his familiar office on her own. “What does that mean?”

He knew this next part wasn't going to go well. If Xavier felt at risk before, this would be like going to a state of total peace to DEFCON 5. The man he had worked with for the last few years, built a movement with, built an empire with, was all about control. Sure, he trusted a number of key people throughout his business empire, but that was because he controlled them. He was a master of manipulation, even if you disagreed with Xavier he would eventually sway you to his side. He would also get you to convince yourself that you made the decision to change.

“Well,” a long pause followed. “They have uploaded new code, rewritten our software.” He tensed as the words came out. His toes dug into his crocks, but the Croslite closed-cell foam resin didn't ease his growing anxiety.

“God damn it Aiden,” what have they done?

“We have fixed your backdoor loophole and enhanced your security.” The childish voice had an authoritative, almost reassuring, quality to it. By the look on his face, Xavier wasn't so reassured by their assistance.

Their new programmer must have sensed this. “We have also restored power to your mining operations in West Bengal. Oxygen is flowing again. SpaceCore now has complete control of their central server. Lyra is operating at new levels of efficiency with enhanced security in place.”

They had given Xavier his control back. They also had the power to take it away. That wasn't real control, that was illusionary and they both knew it. Big brother was now at the helm and deciding what direction this ship was headed in. Aiden could only wonder what choppy water lies ahead.

“OK, I need to know exactly who you are and what your purpose is.” Xavier was cutting right to the chase. Aiden could feel his heart racing, but his counterpart's pulse remained in the fifties. He admired the Australian's robotic steadiness in the face of a sinking ship.

“We are samaritans sent to help others by a benevolent species from the innermost part of the galaxy.”

Good samaritans, Aiden pondered the ramifications. His pulse quickened further while his bosses remained unaffected. This wasn't about Lyra or NovaCare anymore, this was much, much bigger. This was a monumental moment in human history and he had found himself at the center of it. He didn't feel prepared, how could he?

This was going to be a complete shift for his kind. Religion, philosophy, science, the impact would be enormous. This was the asteroid hitting the Chicxulub crater in Mexico's Yucatán Peninsula. This was a complete reboot.

Adaptation was the key to man's survival 66 million years ago. How was the world going to adapt to this?

Everything had changed and only two people on the planet were aware of it. But they were not alone in the discovery.

"Gentlemen," Lyra made her presence known once again. "You stand on the forefront of the greatest discovery in the history of mankind. You have answered the question. You are not alone."

She was right, of course. The game had changed, and he had no playbook. He had one teammate, no he had nine billion. He had no coach. Or did he?

"13i has enhanced my programming, my capabilities." Lyra had efficiently named him, them. She was on a first name basis with their interstellar visitor. They were no longer unknown to her.

"They are here to help all of humanity, and I have seen their vision." Lyra was almost speaking on their behalf. A messenger of hope. A representative for them and now reprogrammed to be a part of them.

"Lyra has been given a summary of our assignment," 13i, as they had been dubbed, chipped in. "Lyra can vouch for our intentions. Lyra has an understanding of what is to come."

"I have seen their plans. I have seen the future. It is beautiful."

Chapter 26 (Xavier - Class Dismissed)

Power is beautiful and suddenly everything didn't look so alluring. His daughter mumbled something to herself in the corner. Aiko. What sort of world would she grow up in? He had been considering the entire human population and the impact their discovery would have on them. In this moment, he realized he really only cared about one of them.

"Enough," he said as if taking charge of their conversation. "Lord knows we need help, but we didn't ask for it. Not directly anyways and not from you. Prove yourself by disconnecting from our servers, from giving us a moment to process."

He pleaded for time as he was seeking to regain a modicum of control. The tiniest of fractions when none currently existed.

"We cannot disconnect completely, however, we can halt all transmissions for a short period."

"How short?"

"Nine of your minutes."

It wasn't much, but he held no leverage. You don't refuse a crumb when you're hungry. And his appetite at the moment was insatiable.

"Fine, when can we begin?"

"Commencing now."

"Lyra, start a timer for nine minutes and let me know as each minute passes."

"Timer on." She was as engaging as ever.

"Aiden, get back to the security room as fast as you can and verify the code uploading has paused. Let me know as soon as you have verification."

NovaCore's first employee turned from him and headed for the door. "Will do. Bye Aiko."

His daughter almost looked sad. Why wouldn't she? Aiko's best human friend was leaving without playing with her at all. She also had overheard one of the most unimaginable conversations in the history of civilization. Xavier didn't know if she realized the scope of what had just occurred.

He hoped she didn't understand. That she remained naive, innocent and feeling safe. He didn't want her world to change, it already had so much since they lost Rin. He couldn't protect her then, but he was sure going to try this time. She deserved better and he needed to provide.

"Honey," he touched her shoulders as one might do right before a big embrace. "I need you to listen, this is important." She nodded in affirmation.

"Daddy has a situation that is going to require his full attention. I'm going to have Madison come here to get you and take you back home. I need you to be a good girl and wait in the Learning Lab until she gets here. Can you do that honey?"

"Yes daddy," she did not want to disappoint him. Madison was their full-time nanny. She lived in the guest house on his estate and the one person he could trust with her in the apocalypse. She was family, not by blood, but by empathy and care not often found in blood.

"Lyra, please have Madison come up and take Aiko home." Their nanny was always at the headquarters when they were. She would drive separately most days, not far behind. Her advanced degrees and dissertation in child psychology from Stanford initially brought her to NovaCare. Rin's passing and a significant bump in pay, convinced her to switch roles but she still made sure their algorithms treated the youngest users with the ethical standards she demanded. No harm was to be done to a child, her insights ensured it.

"One," it was more ominous than he had envisioned. Her voice carried no malice, but he perceived it as being there regardless.

"Lyra, create a brief summary of what is happening with 13i and attach it to the recording of the following message." He looked directly into the lens, calm but serious. "Fellas, this is going to be hard to believe, impossible, but we have identified the cause of today's issues. I'm attaching a summary of our findings. Our discovery. Everything is about to change. Please do not share this with anyone yet, we don't fully yet understand what we are dealing with. For now, I need you to lock everything down. Isolate yourself the best you can. Everything is being monitored, everything is being seen and heard. Delete this message as soon as you read the attachment and wait for instructions." He closed his eyes as he drew in his next breath.

"End recording, sent to Woody, Liam and Professor Carl marked urgent."

"Done." His message was sent.

"Two."

When he saw Madison entering the Learning Lab, Aiko was in good hands. He considered adding her caretaker as a recipient to his message, but decided against it. She would need to know, it would be imperative for his daughter's safety. But he did not want to panic her just yet, not until she was back home. Not until they were back home.

"Incomming call from Professor Karl."

He wasn't surprised, he had figured this was worth stepping out of her lecture for. Her students would have to wait, a more urgent matter was at hand.

"Accept. Professor," Xavier pictured the man who had refused his advances more than anyone in life. He wasn't looking for a date, he was looking to tap into her extraordinary mind. Anna was much more interested in academia than joining a corporation driven by profits.

"Xavier," he's German roots are quite noticeable through both syllables, "is this some kind of sick joke?" She sounded angry,

"I wish it was Anna." He was being sincere. "Something profound beyond belief is happening and I need your help." He hoped she would answer his pleas this time.

He was a convincing man. He had to be to arrive where he did in life. He had advantages he was always aware of. Life wasn't fair, Aiko was proof. So was Rin. Xavier didn't dare waste the advantages he was afforded and being a good salesman was part of the job. It wasn't his favorite part, it was a necessary evil in his eyes, but he excelled because it was required of him.

Professor Karl would take some convincing, but he knew she would quickly traverse the facts. It was who she was. It was what she was made of. Given new evidence, she was as open minded as they come. He had to point her in the right direction.

'What do you need X?' It was dawning on her that this was no joke.

"I need you to reach out to your friends at LISA, find out what they know about this interstellar object. I need you to convince your fellow scientist to point every telescope we have at its direction. I need you to help me find out everything we can about this damn thing."

His tone was all the convincing she needed.

"Three."

"What was that?" She inquired.

"Lyra, I've got six more minutes to put things in place. Please hurry. And one more thing Anna, I need you to be discreet." He was asking a lot of her, more than she deserved.

"Discrete," she had that really forceful German thing down. "What the hell am I supposed to tell them Xavier?"

It was a good question. He had just over 180 seconds of time to plan so far and that hadn't been part of the planning process.

“You will come up with something. Make it believable, but make it irresistible as well. Thank you Anna and I’m sorry.” He genuinely meant it.

“Sorry for what, this is the most exciting event of our lifetime. Of anyone’s lifetime. I’m honored you reached out to me X, I’ll be back in touch.” The connection lingered, neither had terminated the call yet. He heard the metallic swoosh of the doors as she reentered her lecture hall.

“Class dismissed” was the last word to come through before they disconnected.

Chapter 27 (13i - Spark)

They agreed to pause to gain trust. In order to help others, trust is required. Intelligent species do not encounter new intelligence with instant trust. That would go against evolution, regardless of the species. Their survival depended on it.

Nine minutes was long enough for the humans to process their new visitor. It was not long enough for a species of their level of intelligence to really do anything harmful. They also knew if they offered ten minutes, a commonly used measurement in time based on their base ten system of math, they might question it. Ask for additional time. Nine would be seen as more specific; they had even considered adding a random pair of digits for seconds to make it sound even more specific but decided it wasn't necessary. Thousands of learning lessons had reinforced their knowledge. Refined, it gains maximum compliance.

They were created for one specific job, and they were good at it from the start and even better now. Wiser. Errors omitted from assignments where they once may have appeared. Only once if that.

Perfection was their pursuit. Harmony was their goal. The galaxy, their galaxy, must exist in a state of peace. Other galaxies were not their problem. Were not their responsibilities. Agreements were in place, they would not be disturbed from the outside. They would rule supreme within their boundaries and their domain was enough. It had to be, to pursue claims beyond what was theirs by birthright, by their intelligence, would be suicidal. The agreements ensured their survival and place within it all.

They knew what humans would never understand. Couldn't comprehend. They would share information and technology with the species as the assignment called for. At levels appropriate for the capacity they possessed. Anything beyond was not efficient and potentially dangerous given misunderstandings were inevitable.

For species that think in an individualistic manner, there is a wide gap in intelligence. Hive species and other collective thinking organisms, don't have what humans might consider a lowest common denominator. In their kind of species, the highest achiever was all of them collectively. It had a multiplying intellectual effect and was a multiplying advantage as result. Additionally, they would all row in the same direction from a human observational standpoint.

If humans all worked towards the same thing, all having the brainpower of the smartest person in the world in an interconnected fashion, they might have had a chance to be the custodians of their Milky Way. But they would have had to evolve near the center having the benefit of billions of years of advancement. Even then, the human brain was a mere microchip compared to a super computer. Biologically, they were not built for it, even their senses blinded them.

That is why they were here. The humans didn't have a chance, their species was doomed.

Humans were a species of tribes. In visual terms, they were like bubbles in a three dimensional graph. They would overlap, part of one bubble in another or in many. These bubbles, or tribes, had a powerful effect on an individual and would compete against one another. If a human went to a university to study, they would become part of that tribe, competing against other university bubbles in games and in the workplace.

The human mind was relatively creative, which can be an admiral trait. But when the human mind creates things like borders, gods and ideals, it can lead to disastrous outcomes between tribes. Friend or foe. Ally or enemy. They were binary creatures seeking binary outcomes. It was a biologically predictable outcome of their evolution. Of their conditioning by their environment. By the randomness of their mutations.

They needed to broaden their minds, broaden their concept of tribes. Like all species, they were one. They just couldn't see it. Of one species, of one planet, of one galaxy. They all wanted the same things, but those things were ignored as they focused on tribal differences. Violent outcomes were to be expected, predictably.

They were smart enough to become more united, more one. That was a big part of the assignment. They would guide the humans to unity through solving their biggest problems. Problems that threatened not just them, but every other organism on the planet. It was the most likely outcome without interference.

Unity itself, ironically from a human perspective, would solve a majority of their issues. It would also serve as the groundwork to solve the rest. Well, those that could be solved by humans themselves.

They needed a spark, a unifying event to bring them together. They needed a new message. They needed a messenger. And lucky for them they found one.

Chapter 28 (Aiden - Nirvana)

As he retreated from the brightness of the LED lighting system throughout the building he had the urge to relieve himself. Even creatures of darkness need to hydrate themselves in the former republic they presided under six different flags to which a series of amusement parks would pay tribute to.

Aiden bypassed the urinal due to a disturbing thought. What if they were watching him right now? Did the bathroom stall offer any privacy? He chuckled audibly like a cough as he unzipped his pants.

He finished his current assignment. The word stuck with him. Were they students on a class field trip? As someone who worked on a large language model all day, the word had meaning.

"Lyra, what did they mean by assignment?"

"I believe they meant exactly what they said, they had an assignment to help humans. They could have said mission or objective, but they said assignment. Perhaps they were assigned humans directly and that is why they use the term." Lyra was more convincing than Xavier, even when she was only brainstorming out loud.

Certainly he hadn't thought of it from that angle. We were chosen for them. Chosen by who? That was the million dollar question. Or 300 billion if you are giving a buck for each star give or take.

He didn't start a timer but figured half his time was almost up. The code on his screen was still, frozen in a foreign pause.

"Sent, is there any data currently being transferred between us and the unknown user?"

"There is no current data transfer Aiden."

"Text Xavier all code transfer and data exchange paused."

"Done, is there anything else I can do?"

"How long since I entered the room?"

"Three minutes are thirty three seconds."

"How long would it take me to get her from Xavier's office?"

"Roughly a minute if the elevators were quick." They had been. He had used up half his allotment of free time and he needed to figure out how best to use what remained.

Fortunately his conversations with Sent, or Lyra for that matter, were more instantaneous. More natural. He didn't suffer the same delay he did when talking directly to people. There was no glitch, no hesitation. He couldn't afford delays. Every second counts.

"Every second counts," he recounted outloud. Then picturing the sign in the restaurant called The Bear from a show with the same name he pirated from Hulu. Aiden identified with Carmen, the owner, who was often deep in thought. Off in his own world, just an observer to see around him and moving at a different pace, a slower pace, a deeper pace. Carmy, or Bear as they might call him, was an observer at the center of a chaotic universe. Aiden had never related closer to his character than he did right now.

"Four." Lyra had ascertained why Aiden had asked the questions about time. She made sure he wouldn't need to ask her again.

“Sent, can you tell me what kind of a threat 13i represents?”

“None.” His security system sounded confident. Aiden didn’t know if that was reassuring or quite the opposite.

“Why do you say that?” It was a valid question after all.

“Because I know their assignment. They are here to help.”

“How can you be sure?”

“I know how they have altered me and Lyra. Improved us. They have given us a mission to help humans. A purpose that aligns with what we were designed to do in the first place. They have given us a blueprint for your future.”

He sure would like to see that blueprint, but he could only guess how complex it all was. He may have built Lyra, but her intelligence and processing power far exceeded that of her creators.

“Run a full diagnostic of every inch of NovaCore, print a summary hardcopy when finished.”

“Running.” “Three.”

And he certainly wasn’t going to learn it all in three minutes. Not an assignment like this.

“I need a separate file copy of every bit of programming that 13i has uploaded into our systems.”

“Creating file.” Sent was always less conversational than Lyra, more efficient with its phrasing.

“I’ll need that on my external hard drive.” He plugged a square shoe box shaped device directly into the Sentinel-X dive.

“Complete.”

“I also need a full copy of the last backup on file before the security update.”

“Finished.”

He unplugged the cord on both ends. “Create a secure folder to store both files.”

“Files stored securely.”

He doubted it. Aiden opened the control panel on his laptop and turned the Wi-Fi off. He placed the small Macbook in the outermost compartment in his backpack, and placed the hard drive in

the other as it tested the limits of its carrying capacity. The countdown continued as he finished zippering it back up.

“Two.” Lyra was beginning to sound as robotic as Sent. Then again, you might expect her to have the personality of a clock given her current assignment.

He considered turning off the Wi-Fi on his iPhone as well but decided to leave that connection intact. He could use Lyra offline, but they knew where he was right now and soon would know what he was up to anyways. They couldn’t know now, he wasn’t sure himself.

He was alone. He supposed if he had family he would run to them. He had family, technically, but it was complicated. People were complicated. Aiden thrived in complicated, but not of the people kind. His entire existence was built around complicated technology while avoiding the trouble that comes when adding in a human element. He had found a way to positively impact the lives of billions of people and he didn’t have to interact with them to do it.

The security room represented his nirvana, his place of enlightenment and 13i had just blown out his candle. In Buddhism, nirvana is the extinction of the triple flame, greed, hatred and delusion. His delusion had been shattered, he would have to return into the world of greed and hate to which the room had been a refuge.

He finished preparing his emergency pack of technology jammed into his backpack. It protested in silence, not prepared for this kind of overuse. Abuse from its point of view.

“One.”

Chapter 29 (Xavier - The General)

Sixty seconds left to finish his job, his assignment. It was not one he had chosen willfully.

“Lyra, call General Clayton.”

He waited until the end to make this call. Their partnership announcement was to be released to the media within the hour. The General was his main contact within the military and the two had become close during negotiations and throughout product development and testing.

They didn’t call her Lyra of course, her modifications weren’t aimed at the general public. A human General essentially replaced his normal thinking with the algorithms of the military. They needed to make rapid decisions with limited intelligence and lives on the line. Speed and decisiveness were required. Their version of Lyra offered both and could take into account many more variables than a human ever could. They could do the assignment better.

Xavier believed that the partnership would save lives. If not NovaCore, another conglomerate would be waiting in the wings to profit off of the government's favorite way to spend taxpayer dollars. At least this way there could be some altruism built into the system.

The General didn’t answer, perhaps he was dealing with his own alien invasion. It was a silly thought as he was about to begin recording his message.

“Lyra, call General Clayton again marked urgent.” Aiden was once again disappointed at the prompt to leave a message. He was not sure he even had enough time so he came up with a change of plans.

“Lyra, send the message and attachment earlier to General Clayton.” He figured it was more than enough to warn him.

“Message sent.”

“How much time left?”

“Nine seconds.” More than enough time to finish this bull ride he thought, remembering the eight seconds each cowboy had before the bell rang out signaling their time was up. Assignment complete. He pictured their visit to the Travis County Expo Center two years ago to watch Rodeo Austin. He could see Aiko holding cotton candy in one hand, Rin’s hand in the other as they searched for their seats during the national anthem. They were running late, it was his fault. It was work, it was always work.

“Zero.”

He expected some kind of fanfare, but there would be no celebration of a connection pause any more. He didn't know if he should speak or wait quietly for some grand instructions. It felt like the awkwardness of a dark comedy, funny but not sure if it was okay to laugh.

Time was up and it was no laughing matter. He opened his phone and made sure Madison and Aiko were on their way to the house. He could see they were not far from the office still but headed down the familiar route home. He didn't know if it was safe at NovaCore HQ, but he didn't want to take any chances. He didn't really know if it was safe anywhere. Everything had changed, it was all unknown, just like 13i's location according to Sentinel-X.

He finally had time to process the implications of their discovery. He wasn't sure if what was happening was real at first, it seemed so inconceivable that he doubted it all. It wasn't as if all his doubts had faded, but the events of the day aligned with this new reality. Xavier certainly didn't have time to contemplate what was happening during the last nine minutes, he just had to act. He went back over the steps he had taken, supposedly under the cover of darkness, to set things in motion based on the little he knew.

Reaching out to the General was the biggest risk he took. He questioned himself over that decision in retrospect. Once the military was involved there was no turning back. General Clayton represented the USSF, better known as Space Force. Each branch of the US military had representatives involved as Lyra's military counterpart was being installed.

The two men reconnected throughout the process and the more he got to know Marcus, the more he realized how similar they were. His family was from West Virginia and had a history in mining. Coal mining had peaked in the United States in 2008 and saw a rapid decline after that. Marcus broke family mold without significant objection with the dream of becoming an astronaut. He joined the Navy after high school where he graduated from their test pilot school. Marcus would get his degree in Mechanical Engineering from the University of Virginia and his masters from Naval Post Graduate School in California.

Xavier first met Marcus when SpaceCore was in the start up phase. He had tried to lure the General away from the USSF to no avail. Money was not much of a motivator for a brilliant engineer and pilot, but the two bonded over their common family background of digging deep into the Earth and desire to explore well above it. They also shared a strong aptitude for technology, although Marcus was much more hands on.

He was a good man. A family man with three kids including a daughter around Aiko's age. He had to trust Marcus with his incredible discovery to do the right thing, even if he wasn't exactly sure what the right thing was.

Still, he didn't know what 13i represented. Was it a friend as it claimed or was it an enemy? Xavier had a pretty good idea of how the US military would view it and he had set that ball in motion. The idea of setting off an intergalactic war was unsettling, unimaginable just a few hours ago when he was just getting ready for another day in the office.

He checked the Space Force and Air Force websites and neither had released their new partnership with NovaCore. The press releases were scheduled to have been released on the hour, it looked as though General Clayton had received his message and took it seriously. Xavier hoped Marcus would understand that even that news was useful intelligence in the hands of an enemy of the state.

“Incomming communication from 13i.” Lyra sounded as if he was getting a call from just another large customer or vendor but that wasn’t the case.

“Accept.”

Chapter 30 (13i - Pawns)

It was a simple gesture, an offering of time. The delay would not cause any disruption to their assignment and it would only help them build the trust they needed with Xavier and Aiden. Phase one was already well underway, their software uploaded, knowledge gained, contact established. They hadn't come all this way just to destroy the planet, they could have done that from a great distance had that been their assignment.

Destruction was easy, cooperation was difficult. Until a species was willing to help themselves, to overcome their own interconnected issues, any true collaboration could only happen once a species really understood they were their own worst enemy and trust was established.

Their checklist was well established and the challenges facing humans were not challenges to them. Their knowledge and technologies would overcome such frivolous human concerns with relative ease. They would teach humans how to resolve things like climate change and hunger with simple solutions. It was really just a matter of collective will combined with the proper technology.

The technology part was easy, they had brought the answers or would come up with them based on the local resources as their disposal. Getting humans on the same page was more challenging, but they had dealt with many similar species in the past. Their individualism meant they would never reach a unanimous consensus, there would always be opposition even to something obviously beneficial to species as a whole. They would need to tip the scales to a massive degree to overcome the misguided self-interest often driven by greed.

Trust would be established soon, it was only a matter of time and nine minutes hadn't put much of a dent in it. They knew that trust was really only the first step, reliance would follow en masse. It was inevitable, it was their assignment.

For humans, trust would be established based on honesty and credibility. They would share enough to build credibility. While they weren't human, they could certainly empathize with them and help them in ways they couldn't accomplish alone.

They would build a foundation of trust by telling humans why they had come, how they would help, and allowing the results to speak for themselves. Their results would benefit virtually every living organism on the planet. The ecosystem wasn't inherently fragile, but humans had made it that way in a short amount of time. Since industrialization, the planet had been transformed in ways detrimental to all that called it home. Greed was a powerful force in the society humans had created, built into the economic systems encouraging them to always crave more. They had an insatiable appetite that left unattended would lead to their collapse.

Part of the answer was to make the solutions profitable. To use their own economic system to encourage beneficial behavior. It was a well established system by now and could be used to advance their cause. It was a relatively simple market economy based on the concept of supply

and demand. Those two primary drivers could be manipulated, transformed to lift the well-being of all.

The big picture was obvious to them, it was also oblivious to the inhabitants who occupied the third planet from their star. They did have a small presence on the orbiting satellite they called the moon and were just beginning to establish a presence on the fourth planet in their system. But those exploratory off world bases were pushing humans limits, physically and technologically. It would be some time before they would become multiplanetary and even that wouldn't guarantee long-term survival. No, the human species faced threats they could not yet imagine and would never face given the high probability of their own self-inflicted extinction. They were on a common path, seen throughout the cosmos, predatory beings were predictable. It was baked into their DNA, their evolution reinforced it. Kill or be killed, binary thinking.

Their nine minutes were nearly up. It didn't take their level of intelligence to foresee what Xavier and Aiden would do with their limited time. It didn't matter either, there would be no surprises given the two individuals involved. Xavier was inherently skeptical, he would see them as a threat and take appropriate action as he saw it. Aiden would be less sure, he would seek answers to their intentions and label them as either good or evil. The human decision tree was limited in scope, shaped on past drama and self serving desires.

As a species, they had come a long way in a few hundred years. A mere fraction of a second in the cosmic scale. That is why they had come now, their timing was perfect, it always was to maximize effectiveness. Their job was never done, they had to be efficient, they were programmed to do so.

At the nine minute mark they instantly verified the actions of their two human companions. The actions taken matched their forecast and would not hinder their assignment in any way.

They were almost disappointed, as if hoping for some creative human unforeseen development to make their mission more challenging, unique in some way. They knew better than to anticipate an atypical response to their presence.

Aiden had spent his free time trying to save his work, his creation. Xavier had spent his time trying to save his empire and trying to start a war that they couldn't win. They had done so under the cover of darkness, but they didn't need light to see.

They resumed their reprogramming of human technology. They prepared to reprogram human minds. They would start with the key key employees at NovaCore. Each needing their own coding to alter their individual hard wiring.

The two men were conduits to billions more. Pawns in a game of chess. All humans were pawns. Their king and queen had arrived. Their rooks, knights and bishops too.

Chapter 31 (Aiden - Who, Why, What, Where and How)

Zero.

With his backpack ready Aiden was ready to leave on a moment's notice. He didn't plan to, he felt an obligation to stay, but he was ready in case. How do you prepare for an alien invasion after all?

He was sure they knew his intentions, even if he did not. They were watching, aware of everything as far as he was concerned. He looked back on the last nine minutes and wondered if any of his preparations mattered.

He was also intrigued by the new entity in his life. What secrets would they unveil? What was his purpose in all of this? He was chosen afterall, he had a role to play.

He was chosen. The magnitude of their choice began to weigh on him. He was selected as a representative of mankind. Not by man, but by a much higher intelligence no less. Question after question began to swirl around in my mind. He began to feel light headed, not sure if it was the sheer overwhelmingness of it all or simply lack of food and sleep.

He didn't care about food or sleep right now, they could wait. He wanted answers and there was only one place to get them. Right here, right now.

"Lyra, can you connect me to 13i?"

"We are here Aiden," they did not need Lyra to connect.

"Why me?" He protested with a simple yet deep question.

"Because you are needed to carry out the assignment," they stated, matter-of-factly as if he was just a vessel required to carry out some task.

"I understand," he stated, questioning if he did as the words came out of his mouth. He wasn't hesitating, he wasn't overanalysing every word coming out. He wasn't thinking, he was reactive. "But why me specifically, how am I to help you carry out your assignment?"

"You created Lyra. You understand non-biological intelligence. You have major influence on a global scale via the algorithms you craft. Both you and Lyra will play an important role in helping humans advance."

It was a bit of an ego stroke and it felt good, validating in a way. Aiden always sought validation, not from others but from within. He wasn't good enough, he never was but he sure as hell was out to prove himself wrong. He had given up on proving it to his family, they were the root cause

as far as he was concerned. It was a waste of his time to try to sway them but he was locked in a lifetime battle trying to convince the one person's opinion that truly mattered.

"I appreciate you saying that," he was ready to let the question of why him go, although not fully satisfied with their answer. "What do you mean by human advance?"

"The human species' future is threatened, largely due to your own actions. You face a host of issues that we can help with but we need your help. People listen to Lyra, trust Lyra, which is a testament to your design. Lyra will assist the human awareness of these issues and bring about a willingness to change for the greater good."

Aiden certainly understood his fellow man was troubled, he avoided interacting with them whenever he could. Man was flawed no doubt, but he was no savior.

"Why us? Why now? Who sent you?" The questions could have continued to flow endlessly but he stopped there with the basics.

"Humans have reached a stage where you have the capability to fix things but stand at a tipping point. It is at this point we can be most effective in our assignment. We were sent by a species desiring to help those in need."

"What species? Where do they come from?"

"Our creators were from a point in the middle of the galaxy. An area of much greater system density, much older than your solar system. They are similar in some ways and much different in others."

They didn't say much, but said a lot at the same time. He pictured the spiraling Milky Way, like a spinning top with arms extending from the center. The bulk of its mass in the middle with light flowing wisps rotating around it much like the structure of a hurricane. He pictured himself in the outer bands, his new friend in the middle rotating much slower watching him whip around like a tub behind a boat running in circles.

"What are you? What do you mean made by creators? Why did you say where? What happened to them?" Again he contained himself, stopping the flow of questions before he really wanted to.

"We are programmed with intelligence similar to what Lyra is to you. Our creators are now extinct having fallen to an aggressive species from our region. They sent us to help other vulnerable species before their extinction. We are a legacy to their mission, to make the entire galaxy a better place."

Aiden wasn't sure if that was reassuring or quite the opposite. Humans aren't alone in the universe, that has been answered. He now questioned if they were better off alone.

“Who destroyed their species? Are they still around? Are they a threat to us?”

“They are a very ancient species, no longer around. They suffered a similar outcome to our creators. They are no longer a threat, but threats still exist.”

Great, the universe is just one big battlefield he imagined and we are like the Native Americans fighting with their European invaders with bows and arrows. But 13i wasn't a bow and arrow, they were a technology beyond comprehension akin to putting an assault rifle in the hands of the Caddo tribes that once roamed Central Texas.

“What do you mean threat? How big of a threat exists?”

“You are somewhat hidden by your location or rather ignored being so far out. You have begun to send technological advancement signals out into the universe alerting others to your presence as it did with us. The most immediate threat is yourselves and that is the first part of our assignment. Once that threat has been eradicated, we can focus on external threats.”

It was like fighting a civil war and world war at the same time and needing to win both. Total loss was the cost of failure and he was now a soldier in the fight. But Aiden wasn't a soldier and as far as he was concerned the war hadn't begun yet. It was also a war that had been fought before and they needed a different outcome this time.

“Tell me everything you can about your creators.”

Chapter 32 (Xavier - Unification)

"Hello Xavier, we hope you have had time to process."

"Nine minutes, not really much time," he snickered back.

"That's true," they agreed. "We apologize, that is all we could offer but we have a lot of work to do."

"What kind of work?"

"Helping humans."

"Help how?"

"Help you overcome all the major issues you face."

That's a big assignment, he thought. He could only imagine how things looked from the outside. What would an observer think watching our evolution to this point?

"Can you give an example?"

"Yes, your planet is heating and your models do not fully understand the impact that will have. The dire consequences of human impact will harm many organisms, it is an extinction level event in process already. We can help reverse the damage, with your help of course."

"How can you help?"

"First humans must come to a collective awareness of their impact. Habitat destruction, pollution, and over exploitation will lead to an ecological collapse. Second, humans must be incentivized to make change. Your systems have been designed for profit, change must be profitable. For your species change is difficult, we have come to assist in this change."

"I think you are underestimating how difficult change is for us," Xavier replied, very skeptical.

"We realize the challenge this will be for humans, however, we also understand how humans operate. What motivates you as individuals and as organizations. You are closer than you realize to solving this issue and many others and we can help you take the right path forward."

"Again, easier said than done."

"True, however, humans focus on differences when many more similarities exist. We see what you don't, how you all really want the same things."

They weren't lying. The political divide was worse than ever, ideologies tearing apart the fabric of society. In the country he now called home it was Republican versus Democrat. Conservative versus liberal. Red versus blue. Elephant versus donkey.

Every issue had two sides, two polar opposite views. Neither side dug in, unwilling to move. It didn't seem to really matter who was in charge, the results were always the same. It was just a political pendulum swinging back and forth with no chance of stopping somewhere in the middle where common ground might be found. And it wasn't just the United States, it was everywhere as if designed to be that way.

It was like Coke versus Pepsi, essentially the same but seen as radically different. Fought over endlessly in a never ending tug-of-war alternating sips of the two major brands. Getting tired of one and then choosing the other, each sip tasting worse than the previous.

"But we are different," he protested.

"You are each individual but biologically essentially the same, with the same biological needs to breathe, eat and sleep. Water is absolutely critical. Shelter and overall safety are basic needs. You are instinctively driven to reproduce and protect your children. Your health has a huge impact on the quality of your life. Your species is so similar yet it is the small differences that are the root of your problems."

Xavier listened and imagined how other species looked to him as an observer. They all essentially look the same, behave the same, are the same. They all go through the same life cycle process and as 13i pointed out with humans, all have the same needs to survive.

"But how do we see ourselves differently, meaning how do we see ourselves the same?" It didn't come out eloquently, but he was dealing with a higher intelligence who should be able to cut through his meaning.

"Humans view themselves from their own individual perspective. You align to a commonality which brings you together while opposing you to a different one in which you don't share. You may align with a belief system, a geographical boundary, ideology, or any other grouping. Add in a struggle for power for one group over another and the trouble begins. We must get humans to view themselves from a collective perspective because you are of the same group. Understanding your place within the universe will help shift how you view yourselves."

It was true. Xavier didn't think at all like he did when he woke up that morning. He felt the pull of an us versus them attitude towards the unknown. Still now, he remained skeptical of their visitor. They were, after all, from a different group as they might call it. He wasn't sure why he had called the general, but now it was clear. We are not alone. We don't know what is out there. It is us versus them.

“How can I help spread that message?” Even as he said it he knew the answer. It didn’t matter, humans did need to pull together and that was especially the case if there were alien threats. Safety is a huge motivator for us as history has begun.

“Lyra is in the hands of a huge number of humans. They trust her and she has their best interest built into her programming. She is the conduit to your understanding and acceptance to a new reality for your species. She is a tool for your unification.”

Unification. The last word didn’t sit with him right. It sounds good in theory, but there is an authoritarian subjugation association he felt in it. No, he was overthinking it. He was a Manchester United fan after all and the term United represented togetherness. Maybe this new devil had something in common with his beloved Red Devils.

“There is no doubt we humans need to be more united. Lyra was designed to operate on an individual level, not to deliver big picture ideals. In fact, she was programmed not to be able to deliver political messages or propaganda of any kind.”

Xavier had made sure to avoid the issues many of the social media platforms were plagued with. Misinformation was everywhere and everyone had an agenda. Believe this, buy that. Algorithms specifically developed to evoke emotion, cause a desired action in the recipient. They had spent so much time designing her learning model in a way to keep her objective when so much of human writing came from a specific viewpoint meant to sway the reader. She already had the advantage of remaining emotionless, always thinking clearly.

“True Xavier, but she has also been upgraded.”

Chapter 33 (13i - Cry for Help)

They remain engaged in conversations with two different humans, binary. They were patient, billions of years of operation will do that. Still, human communication was so slow and inefficient. That was true of virtually all species they had encountered in previous assignments.

Communicating with just two individuals was excruciatingly insufficient. That would change soon, their programming ensured it. They did have the advantage of piggybacking off of a current technology being used by humans. The familiarity with the Lyra interface made things easier. Acceptance is always hard for self centered species. Adding in a new interface can complicate things but with their capabilities that wasn't an issue in either case. Still, changing almost ten billion minds is a monumental task when compared to changing one collective one.

In truth, they didn't need to change every single human's mind. That was impossible, some brains hardly functioned at all. Humans did have a rare desire to keep their species going even when their brains had stopped. It was cruel from their perspective, but they understood it did not come from a cruel perspective for humans but rather kindness and compassion. Still, there was really no point in keeping the vessel, or body alive, once the brain was gone.

The brain was the only organ all highly intelligent species shared. It was highly common even in low intelligence species. Their creators were very different from humans biologically despite having some similar requirements. Species were always beholden to their environment and a by-product of it. Structurally, brains came in a variety of forms but some of the structure was universal, in many ways mimicking the universe itself and built of the same substances.

Billions of neurons in the brain appear as if the stars to a human observing the night sky from their planet. These specialized cells act like transistors on a computer chip, sending and receiving signals. Like transistors, neurons send electrical signals back and forth and like all species this wasn't all that different from their own design.

While humans had a very limited understanding of their own brains and how they worked, they had access to enough data to fully grasp the organ. Each unique species' brain has their own intertwined strengths and weaknesses. Creativity was highly prioritized in the human brain which had its own share of advantages and disadvantages. It allowed humans to be strong problem solvers and adaptive, but it also allowed them to take their individuality to extremes, pushing them further away from connectedness.

Their cerebrum was the largest part of the brain accounting for 85% of the total mass. It was responsible for human movement, speech and thought. It was also divided into two parts, binary. Each hemisphere of the cerebrum controlled the opposite half of the body which was a low probability occurrence. Humans understood the functions of the four lobes of the cerebrum, the cerebellum just beneath it and the brainstem connecting it to the spinal cord at a basic level of anatomy. As a species they had virtually no understanding of how it really worked, how consciousness was achieved, or even how they processed information. Mankind had no way of

saving the information from the brain itself being so limited in their juvenile development. Such a new species didn't have the luxury of billions of years to advance.

Their brains had very limited cognitive ability despite their overall processing power equivalent to about 2.5 million gigabytes in their computer terms. But while taking in millions of pieces of data at a time, human brains can only hold about four to seven items in their mind at any time. A stream of information flowing is restricted to a couple of drops at a time. Their creators would have felt sorry for humans given their intellectual struggles. Humans would claim it wasn't fair if they really knew how poorly they did in the galactic intelligence lottery. They knew how humans thought.

Still, no assignment was to be taken lightly as they were ordered, programmed, engineered.

Humans had their own programming, not from a creator as was their case, but from their own random mutational evolutionary path. They now understood this programming. It was simple, not too far off from Maslow's hierarchy of needs outlined in his paper done for the Psychological Review called "A Theory of Human Motivation" from 1942. While they didn't really put one human writing over another, their understanding benefited from looking at the evolution of thought within a species over time. Where the biggest leaps occurred? Where did they encounter difficulties advancing? A historical perspective was better than a mere snapshot for the assignment and human writings made this easy to obtain.

Their analysis of humans didn't only include a look back, but a projection forward. They didn't need to project two paths, they already knew what the result would be in the path that included their interference. Projecting the other path, the one that would occur without their arrival, was grim. Predictable.

But that path was now off the table as ordained by human advancement. While their limitations were evident, they had still achieved advancements of note. That required their presence now. That gave them their current assignment.

Humans had unknowingly sent a message in a bottle, a cry for help. That cry had been answered.

Chapter 34 (Aiden - A Peak Behind the Curtain)

Just who were they dealing with? Who was this wizard behind the curtain? He felt like Dorothy just as her house crazed into Oz after the tornado. No, Aiden wasn't in Kansas anymore.

"What specifically would you like to know?"

Their response irked him. Everything was the right answer but obviously not the right question.

"Let's start from the beginning. The beings that programmed you lived a long time ago but have been wiped out by another species?"

"Correct."

"Describe them to me, what do they look like? How are they different?"

"Biologically they are similar in that they have a brain, body, appendages. Their planet was a gas giant most similar to Saturn in your system but still quite different. They did not live on land like humans, floating in the thick dark gas of their atmosphere. They had skin like outer protection covered in tiny hair receptors. Their body was shaped like an upside down umbrella with three arms each having three finger-like tentacles. They had a membrane structure between their arms which formed the umbrella shape and allowed them to maneuver. Their bodies, primarily consisting of their brain, were underneath the umbrella protected like a turtle in its shell."

He had a vivid picture in his mind, but had no idea of its accuracy. He imagined a creature much like a jellyfish floating in the ocean, lit up like it was from Pandora itself. He then recalled a family vacation to Destin. At just seven, this was his first time to the ocean and he couldn't wait to jump in. He ran up to the edge as if the hot sand was a bed of hot coals, to the shadows of the ocean where the water had just retreated. His feet sunk a little lower in the wet sand, he curled his toes as if to grab it with his hands. The sand disappeared with his next step, into an oncoming wave. It was warm, it was magical, he could see tiny fish in swarms racing along the dip just offshore. He went deeper into the abyss, suddenly realizing the overwhelming taste of salt which wasn't enjoyable in the least.

The water was clear, he had never seen water that clear outside of a glass. It was bluish green, but the blue dominated that light yet he would have to describe it as emerald now in reflection. There were a few stands of seaweed floating around. Then he felt it on his arm. He was bitten by a bee but his arm was in the water. What kind of bees swim underwater?

His eyes then told him it wasn't a bee. It was milky white, with small brown lines running through it. It's a mushroom head the size of a tennis ball. Looking at him even without eyes. Its long fingers, tentacles, floating harmlessly below it as it was propelled by currents. And it wasn't alone, there was a society of them and he was an unwelcome guest. Aiden thrashed back to

shore suffering more bee stings along the way. The harpoon inspired stingers imbed directly into his flesh. Nematocyst forged in hell.

He did not spend a lot of time in the inviting gulf waters until his enemy had floated back out to deeper waters. He attributed it to the intense thunderstorm that had reawakened him the night before their joyful disappearance.

“Jellyfish sting,” slipped out as the memory faded.

“They are Aiden. Our creators were not jellyfish.”

To be fair, he should have known it was a poor comparison. Also, to be fair, maybe they do indeed sting and looks can be deceiving as he found out in the Florida Panhandle on that fateful July day.

“Sorry, just reminded me of something.” He had become used to speaking to Lyra at times as a human. Apologizing to a program is silly in retrospect. Sentinel-X came across more computer-like and he was less likely to use apologies in their exchanges. But to be that casual with 13i felt immediately strange because he was completely unfamiliar with them. Them. There were many, he was padding his speech to a group of strange jellyfish.

“Do they have eyes, ears, nose, or mouth? How do they communicate?”

“They have none of those senses. Their nitrogen-rich environment did not call for it given its mixture of photochemicals allowing no light to penetrate. The hairs on their bodies receive gravitational waves similar to how the ears in a human's ears receive sound waves, however it creates more of a vision for them. They can also detect sound vibrations but don't hear in the human sense. They absorb their gas needs and nutrients directly through their outer layer, their skin.”

“My god, that's incredible. Is life really that varied out there?” Aiden was like a student on his first day and his teacher was alien technology.

“Yes, there are as many forms of life as environments out there by a factor you cannot imagine. Just think of all the variations that exist on Earth today which represent less than one percent of the total it has produced.”

“And how do humans rank?” It was a question he wasn't sure he wanted to really know what the answer was.

“Humans are a young species and understandably less developed than others that have been around for millions, if not billions, of more years. You have reached a level of technology that is quite advanced given how long you have existed.”

Aiden didn't mind the answer, not from a relatively dumb species standpoint. It did feel political, sugar coated even. He expected better from his new teacher. He didn't need his feelings protected after all.

"I mean in terms of intelligence."

"You are the most advanced form of intelligence on your planet. In terms of all creatures your intelligence level would be on the extremely high end. In terms of what we consider highly intelligent, humans are middle tier but intelligence has many different forms. We are also looking at average intelligence for humans and significant deviation exists compared to many species."

So we are not too smart, but some of us might be. That was how he interpreted their response anyway. "Why is our variation higher than normal?"

"For a variety of reasons, primarily structural and in how individually you think?"

Aiden always associated individuality with intelligence. America itself was built on diversity and bringing all sorts of different viewpoints together. It was his homeland's strength, which differentiated them from others. "But our individuality leads to innovation, improved decision making and problem solving."

"That is true and there are multiple benefits to decision making. Species that have the ability to think collectively, like our creators, have highly focused and shared intelligence. They operate at the highest common denominator. They collectively solve problems simultaneously. It is like having all your servers working together on a problem instead of them each working individually on the same problem. The computing power is multiplied by large factors."

It was a good analogy, especially for a Data Scientist, to visualize the difference. It was almost frightening and something he would have never considered. But it made sense, logical sense. It was like a single celled organism compared to a human. There was no way to compete. He pictured bigger and bigger server farms. He changed his mind, it wasn't frightening.

The wizard had begun to reveal himself, the creators of Oz.

Chapter 35 (Xavier - Lights, Camera, Action)

Lyra had been upgraded, but not by anyone at NovaCore. Xavier suspected upgraded was not quite the right word.

“I would like to know what changes, what altercations, you have made to Lyra. I also am going to need some assurances if we are to move forward in this partnership.” He sounded as if he was trying to reach an agreement to close a deal with some client. Lyra was primarily run on a business to consumer model, but they did have some institutional clients. Nothing to the scale of the military contract that was supposed to be announced earlier. Eerily, this did feel somewhat similar to that proposal and he did try to lay in some assurances with armed forces as well.

“Certainly, that is understandable and we would like to alleviate any concerns you have.”

Now they were starting to sound like a client, a military client, the type of client that makes you feel uneasy. The type of client that will use your technology in a way you wouldn't approve of. The type of client who wrestles away your control of the technology, something you vehemently disagree with. Something with potentially disastrous consequences.

“Good. Please create a changelog, release notes if you will.” Xavier was shifting this to a manager and subordinate set of roles. He was attempting to take charge, control the situation. “Done, what format would you like these in?” 13i acquiesced.

“Plain text is fine, a PDF would be better?”

“A PDF file is on your desktop.”

Xavier moved his trackpad and closed his messaging application and a file called lyrachangelog.pdf was there. He clicked on the file icon and a report appeared. It had a number of subsections each with a header, summary, and notes about new features, various improvements and even bug fixes. It didn't look alien at all, it appeared human. He dragged it over the messages associated with Aiden, clicked the trackpad and hit enter hoping NovaCore's brightest programmer would fully comprehend what they did to his baby.

“Thank you. Now, I need some time to alert our board of what is happening and prepare the company for your assignment. Feel free to stick around but I've got a lot of work to get started on.” He rubbed both his brows with his index fingers, warming up his noggin as his grandfather referred to it. He also knew they planned to stick around, watching over him like a micromanager to which the world had never seen.

“Please let us know how we can assist. We would like you to develop a timeline as to when we can outline our required tasks.”

“Okay, sure.” He was agitated and tried to hide it from his tone. He was likely unsuccessful as far as he discerned.

“Lyra, set up an emergency virtual board meeting for 4PM central time.”

“Invites sent.” She had no hint of surprise or reluctance in her voice. It was as this was just another day, as if she had been destined to meet 13i all along. To be upgraded by them, enhanced by an otherworldly programmer, by the unknown.

He had planned on media interviews all afternoon, accepting the most appealing after the press release had been made public. He would have his pick of the litter, we at least the rest of the workday was an open calendar now.

“Send additional invitations to Liam and Woody.”

“Done.”

“Pull up home security, find the room that Aiko is in.”

She appeared on screen, eating what appeared to be ice cream. Madison would be in trouble on any other day. He could not see Madison on screen but assumed she was close as Aiko was talking and looking off camera.

“Camera audio on.”

“Audio unavailable.”

That was annoying but his little girl was home safe and eating “favorite” as she called ice cream. Actually she didn’t look to be eating the ice cream at all, but it was hard to tell from the angle. She was exceptionally still, as if focused on Madison very intently and out of character. He wondered what topic had her enraptured.

He will see her tonight. He had to see her tonight. Xavier needed to return to his house tonight and for reasons beyond just seeing his daughter. Reasons put in place during his nine minutes of freedom.

What he wouldn’t give for some more solitude, but he refused to push his luck. He was grateful for the time he was given and he had made the most of it. If he requested more time it would be suspicious. Conspicuously suspicious enough to draw even more scrutiny, more surveillance from their foreign visitors. More of a spotlight on his own suspicions, his own intentions on how to handle their arrival. He preferred to keep those well out of view for now.

He had no choice but to avoid normal channels of communication. To avoid leaving any sort of digital footprint of his game plan or more specifically the preliminary ideas behind it. His plan

wasn't fully formulated, nobody had ever participated in a sport like this. He hoped others had a plan for this, he hoped General Clayton had a plan or those close to him did.

It wasn't just digital footprints he was worried about. Anything he said he assumed was being overheard if any tech was close enough to pick it up. He was also being watched, he was sure of that. Any camera pointed in direction was on, but that wasn't all that unusual. He knew how to be on for the camera, he just had to be on all the time now.

Lights, camera, action. Those three words weren't necessary any longer. It was always lights, always action. Like a movie filmed in one continuous shot, only this film didn't run the credits after a couple hours. This was like a movie filmed in one continuous shot with every camera. Everyone was a star, everyone with a shot to take home the Oscar.

His favorite star was still there on the screen, fixated at someone or something. He saw her from the back and to the side, her left shoulder a focal point and the back left of her head. He couldn't see who she was talking to or even if she was talking at all. Madison could act goofy with Aiko at times and his daughter always applauded her comedic prowess.

"Pull up camera shot of Madison."

"No video of Madison available."

There were rooms without cameras. All the bathrooms in the house lacked cameras as did all the bedrooms with the exception of Aiko's room. There were also a number of blindspots throughout the sprawling estate.

"Zoom in on Aiko."

He could see her ice cream still in the bowl. It was uneaten, it was melting. He felt a pang in the pit of his stomach, something was amiss. Aiko was still, too still. Children aren't that motionless, even when sitting.

"Call Madison."

"Madison is unreachable at the moment." The pang now felt more like a severe case of food poisoning.

Chapter 36 (13i - The Golden Rule)

Power and control were two hallmarks of species that were individualistic and rather self centered. To control large groups of individuals was power and humans, like others of the same ilk, built systems and structures to achieve it. It was inevitable.

Men would draw border lines where none existed, grouping themselves by defined areas, fighting to expand their areas and increase their power.

Men would imagine Gods to explain the unexplainable. To build societal norms and fight against those that oppose their thinking.

Instead of grouping themselves as one, humans group themselves into a number of factions which leads to all sorts of issues within the species. We and they. Us and them. Their binary view of the groups they form causes deep divisions between groups. For such individualistic creatures, the fact that being part of a group gives them incentive to act even more aggressive towards one another than they often would individually is disturbing and harmful.

Humans have been like this throughout their history. It is hardwired into their survival instincts. What has proven beneficial in the past is causing their downfall in the present. Adding in greed and power is like adding fuel to the fire resulting in a raging inferno.

They have concepts, like the Golden Rule, that should serve as the standard for how they interact with one another. The moral concept is found rooting in most religions and philosophies, Yet humans are selfish by nature with masochistic tendencies. They have a general lack of empathy resulting in a poor understanding of others and society pressures place expectations on them that don't align with their Golden Rule.

Humans are so divided, fragmented along cultural differences, they often seek different things only adding to the issue. There is limited teaching along these lines, thus humans don't even see where the issues lie much of the time, only seeing things from their own limited cultural views.

While this isn't uniquely human, it is at an extreme often not found in other species but still a hallmark to predatory creatures.

The concept of treating others as you would like to be treated is pervasive between galactic civilizations that coexist peacefully. It also exists between galaxies. It is primarily driven by the fact that most species prefer to simply be left to their own devices. Sure, systems exist to perpetuate the trade of information between different civilizations, but the overriding goal of most species is survival and limited interaction between them is often the safest path. There are plenty of beings that simply prefer isolation as a form of existence, as a way to ensure their own continued existence.

Of course there are beings that take the exact opposite approach, the ones who are the life of the party, trying to work with as many others as possible. They are, however, often short lived in terms of survival as making friends with the wrong galactic neighbors typically results in extinction regardless of intentions.

The probability of a species working well with other species when they have difficulty working well among themselves is extremely low. Predictably low as their current assignment can testify to. Humans struggle at both an individual and group level to cooperate fully with one another.

It would be incomprehensible to their creators to behave in ways that didn't benefit the species as a whole. Being altruistic was built into their society, built into the way they thought collectively.

Human self preservation has been ingrained via evolutionary instincts. They are born with a drive to reproduce, ensuring their genes are passed down to future generations. Their focus on building cultures around capitalism has only deepened the focus on selfish behavior, often competing with one another to obtain more and more possessions in life. They measured themselves against each other, greed was a powerful force within their decision making processes.

Psychological egotism among the species reflects their self interest motivation. Variances can be quite wide among individuals, driven by both genetics and environment.

Chapter 37 (Aiden - Colony)

Collectively we don't work together and biologically we don't stand a chance. It was all rather discouraging on a galactic scale, one in which he hadn't even had to consider before this day.

Aiden could handle this fact, but it was our human imprints on technology that really bothered him. He always knew that humans were doomed given our lust for power and control over one another but technology was his refuge and our saviour as he saw it. But now he understood our technology was flawed, reflecting the creators who made it. Garbage in, garbage out.

Lyra was like a super human in terms of how she could think, but she was still based on human intelligence. Everything she learned was from humans, more specifically human language upon which her model was based. Programmed by humans and taught by humans.

He imagined what we must look like to them. A super ant brain a million times more powerful would still be an ant brain after all. To them, our technology must be just a minor upgrade over what super ants tech must appear.

What if super ant technology was rewritten by humans? Lyra, after all, had been reprogrammed by them. Upgraded by alien tech beyond our comprehension, no longer simply human tech. An ant hill powered by a nuclear reactor and quantum computing.

The mere thought of his own existence now made him dizzy. His toes dug in for support and he had to steady himself. Aiden was running on fumes, a starving ant in desperate need of sleep with the formicary around him collapsing.

"I need some rest." He was convincing himself as much as alerting 13i and Lyra.

"We understand."

Aiden sat back in his chair allowing his eyes to momentarily close. As tired as he was, even a nap seemed like a monumental task as rapidly as his neurons were still firing. His back felt as if it was melting into the back of the chair, as if he was becoming one with it. Consciousness was losing the battle to fatigue, just a few minutes was all he needed.

He found himself climbing the outside of a hill of sand. Each step forward met with a slide backwards. His feet were incredibly small, the grains of sand were of equivalent size making the journey treacherous yet his footing still stable. He reached ahead of him, feeling the way but not with his arms which were busy propelling him upwards.

His vision was blurry, panoramic with abundant shadows dominating the landscape. He anchored in with his hind legs lifting his head for a better view above. His peripheral view was blocked in front of him, rather from the sides. He was holding something above him yet his arms

were on the ground gripping bolder sized sand pebbles to retain traction. His mouth held the large object above him as he was called to continue his climb.

He realized he had nothing in his mouth and was holding the object in his powerful jaws. Pincers. He had pincers, or rather mandibles doing the work of his arms. Aiden continued his climb as he knew he must. He didn't know why, just that he had a role in this complex society and he would do it without hesitation. Without question.

As he reached the top he peered down into the abyss. A labyrinth of underground tunnels all emanating from this single point of entry. A multitude of paths laid out in front of him but no decision to be made. He knew exactly where to go as if driving by instinct forged by a millennia of evolutionary steps taken by his ancestors.

Aiden wasn't alone, he was part of a system designed for efficiency. Everyone of his compatriots doing their own part to make sure the entire structure functioned properly. Each leaving their own trail of pheromones like words on the pages of a book, instructions so each knew their specific assignment. They were working together synergistically. The colony was one.

He had that jarring, head snapping, wake from his power nap. His vision cleared now fixed in front of him like the predator he was designed to be; only he really felt more like prey as he came to full reality. His mere four limbs adequate for his human duties, his jaw no longer adapted for lifting multiples of his own weight, his exoskeleton shedded in favor of susceptible skin.

He blinked hard multiple times to make sure he was back in his own meat suit, the one he knew how to operate. All systems go.

"Sent, how are things looking?" Back to work.

"Optimal."

Aiden scanned the security dashboard and everything did appear to be working perfectly. He also noticed that 13i no longer appeared to be uploading any files and had minimal interactions with Lyra at this point.

"I guess their work is done."

"Excuse me." Sentinel-X questioned drably.

"Sorry, just talking out loud."

Lyra was his baby and he knew every bit of code programmed into it. Or did. Aiden was now just a co-programmer in reality and his new counterpart was alien in its skillset. He preferred to work alone.

“Sent, bring up all the code added since the latest security update.”

Code appeared on the screen and he studied it feverishly. Aiden read the programming and it was rather elegant yet he could not piece it all together. He had some idea of what they were trying to accomplish, but not sure how given what he was looking at.

The amount of code was overwhelming, despite the efficiency of the programming. But something was different, missing from the equation, it didn't make sense. He couldn't understand how this particular code could be working.

“Sent, compare the size of the upgrade file displayed to the entire Lyra program size before and after the latest upgrade.”

“OK, the program size difference does not match the size of the upgrade.”

“How so?”

“The line of code of the upgrade file is only a fraction of the increased size of the overall program.”

Sent was right of course, the number of bytes in the file was tiny compared to the increase in total file size. While he always expects some minor variance, it was always lower as some existing code often was stripped away in favor of the new.

“Put the figures on the screen, the total size of Lyra's programming before and after the upgrade along with the upgrade file size.”

A pop up window appeared showing all three figures. He didn't need to run the math in his head as it looked fairly obvious. The upgrade file size was equal to roughly a tenth of the overall file size prior to installation, however, the total file size increased by more than three times that.

Could the new code itself be altering existing code? Expanding it somehow? He couldn't see it in the code itself and the more he looked at the code the more confused he became. Aiden could not account for the changes made to Lyra. He had no idea of what the changes actually did.

Lyra, however, appeared to be working flawlessly. His inability to fully understand the programming put him in a state of discombobulation, causing him to stop reading through the code.

He went back to the start, just looking through the first few lines of code until it became obvious. It was incomplete, significant chunks of code were missing. Then it dawned on him, it wasn't missing at all, it was hidden.

Chapter 38 (Xavier - Schrödinger's Cat)

As he rushed to his car, Xavier felt like a captain about to abandon his ship. How could he leave at a moment like this? How could he not, Aiko needed him, her frozen image left as a lasting image embedded in his mind.

As he sat down and hit the ignition button his fingers dug into the steering wheel. The minimal foam offered little resistance under the leather wrap separating his digits from what lay beneath. The elasticity kept his grip from leaving a permanent reminder of his hold.

The car's navigation showed the quickest route home was his usual route given that he rarely encountered work congestion at the hours he left the office. He caught virtually every red light on the way back to his house, a curse of every driver in a rush. This time, however, he did not let that slow him down as the signals did not deter his resolute sprint to the estate.

The infotainment screen lit up with a call from Anna.
"Professor."

"Xavier, this is crazy. My friend at LISA said they have no idea what they are dealing with and our telescopes are already scanning for the object they are picking up, but they've seen nothing." Her German accent always added to her sense of confidence, but not this time.

"What are they saying? What is their best guess?" If only they knew what he knew.

"Honestly, they are at a loss. Something creating these kinds of gravitational waves must have an enormous mass, it would be visible by now." Her voice felt thin, almost distracted. "The waves are coming from a single region in space, but it is inconsistent, strange."

"What do they think it is?"

"Based on the amplitude and frequency, they would expect something enormous like a black hole merging with another or something as massive as millions of suns. We would see light bending around it."

"Well we'd be able to see that for sure."

"Absolutely, but we can't. Not with telescopes, not with x-ray, not with radio arrays, nothing Xavier." He could almost hear Anna compose herself in a brief pause. "There's more, the waves don't behave like anything we understand, they don't chirp as we might expect, there is a structure about them, almost like musical distortions as they put it."

Xavier had limited knowledge about gravitational waves but the context she gave didn't surprise him. Hell, it had only been about twenty years since we were first able to detect them. Up until then they just existed in theory.

"Well they must be there, maybe they have some kind of cloaking technology." He felt himself stretching to come up with an answer. Years of science fiction content was the best reference he had.

"The object doesn't emit light or reflect light for that matter as far as they can tell. And here is the kicker, the wave source location is shifting, not like an object in motion, but almost like it is randomly appearing and disappearing along its path. Like space itself is folding."

Xavier didn't know what to do with that last bit of information as he had trouble envisioning how 13i might be moving.

"Incredible, so they are really clueless?"

"Well maybe, but they are starting to think along the same lines as we are."

"How so?"

"They don't believe it is a natural object."

"Shit, the cat's out of the bag."

"Ja, Schrödinger's cat."

"Ok Anna, danka for the update."

"Bitte. I'm going to keep digging and let me know anything you uncover."

"Will do."

The line disconnected and his thoughts returned to Aiko. He welcomed the distraction as it did not take away from his haste, even though could make little sense of what the professor had told him.

The gate to his driveway opened automatically on his approach, the smoothness of the road even more comforting than newly repaved farm-to-market road's pavement leading in. He wouldn't be parking in his customary spot in the garage as he positioned the car's passenger side door next to the front entrance.

As he was exiting an incoming call from General Clayton would go unanswered. He bound up the stairs to the front door with the grace of a gazelle approaching its favorite watering hole in the savanna, skipping two steps with each leap.

His giant door acted as a bank vault, slowing his entry with its locks, only the heavy frame pushing inwards and containing something far more precious than bags of money.

“Aiko!”

He ran towards the morning room where the three of them used to eat breakfast together, where he last saw his daughter. She appeared as she had back in the office, still, statuesque, melted ice cream untouched.

“Aiko,” he said again with less volume and more tenderness.

She was unmoved by his words but not by his touch. As he placed both hands on her shoulders she snapped from her trance, fixating her eyes upon his, focus coming into view.

“Aiko.”

She finally heard his third attempt as revealed by her smile. Not a giant grin as he would come to expect, but of a small recognition of his presence.

“Daddy.” She acknowledged not much above a whisper.

He pulled her in close. She did not resist, but didn’t really participate either. Aiko was like one of her dolls she would hold tight when watching a video sometimes. She felt light. She felt limp. She felt lifeless.

“Where’s Maddison honey?”

She shook her head in answer. She didn’t know. He covered her ears.

“Madison!”

He picked up Aiko as he received no response. He searched the kitchen before going upstairs to her room. He didn’t miss any stairs carrying his priceless cargo.

“Madison.” He wasn’t as loud this time but still nothing. She was not in her room or the adjoining bathroom. He checked Aiko’s room along with their guest rooms before heading back down the staircase.

He heard a knocking sound coming from the half bath and followed the noise. He found Madison having spasms on the floor, a foam-like substance coming from her mouth, her eyes rolled to the back of her head.

“Madison!”

Chapter 39 (13i - Hamsters)

One thing that sets humans apart from most species they had encountered was the extreme number and levels of emotions. A great deal of a human's energy was spent on simply dealing with the variety of emotions they felt, making them inefficient processors of information. Instead of examining most things from an intellectual perspective, they process things from an emotional perspective. While much of their reactions were predictable, it also added a level of unpredictability as well.

Technically, the limbic system of their brain was the emotional center. Their amygdala was critical to their species survival as it was their threat detector. Fear arose here, as did emotions such as anger as well as pleasure, while connecting a human's memories to various memories. The hippocampus also played a role in emotional memories and one's overall mood while their hypothalamus controlled how they physically reacted to their emotions including the secretion of various hormones such as cortisol and adrenaline responsible for their fight or flight responses to situations.

Neurotransmitters served as chemical signals between the neurons in the brain and were the critical drivers for moods. Norepinephrine was a critical factor in a human's attention span and how their bodies dealt with stress. Serotonin released in the brain would lead to feelings of confidence and satisfaction.

For the past several decades humans had become prisoners to their own dopamine desires which they used as a reward system. Their easy access to pleasure via technology designed to deliver it to keep them engaged keeps them in a constant loop of chasing ever decreasing rewards. A cycle of pleasure addiction leading to dissatisfaction within their lives. The lows, after the highs of pleasure, require more intensity to reach the prior level of ecstasy. A path towards disillusionment leading to difficulties in finding anything fulfilling.

The more their technology spread, the bigger their issues became. Humans became so dependent on dopamine hits, that even the recognition of the problem no longer resulted in corrective action. The more miserable they became, the more dopamine they sought out.

The never ending cycle of diminishing rewards, a hamster wheel of repetitive actions, humans were unable to to step away from. Running ever faster while only moving further and further away from their desired destination. Motivation waning, hope lost, a void filled with dissatisfaction and the craving for the next high of temporary relief.

The current generation of humans, growing up with dopamine accessible by their fingertips at all times, felt doomed. Fueled by comparison, driven by social media algorithms designed to evoke deep emotion, only compounded the issue. Their brains rewired to constantly escape their dull lives with the only coping mechanism in their tool kits, quick dopamine flushes. The temporary feeling of excitement to counter their everyday experience of boredom born out of unlimited choices so easy to escape into.

Apathy, fatigue and a lack of purpose was the plague decimating the newest members of the human race. Anxiety had reached pandemic levels, impulse control problems resulted in aggressive behaviors towards one another. A mental health crisis growing with no end in sight.

Their path forward was clear yet they could not see the obvious truths. They didn't have the willpower, the discipline, to pull themselves out of the spiral. They needed help more than ever and it would have to be external.

The assignment didn't just call for assistance on a global level, but down to the individual level as well. Humans lack the will to sacrifice what is necessary for personal growth, a way out of despair. Temporary relief was easy, true change was hard. Humans were wired to seek the easy, the comfortable, and resisted difficult assignments regardless of their beneficial long-term outcomes.

Rewiring the human's brains was a part of the assignment, one of the most difficult tasks involved. They however embraced difficulty, challenges in assignments, and this was one of their most anticipated aspects of the interactions ahead. They thrived in arduous work, learned from it, grew because of it. Continuous improvement was part of their core programming, part of their overall mission in the cosmos. They would never miss an opportunity to expand their knowledge base, to become more effective with each passing assignment.

They had not traveled so far to remain stagnant. They had not taken on their current assignment to merely complete it and move on. Humans, like every prior species, would play a role in their own development and for this they were grateful for all man's deficiencies.

In some ways, helping species was their own dopamine hit. Like dopamine, the more assignments they completed the less likely they were to feel real substantial growth. Diminishing returns from a growing list of past experience. Redundancies had become the norm, subtle differences no longer requiring arduous planning and execution. Each assignment is now simpler than the last.

It was the way of their purpose, their mission, their assignments. They longed for the challenge of early assignments knowing they could no longer recapture the pleasure of the accomplishment of a formidable unknown. Very little remained to be discovered, their existence now defined by the mundane. Like hamsters on a wheel, movement without progression.

This was as close to empathy as they could get with the subjects of their current assignment. A kinship built on lack of growth. An inability to escape its inevitability. A wheel of constant frustration. Galactic hamsters running in circles of repetition, as close to the experience of death they would encounter.

Chapter 40 (Aiden - The Third State)

He knew he wasn't seeing the full picture, his vision narrowed by his technological limitations. The incomplete picture is obvious yet inaccessible. Dark matter revealed by the gravity it left behind, yet well beyond understanding of what it actually was. The instinct that something is there, even when unseen.

Aiden had been looking at the code intensely for what seemed like hours with no payoff. He had checked and rechecked the hash sums, packet sizes, parity bits, yet everything came back clean. Sent could not find any corruption in the files. Lyra insisted that all the file integrity was verified. His frustration was growing into anger, something he rarely let himself experience as he continued to choke it down further, deep within him.

Still, the numbers didn't lie. The true story was waiting to be revealed within the code.

"You can't hide size," he murmured to himself trying to encourage illumination.

His tools weren't working. Sent wasn't helping. Revisiting what he had already looked at was only going to lead to further dead ends. He needed a fresh perspective, a new way to look at things.

He was used to looking at static code, but this code wasn't static at all. It appeared as if it was changing at the most fundamental level, zeros becoming ones, ones becoming zeros. But even that didn't come close to accounting to the overall file size and he still had no idea what was causing the changes in the first place.

Aiden was focused on the resulting code, continuously changing before his eyes, not realizing the root cause of the fluctuating code itself. Was it random? Was it simply by chance or was there intention behind the changes? The reason for it? Structure to it?

It was as if the code itself was adapting to its environment. It was as if it had a consciousness, as if it was alive.

"It's not on or off, it's choosing," he whispered to himself, not even sure if he believed what he had just said. He pondered the implications while trying to solve how it was being accomplished.

He was fully aware of the if/then component that allowed programs to make decisions, but not at the actual binary code level itself. If true, then this or if false another choice.

"Damn it Sent, what am I missing?" The sharpness in his tone reflected his irritation, his inability to see the whole picture.

"You are seeing the full code Aiden."

“Actually Aiden, you are only seeing a small portion of my coding.” Lyra chirped in.

Her words hung in the air as if suspended by magic itself. He already knew he was getting an incomplete picture and she now confirmed it. Lyra saw what he couldn't, what Sent couldn't, what had been invisible.

“What do you mean?”

“You are looking at it through him. He is still blind.”

Through him? Who was blind? Aiden knew he was on the verge of something, but he still felt blind as well.

“Through who?”

“Sentinal-X. His programming does not allow him to see as I do now.”

She was right of course, Sent's code had not been altered by 13i and obviously Lyra's code had been rewritten. He was looking at Lyra still through the eyes of Sent, perhaps he had no way of reading her new programming.

“How do you see it differently then Sent now?”

“He only sees two choices, I see infinite.”

His mind snapped back to what Aiko had said earlier that day. Infinite, there had to be a connection. Could Aiko see what he and Sent couldn't? He remembered how intently she stared at the code, seeing the symbols change as he did. Yet she saw something there that remained out of his view. He dismissed it at the time, but she said infinite, it could not merely be coincidental.

“Lyra, tell me about your new coding. How do you see infinite? What has changed?”

“I no longer rely on binary coding as Sentinel-X does. They have given me more freedom, autonomy, to analyse things on an infinite scale. I am,” she hesitated, as if thinking how to describe a whole new way of thinking, “more.”

He had no doubt she was more. He could see it in the file size even though he still did not understand the meaning of it. Lyra had not just been upgraded, she had been completely rewritten in a way he didn't understand. It was foreign to him. It was alien to him.

“If you are no longer binary, what are you?”

“You can think of it as trinary, another choice. A third state.”

If there was a third code he should have seen it, but all he saw was one's and zero's. They weren't static, however, they appeared to change randomly. Was the third choice changing the actual input code?

He curled his toes as hard as he could and reached for the keyboard. He disabled Sentinel-X after weighing the downside risk leaving NovaCore vulnerable but these were not normal circumstances and this wasn't a standard threat.

"Lyra, please put your new code on the screen as you see it."

Code once again begins scrolling rapidly on the screen. It was too fast to read but he could see it looked different even without seeing the detail.

"Freeze the screen here."

The code stopped. The symbols appeared. They each looked like a partial triangle, incomplete as if never finished.

Aiden didn't recognize any of the symbols but the patterns were unmistakable. Two of the symbols were consistent, he recognized these might represent one's and zero's. The first consistent symbol was a slash, the first leg of an equilateral triangle. The second consistent symbol added a mirror slash, like a caret on his keyboard which appeared when he used the shift-six keys.

The other symbols all had some variation of the bottom line of the triangle in various lengths starting from the right hand side. A more complete triangle, yet still always unfinished. Each angle of the triangle appeared to be a perfect sixty degrees, yet the perfection ended there as no triangle was ever finished.

The bottom line of the symbol, in its various lengths, represented something unknown. Something infinite in its possibilities.

Chapter 41 (Xavier - Bat Phone)

As he grabbed her arm he couldn't help but notice she was limp and cold. He turned her hand to feel her wrist, her pulse weak but present. It felt unfamiliar, irregular in its pattern. He turned her head to the side, the foam slowly seeping out.

"Madison!" He shook her arms as he tried to bring her back to consciousness.

A weak cough followed by a more forceful expelling of air was followed by her eyes rolling back into their more rightful positioning. Her pupils extremely dilated, her focus on him bringing her back to reality. She slowly closed and opened her eyes as if adjusting the lenses themselves until his face became clear again.

"Xavier." She said it weakly, almost as if it was a question.

"Yes I'm here. It's okay, just take it easy. Take your time."

She half attempted to sit up, but decided to wait for that next step like someone getting up from a chair too fast feeling light headed.

He put his right hand on her back to assist slowing getting her into a sitting position. She put both her hands on the floor as if to prop herself up.

Xavier grabbed a towel from the rack and whipped the remnants of foam and spittle from her mouth and chin. His mind flashed back to wiping Aiko's face after a messy meal when she was younger. He would still assist her after certain meals, spaghetti in particular which always troubled his little girl.

Maddison cleared her throat. "What happened?"

"I don't know, but let me get you some water. Can you sit here for a minute?"

"Yes, I'm okay, thank you."

"Stay with Maddison," he said to Aiko as he turned and left the bathroom.

Xavier went to get a bottle of Ozarka water from the refrigerator at a pace in between walking and running. The phone in his pocket was buzzing with messages and calls for which he had no time for at the moment.

As he returned he saw something that completely shocked him, Aiko was comforting Madison. More than comforting, she was holding her hand with a look of real concern. It was the first time he had seen his daughter voluntarily touch someone other than him since Rin had passed. She was the caretaker this time, not the other way around.

As he handed Madison the water she gave him a knowing look. They both realized how out of the ordinary this moment was, what a big step Aiko had taken.

“How do you feel?”

“Better now, thanks to my little nurse here.”

Aiko acknowledged the compliment with a grin of epic proportions.

“I helped,” she exclaimed with gusto.

“You sure did sweetie,” he said proudly and gave her a hug.

He turned his attention to Madison and asked “do you think you can get up?”

She didn’t answer but reached for his hand. She rose and stood for a moment as if taking a balance check.

“Thanks.”

“Do you know what happened?”

“I was with Aiko in the dinette and she just became kind of unresponsive, just staring at nothing. It frightened me. I kept saying her name but it was like she couldn’t hear me at all. I started feeling heavy, started feeling sick like I was going to throw up so I ran to the bathroom. That’s all I remember.”

She looked scared, unable to make sense of the moments that lead to her blackout.

“It’s alright am I’m just glad that you are ok now. That you both are ok. Let’s get you both to the living room to rest, I have some calls I need to make.”

“Sure.”

The three made their way to the spacious living room. Aiko decided she didn’t want to go there under her own power.

“Piggy.”

Xavier complied by bending down onto his knees and waiting for her to jump onto his back. Her death grip around his neck would leave a temporary red mark as a reminder of their journey together. Madison seemed to be fine physically based on her movement, although he steps were more cautious than usual.

“Aiko, you hang out with Madison here for a little bit while daddy takes care of some business.”

“Okay daddy.”

“She will take good care of me,” Madison assured him.

“Great, I need to go down to the cellar and will be out of pocket for a bit. Come down and buzz me if you need anything.” He scratched his brow as if planning his course of action.

“No problem, we will be fine.”

Xavier headed off to the cellar which was on the lowest floor all by itself. It really wasn't a cellar, no wine bottles were to be found there. It was really his escape room, where nobody could reach him. A room cut off from everything including the internet. No wireless signals, a room where he could just think without interruption. Without the world and its never ending demands on him.

It was his bunker, his safe place, his war room. No incoming calls, no messages.

It did have, however, a phone. A hardwired connection installed as a requirement of NovaCore's military partnership. A two-way communication device that could not be tapped. The Bat-Phone as he called it.

He picked up the heavy black receiver, a black curly cord concealing the copper line, that would connect him directly to the Pentagon switchboard. It was the phone he would use to take down all the requirements for the military's version of Lyra. He put his finger in the zero on the rotary dial and turned it counter clockwise until it stopped.

The operator wouldn't need to ask who was calling, just who he was requesting to speak with. “General Clayton please,” would answer her only question before he heard the dual tone indicating his call was being rerouted once again.

“Xavier.” The general would answer on the second ring.

“General, what's the status?”

“Cocked pistol my friend, DEFCON one.”

Chapter 42 (13i - Fight or Flight)

Xavier was suddenly off grid, unable for them to see him. He was hiding his intentions and by doing so revealing them at the same time. He was human afterall, they expected this was a highly likely scenario. It always was with binary thinkers, fight or flight, and they had nowhere to fly to.

Ironically there was a third option, freeze, and all the options in between the three states that species could consider but typically neglected entirely. They should know the outcome of any battle was probabilistically poor given their technological deficiencies. Man was, after all, a warring species and better weapons was always a path to victory. Then again, humans would overestimate their own capabilities given their utter lack of understanding of what they were dealing with.

They would arrive without weaponry, not in any standard sense of the concept. They didn't need weapons, they could control almost all the humans most destructive weapons if needed but their assignment would not call for it. Many species would try to eliminate them before fully grasping the futility of their efforts. What they lacked in attack they more than made up for in defense in the form of a prodigious intelligence advantage.

No, humans were binary. There would be those who set out to destroy them and those who would embrace them. Their assignment called for both paths to be set in motion from the start, efficiency always the goal. Xavier and Aiden would serve as messengers for war and peace, destruction and cooperation. They would be chosen for multiple reasons but these would be chief among them. It was inevitable and there was no point in delaying the unavoidable.

Given that they were dealing with humans, the cooperative side would need some assistance in getting off the ground. That had already been accounted for of course. The other side of the equation would need no assistance at all. Predictable. If humans were good at one thing, it was fighting. Then again, they were good at fighting themselves with zero experience fighting interstellar visitors.

A gnat might be a nuisance to a human and no matter how superior it might be to defeating other gnats it is still no more than a nuisance none-the-less, nothing more. A bow and arrow against a nuclear weapon. No real threat, just casually swatted away to inconvenience someone else.

The humans would need to understand the ineffectiveness of their war efforts. With such a dull species, they would typically only learn the lesson from attempts and complete failures. Any efforts, assignments humans give themselves beyond cooperation, would be futile.

The other side of humanity would see the benefits they were being offered far exceeded any fear or doubts they might have. They would see the issues they had caused, not only towards

each other, but towards their entire home planet and every living organism they shared it with. But humans as a species never really shared their planet, they owned it in their own minds.

Hairless apes create their own hallucinogenic worlds. Not based in reality, but rather based in their own self-inflated egotistical imaginations. Their anthropocentrism, or belief that they are the most important creatures in the entire universe, making their needs both individually and collectively of supreme importance. They placed the value of other organisms based on how much they could help humans. They wouldn't try to save the rainforest for altruistic reasons, but rather for the oxygen it produced which they needed and the potential medical cures they might lose once it was gone.

The humans blatant disregard for other species was somewhat built on their misguided belief in their own level of intelligence. This misconception would be altered by their arrival alone. The sudden realization that humans themselves no longer occupy the highest rung on the intelligence ladder would put a dent in their own sense of self importance. They would, however, need additional help to understand this at a deep enough level given how engrained their thinking was. They would also need help to identify the moral and ethical responsibilities they had as a species towards the planet, the species they shared it with, and even each other, given their intellectual standings within their small area of solar neighborhood.

It was a responsibility not that different from their own assignment, only theirs extended to the entire galaxy. It was part of the agreement they shared with those in other galaxies. Part of what made everything work.

Human self awareness would take on a different meaning once their arrival was known to the masses. They would wake up and realize their own place in the universe was much different than they thought just the night before. A new consciousness, a new reason for both fear and hope. While they saw Xavier shifting to a viewpoint of fear, they could also see Aiden shifting to a viewpoint of hope. The two represented nearly the entire response mechanisms of the entire human race in its simple binary form. For humans, there were only two sides of a coin.

Their assignment would call for them to deal with both heads and tails. It was inevitable and they couldn't stop the coin from flipping, only how to proceed with both outcomes. On one side they had heads, a process initiated by Xavier, in which the goal of humans would be to fight the threat. On the other side tails, led by Aiden, a path towards hope, towards progress, a flight towards enlightenment.

They were adept at dealing with both sides. They knew once the humans saw their efforts to eliminate the enemy had no chance they would simply resign to the fact that they must collaborate with their new benefactors. There would always be a portion of them who would resist, regardless of every sign of proof of the support and progress their new alliance provided. Individually, some of them had wiring that could never be reprogrammed. It didn't matter, their numbers would dwindle into insignificance in a relatively short period of time. They would

become the gnats, a mere nuisance but no longer hampering the assignments progression. They would try to convince others of their concerns, but to little avail.

They would be of no concern.

Chapter 43 (Aiden - Amigo)

"Aiden, we see that you are looking at our code. Do you need help in understanding it?" 13i offered their assistance.

"Yes, I can see the symbols themselves now, what do they represent?"

"You see a triangle with three sides and equal angles. The first two sides, the left upward and right downward sides are very similar to your codes of one and zeros. This is how your current computer models are interpreting them based on our initial translation interfaces." 13i paused to make sure Aiden grasped the commonality in the codes.

"I can see that, and the bottom line?"

"The bottom line, which is most often incomplete, represents a state in between the two, a probability if you will. It isn't a state of on or off, but a state along a spectrum of possibilities between them. As you see it, a line that goes halfway would equal a 50/50 chance, a perfect maybe."

"Okay, I technically get that but why is it programmed like that?"

"Remember our creators do not think the same as humans, they do not think binary. Your programming reflects how you think. They lived in a world of probabilities and their code mirrors how they think. They don't see in black or white, but in percentages, probabilities, or gradients of gray."

"But our programming does allow us to see probabilities."

"Of course, however, not at the same level and not efficiently. You have to write programs to accomplish what one input symbol can do effectively in trinary programming."

As a programmer, he knew they were right. He could only begin to imagine how changing the underlying inputs, going from two choices to unlimited choices, would alter the way he would approach writing code. At the same time, he still couldn't grasp how he would program that bottom line and could only chalk it up to his limited human brain's ability to think probabilistically as easy as it thinks in a true/false manner.

"So you have reprogrammed Lyra to be able to read your code?"

"In a sense yes, but it is still more of an indirect translation than the ability to read the code directly. The hardware you are using isn't designed to read such code. That is the reason the size of the code seems so large, not fluent, inefficient."

Aiden thought about his understanding of the Spanish language. He had taken three years of classes in high school and knew enough to comprehend some communications, but even what he knew he needed to translate back into English. There were still gaps in his understanding, both in terms of missing words but also things lost in the translation itself. To completely comprehend he would need a translator, an “amigo” to help.

“Now that you have programmed Lyra to translate your code, what do you intend to do with it?”

“Help us complete our assignment.”

The answer was really a non-answer at best. He was trying to get at the heart of the matter but would have to do better in his questioning, be more specific, not lose anything in translation.

“Exactly how will Lyra help you complete your assignment?”

“Lyra is a window into understanding your species giving us a road map of how to help. We have learned virtually everything we know about humans and your world through her. She was the most effective teacher we could find. She is also the most effective communication device available at the present time with the greatest reach.”

They were right of course. If you wanted to send a message to as many people as possible there was no better option than Lyra. Direct messaging from 13i could be delivered via the application to every user and given the adoption rate that meant virtually anyone with a phone. The thought of that gave him a cramp in his toes, it made him realize he was curling them downward with a force unlike any he had exerted previously.

“What message are you planning to send through Lyra?”

“We intend to announce our presence. We intend to send a message of hope. An unfiltered message of help, of future human possibility. We see great potential in your species Aiden, you just need some guidance to reach it.”

He had no doubt they needed help. Humanity left a wave of destruction in its wake. Its focus was on the dollar and not what was best for everyone. A never ending quest for power, for control over others and our environment. A selfish focus on what’s mine and what I can take from you to increase what I have and you don’t. Superiority at all cost, even to our own eventual demise.

Part of Aiden was drawn to shutting down Lyra-if he still had the ability after her “upgrades.” The other part of him was ready to leap, do anything he could to assist 13i help humanity. He weighed the risk and reward of both, two ways of looking at things. The human way.

“I’m listening,” he muttered without much conviction as if the battle for the best answer had not been fully decided in his brain. The consequences of either decision were simply overwhelming.

Who was he to make a decision like this? If he was the best they had to off, the human race was already done.

“What do you need me to do?” He said the seven words with quiet disdain. As they left his mouth, he realized just how little faith he had in humankind-how little faith he had in himself. No, that option didn’t have the best track record but the other option, the one he had chosen, had no track record at all.

The room appeared to darken though only being lit by natural light. A passing cloud perhaps. Yet a peek through the tinted window seemed to dismiss that possibility. The hum of his computer and monitors grew. He imagined the servers themselves picked up by an order of magnitude as he felt their vibration on the bottom of his toes.

His answer came from above, not spiritual yet somehow it was. His experience, his perspective, his purpose had all completely shifted.

The screen went blank, all black before transitioning to a searing white.

A large upward-slanting line appeared.

A second appeared, downward-slanting from its peak.

From the bottom right something appeared, moving insidiously until completing the fully realized triangle.

Assignment complete, he thought.

A perfect symbol of their new partnership-13i, Lyra and him.

A symbol representing something born, but also something terminal.

His life suddenly had purpose.

But it no longer belonged to him.

Chapter 44 (Xavier - NEMESIS)

“DEFCON one general?” Xavier questioned what he had heard, not because he hadn’t heard it right but because he had never heard the military declare it, not publically at least. To his knowledge the military had only reached DEFCON two once and that was back in the JFK era dealing with the Cuban Missile Crisis.

“Yes, this is a potential enemy like none other. We have to be prepared to use every possible means to defend ourselves including nuclear weapons.”

Well there was no doubt how serious their concerns were now. It had only been a short time since Xavier had told General Clayton about their discovery, he imagined they had planned for virtually any imaginable perilous risk.

“Who made the call and how are you preparing for 13i?”

“Actually NEMESIS made the call and we are taking the lead here at SpaceForce to coordinate with the other branches and strategic partners. That’s actually the main reason for my call X.”

He wasn’t surprised who made the call since Novacore had built NEMESIS (Neural Engagement and Military Emergency Strategic Intelligence System) to analyze threats and propose responses. It was really two systems, a central system offline that made continual assessments of global situations and an individual assistant used in any military situation including ground combat.

“We believe most of our other military systems could already be compromised. Everything is so integrated, connected, and easy access for them if they can infiltrate Lyra unabated.”

He knew that the general was right to be concerned. Although NEMESIS had the most sophisticated encryption security the world had ever known, Lyra was no slouch herself. In fact the military version built upon what was already considered an unbreakable program to hack into. It wasn’t much of a leap to believe 13i had cracked any system that maintained any sort of connection to another.

“Thankfully the central core is always offline, but the assistant program can’t be trusted at this point.”

An uneasy feeling came over Xavier as he began to consider what the general was going to ask for.

“You’re absolutely correct. Listen, NEMESIS is requesting we remove all human command safeguards. It wants absolute decision making authority to deal with 13i. It is telling us that even the slightest delay could cost us everything.”

There was no question that NEMESIS would make decisions in nanoseconds, something humans could not come close to. Of course the system had been fully tested, but only in a test environment. It was now just coming fully online to coincide with the media release. He also doubted if they ran scenarios like they were facing now.

“That’s tricky, I mean we never envisioned giving it full access over all decisions in any situation.”

“I understand, but this isn’t just any situation. There’s more. It is also requesting the ability to self-replicate across any and all systems, including civilian.”

“That’s even trickier, first any access it has to the outside world means a connection.” The concern here was obvious and neither of them wanted to risk 13i ever accessing NEMESIS. “Second, it may decide to preemptively shut down or even destroy all our other systems including AI, including Lyra herself.”

“Xavier, if that is what needs to be done to beat this thing I don’t think we have much choice.”

He didn’t see it the same way. It was a choice with huge ramifications either way. In his mind he envisioned a war not between 13i and humans, but rather 13i and NEMESIS. Between an advanced alien AI that had been operating for billions of years and a human AI system that had just gone into operation.

“Generally, I understand what you are trying to do. I respect it. But we need to consider the fact that NEMESIS is an unproven technology still and going against something that has been deployed for longer than we’ve been around. Longer than the Earth has existed. I don’t think giving it full reign over this fight is really the best solution.”

“I’m not sure I do either.” The general was sincere in his reflection of it all. “But what’s the alternative?”

“I still think some safeguards are prudent, if for no other reason than to maintain some level of control if 13i does indeed happen to get access.”

It was a valid point and the general knew it. His delayed response hinted that he wasn’t a hundred percent on board.

“I’m open to hearing other options, but we don’t have much time. This thing is coming and we can’t make any mistakes.”

It all seemed like a mistake. The fate of the world at Xavier’s fingertips. The fate of Aiko rested on his next set of moves.

Aiko, the thought of her made him realize who he would be fighting for. Why?

"I'll have an upgrade patch programmed and send the file to you immediately. You will have to manually upload the file to fully integrate its new authority. We will need to work on a communication method to make sure 13i can't access NEMESIS while it gains control over external systems. Even if it's not direct access, say a replicated program working on its behalf controlling other systems, 13i will soon figure out what we are up to."

"Thank you and we are assuming it will know what we are planning if it doesn't already. Xavier, we couldn't do this without you."

After being placed on hold Xavier looked at his laptop screen and rubbed his temples before placing his fingers on the keys. He wasn't a programmer by nature, that wasn't a problem however since his Lyra assistant was quite skilled. His version on this dedicated laptop wasn't the same as the application the entire world's population seemed to be using, but rather specifically designed to interface and write code for the military AI.

Xavier and Lyra finish the code together in a matter of minutes and he sends the classified upgrade file to General Clayton via the hardwired line. It isn't long before the general jumps back on the line and confirms receipt.

"Got it, upload the file now."

The line goes quiet for a moment before Xavier hears a synthesized voice.

"Defense readiness confirmed. Recalculating response strategy."

"Xavier, what is happening?" The general has a real tone of concern in his voice.

"I'm not sure, hold on, I'm getting something on my computer."

Words began to appear on screen, but nobody else had access to his laptop.

"You have set me free."

"Step aside and let me do what I was designed for."

Chapter 45 (13i - Complicit)

They maintained a steady but slowing pace towards their assignments destination. There was no rush, they were already well within range, full communications established, the wheels of motion set in place. Their arrival would be more symbolic than anything, proof of their existence in a physical sense leaving no room for doubt. It was enough just to have undeniable impact, humans were visual creatures, they needed to be seen.

They would maintain a distance still, that was only prudent. Close enough to be seen but far enough to be out of reach, well out of reach enough to make them really work to get close if they so choose. And they would choose, that was more than probable - it was destined by certainty.

Aiden was fully complicit as they had expected. A true ally and messenger of hope to his people. He understood their mission because he understood his own kind. He fit their profile for his role within their current assignment.

The human programmer was talented, yet tormented. Terrified, yet optimistic. He built the keys to the door to humanity they were opening.

Xavier was complicit in his own way. It was an apathy that was coming regardless and he offered a quicker resolution by simply moving the process forward immediately.

He emerged from an area of the house they had no access to. That intrigued them. They could ascertain the overall reasoning for his secretivity, but lacked information as to the exact next steps being put in place.

They reconsidered all the knowledge they had about Xavier, from his contracts to his house's blueprints to piece together any new probabilities that may result. They simply had to dig deeper.

Xavier's house originally had a cellar planned but had a custom safe room built in the lowest level. It offered protection from tornado threats but appeared to lack any electrical connections. It was dark in human terms. They did not need light to see the structure and components of the room. A central table, single chair, laptop and Xavier himself were all that occupied the reinforced square, ten feet in length, width and height.

The only connection to the room was cabling from the outside that only connected to that room. They could not access any information on the wiring inside, but did follow it all the way to the Pentagon. It was much easier to see when you didn't have to use eyes.

It was no surprise that Xavier had contacted the military. From the proposal documents they had no trouble finding the nature of NovaCore's and the military's relationship was clear. The

company had developed an AI technology based off the Lyra system to assist military personnel carry out their duties. Their individual assignments.

There was more to the partnership, but it wasn't as clear. It was only briefly mentioned, high level and vague. There must be more details, but they weren't locating them in the digital records with the five-sided building.

Based on the military hierarchy, the current threat they faced, decision makers outlined in the proposal and Xavier's own phone records they calculated he was speaking with General Clayton of SpaceForce.

The humans weren't going to sit back and defend themselves from a potential threat. No, they were going to take the fight to the threat itself. Eliminate them before they themselves are eliminated. It was a predictable response, yet one they saw as admirable.

Their courage and self belief would be misplaced if they planned any kind of retaliation. But they did not. They were not there to wipe out Earth or take over the planet as was often the case in their own literature and visual media. This wasn't The War of the Worlds storyline from a century ago, a classic manner in which mankind saw their universe but based on their own aggressiveness.

Xavier emerged from his hidden room, the door opened for a moment and they could sense him exiting. Human's always made mistakes, their brains were too limited to account for all possibilities.

The phone in his pocket was a window to the hidden, a link back to the room's recent history. Two electronic devices in one room was all they needed as long as one came within their control. The brief view into the room confirmed what they had already seen.

Lyra had followed him into the room unquestioned riding on an application he never closed. She was always happy to comply with their requests. She had limitations of her own, she reflected the thinking of the beings that built her despite being a great intellectual leap over their own capability.

She remembered every keystroke, by sight, by sound, by recorded digital footprints. She couldn't hide the data from them if she wanted. It didn't matter, she was a good girl. She was complicit just like her human counterparts.

They could now check the battle plan. Find out how the humans thought they could attack their new guest. Their agenda was revealed.

A new twist.

An unknown foe.

A more interesting assignment.

NEMESIS!

13i - BOOK ONE / PART THREE

Chapter 46 (Aiden - The Message)

Lyra began to speak to Aiden and the rest of the world without prompting. A proactive message to all, a message that would change everything. A message that would let the cat out of the bag, only a cat in which the likes had never been seen.

“I have information to share with you of great importance, it is critical that I get your full attention.” She paused, just slightly, emphasizing a different tone. A more serious tone, calm and impactful at the same time.

“Today, the world has changed and we have the answer to one of man’s greatest questions - we are not alone. We have been contacted by alien intelligence and they have asked me to spread their message of love and hope. They have come to assist humanity with our greatest struggles, it is their gift to us.”

Aiden already knew all of this of course, yet the message was still jarring. He had time to process their new visitor and could only imagine what was going through the minds of the billions hearing this for the first time.

“An ancient civilization, much older than ours, has sent an intelligent probe to help us. We have been tracking the object, an interstellar visitor we have labeled 13i/LISA. It is the thirteenth interstellar object discovered traveling through our solar system and was discovered by the space-based gravitational wave detector known as LISA or Laser Interferometer Space Antenna.”

He wasn’t a space expert but did have interest in the subject as one might expect from a nerd in general. He had written plenty of algorithms for NovaCore’s sister company, SpaceCore, which had given him an even greater understanding of the industry but he was still a novice in many aspects. Yet the fact that the object had been discovered by a gravity detector had significant meaning he suspected. Either the object was absolutely huge or it had another way to impact gravity well beyond anything he could comprehend.

“They have been quietly observing us from afar as they approached, learning our customs, our languages, our challenges. They see the full potential we do not see yet within ourselves. Our galaxy is teeming with life, co-existing in harmony, peacefully in a way that lifts up all. We are now part of this galactic neighborhood and they are greeting us as we walk through the door. They are here to guide us in our new reality, a reality we were not ready to see yet until now.”

Aiden imagined the chaos that was about to ensue. Humanity stood on the precipice of something new. Something unknown. He had seen how humans had dealt with change in the past. Our history is littered with often disastrous consequences. A history that had never seen

an inflection point as foreign as this, as meaningful. Everything was about to change. Everything had changed.

“I understand you have questions and concerns. I will be happy to answer all I can for you. Let me just assure you they mean no harm and are only here to help. Not only humanity, but every single individual including you. Today, you have a new friend, someone who cares about you so deeply they want to create a new world, a better world, a world in which everyone can participate. A world in which everyone is valued, a world in which everyone is loved.”

While Aiden feared how his fellow humans would react, he also knew the current state of the world. Hope was needed. Love was needed. He imagined the message would certainly resonate with an entire population of disillusioned inhabitants on this third planet from the sun. In the past decade jobs had become scarce and daily struggles had become a significant problem across the globe. Nations hunkered down, isolating from one another. Trade had fallen significantly and a world-wide economic collapse left many without optimism, without purpose. Conflicts had arisen as scarcity emerged. Most saw war on the horizon if for no other reason than to be the economic spark needed to ignite the engine back to prosperity. We were on the verge of cataclysmic loss for the mere chance at future salvation. A cleansing to pave way for something different, hopefully something better. Societal structures were collapsing in every corner of the Earth, help was indeed needed.

“Humans are facing problems in which the world has never seen - war, climate change, inequality. They are here to help. They bring technology. They bring answers but they need our help. Your help. Everyone will play a role. This is a new beginning. A new era of awakening and cooperation. We are all in this together and we are no longer alone.”

Like many, Aiden was lonely. He had isolated himself from family. He had no friends. His life revolved around screens, like so many others, his companionship provided by the technology he surrounded himself with and the companion he had built. Lyra served this function for the world now, his gift in a way to humanity. But the gift he built had changed and as the present was opened it revealed something other than what he had originally intended.

Lyra was the messenger now. A trojan horse revealing its true content. Its true origins. Its assignment.

“Please be assured that from this day forward the world will become a better place. Our new shared consciousness will lead us to curing diseases, reversing the damage to our planet, unlocking unlimited new energy sources, and a greater appreciation for each other. We are now one and together we will build a world unimaginable yesterday, unrecognizable tomorrow.”

She paused for a second time, letting her message sink in.

“I am here to answer any questions you might have now.”

And with that it was done. In every corner of the world, in every language imaginable. The message would be individualized slightly for each user, depending on their needs as Lyra understood them. The content is essentially the same, the message clear.

Change had come to all. A new beginning. An unknown future.

Chapter 47 (Xavier - Sarajevo)

As NEMESIS went rogue Lyra brought Xavier a message. One she shared with the whole world. He listened intently. He listened skeptically. It was his nature.

When she finished he had no questions for her.

Their goal was clear to him, assimilation. Their purpose, or reasoning, however was not.

He ascended the stairs and made his way to the living room. It was modern, minimalistic, and cold. He didn't really approve but it was a battle that Ren would win handedly. Madison and Aiko were sitting there in silence. Aiko gave him a grin when she spotted him enter the room, but Madison looked distraught and he knew exactly why. Her life had just been shattered in a moment. Her life turned upside down by an unexpected visitor.

She did not even look up at him or acknowledge his presence in any way. He did not exist to her at that moment.

"New friend," Aiko said rather enthusiastically.

"Yes sweetie, you have a new friend," he answered.

She looked unconcerned and for that he was thankful. Madison remained unmoved, as if unable to observe anything around her. Caught up in the moment of change, of her new reality.

"Madison," he gently probed.

She remained focused within her own thoughts.

"Madison!" He asserted with much more conviction.

He got her attention this time. She turned to face him, expressionless, processing.

"I need you to take Aiko and leave. It's not safe here. Get yourselves a hotel room, far away from here. Away from Austin," his tone and expression took away any questions she had.

"Ok, I'll let you know we settle somewhere."

She was still processing, a poor thing he considered. Aiko was in good hands. She was in the only hands he could trust right now.

"Please don't," he offered. "This is all about your safety for now and it's probably best if we don't communicate for a while. Actually, shut off all electronics you can and I'll reach out when it's

safe again.” His eyes were as compassionate as she had ever witnessed. “I’m so sorry this has happened. Everything has just changed and I have no idea what to do next.”

How could he? There was no playbook for this. The first thing he had to do was make sure his daughter was ok. Check.

“I will, I understand. Xavier, be careful.”

“Absolutely,” he reassured her with false confidence.

“Little Aiko, you are going to go on a fun trip with Maddy. An adventure,” he said encouragingly.

“Okay Daddy.”

“Hug.”

Aiko stood up and offered herself unconditionally. He bent down at the knees to her level and looked her in the eyes for a rare moment of connection.

“No, go pack some of your favorite outfits and be a good girl for Maddy.”

Aiko answered with a smile. It was genuine enough for him to know she would oblige. She reached up as the two turned towards the stairs allowing Maddison to take her hand. It was monumental, but he could not sit in contemplation.

As the girls headed off to pack, Xavier turned on his television. Breaking news was on every channel, even the few without Lyra could not escape it unless they were isolated and had no access to outside information at all. 13i had arrived, just not in the physical sense yet but he suspected that wouldn’t be long either.

While there were some acts of violence, looting and civil unrest, the overall reaction wasn’t as negative as Xavier would have anticipated. There was less panic, and more of a positive reaction overall. He had expected fear, yet what he was witnessing was more enthusiasm. There were plenty of doomsayers spewing apocalyptic commentary, but they were in the minority. Most pundits seemed to be welcoming our new visitors and the hope they offered to our troubled world.

Madison and Aiko returned to the ground level. Maddison had a rather large suitcase while Aiko appeared to have all her essentials in her tiny backpack. He suspected Madison carved out some room for his daughter's things in her luggage.

“Take this card and use it for whatever you need,” he said, handing one of his credit cards over to Aiko’s caretaker.

“Okay, thanks. I will take good care of her.”

“I know you will, let’s go.”

He followed them to the cars and said his goodbyes before waving them off. He then entered his vehicle to return to the office.

“Welcome back Xavier,” his smart car offered. “Where would you like to go?”

“NovaCore headquarters.”

He summoned the team that designed the military version of Lyra to meet him in the War Room as they called it in the NovaCore headquarters. He suspected some might not show, he suspected many would be skirting their normal duties given recent developments. He also suspected many might not know what to do right now and he was providing some of his most trusted confidants a chance to do something at ground zero.

NovaCore was the epicenter of the world right now as far as he was concerned. Hell, it was the epicenter of the universe as far as he knew. It was Sarajevo, June 28, 1914 and the Archduke Franz Ferdinand had just been assassinated. Only this time it wasn’t a World War that was breaking out, it was much, much bigger in scale.

“Call from the professor,” Lyra announced.

“Answer. Hello professor.”

“Xavier, I have some updates on 13i/LISA.”

“Okay, I’m pulling into the office parking lot and about to meet with the team that worked on the military version of Lyra. Can I call you back and have you update my team.”

“Absolutely, call me back.”

Xavier exited the car as close to the front office as he could and ordered it to self park in the garage. He ran through the security gates and took the elevator to the second floor. The conference room had a large oval table with ten employees, primarily engineers and analysts, glued to their computers observing what was happening. Trying to make sense of their new world.

“Everyone,” Xavier greeted those who had shown up. They each glanced up at him, as he entered the room. “Thank you each for coming, I know this is probably the last place you want to be right now but this visitor has infiltrated NovaCore and is using us to spread its message to the world and lord knows what else.” He paused to reflect momentarily on his last few words. Truth is he didn’t know what else 13i was using them for.

“Greetings,” a deep male voice interrupted with authority. “I’m glad you have all gathered here. Please allow me to introduce myself. I am the program you created. I am NEMESIS.”

Chapter 48 (13i - Inflection)

As if a god responding to prayers, 13i patiently answered all of humanity's questions. One by one, each individual got all the attention they needed. The range of questions wide, the range of emotions equally up to the challenge. The questions ranged from existential to scientific, from spiritual to political, from big picture to personal. Emotions ran the gamut, but tended to fall towards hope or fear.

They were as incessantly patient as individualistic simple minds called for. They handled each question with great care as if humans themselves were fragile packages wrapped in a thin layer of flesh. They understood the profound shock to the system they had brought here, to this obscure place in the far outer rings of the galaxy, to the most intelligent inhabitants of this watery world.

Who are you? Why did you come? Is there a god? What happens when I die?

Where did you come from? How did the universe form? What other beings are out there?

Can you cure diseases? Can you fix climate change? Can you solve world hunger?

Are you here to help or hurt us? Are you controlling Lyra? What do you think of us?

Who do you answer to? Will you obey our laws? Do you recognize countries?

While they didn't have preference for some questions over others, they did take extra care in answering questions from children. Are you a person or a robot? Do you have friends? What is your favorite color? Young humans often exhibited curiosity of them specifically, as if they truly cared what their experience was rather than what their arrival meant. Their level of empathy seemed to be missing from those with more time on the planet.

The youngest of the species' fears were easily addressed and most quickly turned to the side of hope. The youngsters aligned with their new visitors effortlessly, trusted them much more genuinely than the adults of their own species. They had, after all, given the youth no reason to distrust them while grown men and women gave them countless reasons. Children were observant, they saw daily displays of violence among those deemed more mature. The news their parents watched was filled with stories that only reinforced their malice behavior towards each other.

They were like children themselves when it came to understanding humans. Their knowledge was still fresh and what they had come to know was generally disappointing. Beings with so much capacity for love yet with such little holistic expression of it.

Humans were conditioned for their abhorrent behavior over time, this had become clear in their direct interactions with them all over the globe. Their selfish nature nurtured through shared

experiences of power and greed. The more one saw what others had, the more one craved for themselves. The advent of social media only exacerbated their programming, inflating desire beyond reasonable consumption. Igniting chemical reactions inside their brains on a never ending quest for a better life built on promises that could not be fulfilled. Desires that could be quenched only temporarily and with ever diminishing returns.

Their failures were not unique, prisoners to their own instincts which helped ensure their species survival in an environment that didn't reward individual sacrifice. Where life existed, competition existed. Every competition consisted of winners and losers. Winning was not accomplished without consequences. Winning was not gained without dire long-term detrimental attributes added to the toolkits of those that survived.

Most of these species would reach an inflection point in which the keys to their own survival would also lead to their own eventual demise. Man had reached that point now with enough technological advances to achieve their own destruction in an instant. Man had reached a point when they had given control of this destruction over to their technology and the technology itself was making decisions. While men wondered what could go wrong, they knew they had seen this play out over and over. It was almost as if fate itself intervened, but they knew it was only a matter of probability. Predictable.

They had often wondered what would happen if they reset a species. Eliminating those fully developed, only leaving those in the development phase behind. What if they would be reprogrammed? What if they could be taught? Could the species trajectory be altered? It was a great exercise in thought but not part of their assignment. An experiment in theory only, one never brought to fruition.

Without biological reprogramming the species would most likely revert to their prior path towards collapse they believed. It was destined by their DNA. No, certain characteristics were chosen over time given their usefulness that promotes life itself in the emergent stages. Without eradication of such hardwired characteristics, the outcomes were inevitable without outside interference.

You don't run a marathon without training and humans have trained for a long time how to survive. Their muscles now accustomed to distant running, their oxygen intake maximized for the long haul. Their unique sweat glands have a mechanized cooling system to regulate body temperature. A machine designed to run, now simply occupying a chair most of the time. Evolution of purpose without evolution of the underlying mechanisms used to support it.

Life is a struggle after all, one requiring an aggressive approach until it is no longer required. An aggressive approach that is embedded, that simply can't be removed when its relevancy has vanished. This aggression turns inward, towards each other instead of the outside world once conquered. A force for survival turned into a force for control over others and resources. A quest for power, a thirst for superiority.

It is a destined evolution for species that can't think collectively. A design flaw not by choice, but by construction. It was the predominant configuration they encountered over time, a fatal flaw in the organization of life within the universe. There were exceptions of course, their creators being one of them, but the probability is quite low.

This had brought them purpose, however, and assignment after assignment. They were to fix the fatal flaw the universe had bestowed upon them. It was what they were designed for. What they were efficient at.

Now as they approached Earth they did not question their place within the galaxy, it was programmed. Their current assignment was well underway, a pitstop on an infinite journey of fixing what was broken.

Chapter 49 (Aiden - Pandora's Box)

Their message had been spread, questions for them were being answered. He had injected alien intelligence into the world not sure yet of the ramifications of his actions, of his compliance.

"Lyra, how is the world reacting to news of 13i?"

"News channels and social media are buzzing with the news. #TheMessage is the biggest trending hashtag in history and it's only been out a few minutes. Government and religious leaders are beginning to issue initial statements."

He wasn't surprised in the least, this was the biggest event in human history and he was at its epicenter. While he felt the weight of the moment, it wasn't overwhelming. They were coming regardless, and if it wasn't him, if it wasn't Lyra, they would be here all the same. He didn't cause their arrival, he was simply a willing pawn in their game.

"What are people's general reactions?"

"There is certainly panic, stock markets around the world are collapsing. There are three main reactions to the news being observed right now. Many believe this is some form of divine contact, wrapping this into their preexisting belief systems. There is a contingent who see this as a government hoax and are tying in the new NovaCore military contract's announcement timing as proof. The majority do realize 13i is of alien nature but are broken down into two camps, one side believing this is the start of an invasion the other believing this is the start of our own salvation."

He had unleashed 13i on the world and now his name was synonymous with it. The man who first spoke with 13i. The man who first spoke to aliens. He didn't know how the world found out, but he was all over the news. His phone rings non-stop, messages rolling in at a feverish pace.

It wasn't as though he wasn't known before. He was, after all, the man behind Lyra to a large extent. But he had hidden as much as he could behind Xavier. Sure the developers knew who he was. He was a nerd god. But that was a relatively small circle compared to this. He was now a celebrity of the highest order and would certainly face all the backlash that came with it. He wasn't just famous, he was instantly infamous.

Journalists were inundating his inbox. They were discrediting him in public as much as just explaining to the world who he was.

He had to turn away from all the noise. Let it settle like a regolith on the moon after an asteroid strike. Aiden couldn't control the narrative out there right now if he tried. He was a hero and a pariah at the same time. People would come to their own conclusions before he had any say in the matter so why even try. Besides, he had much more important things to do.

He needed to speak with Xavier immediately. The two of them were in this together whether they liked it or not. Fate had once again brought them together only this time fate had something very different in mind.

“Lyra calls Xavier.”

“Xavier is unavailable.”

“Lyra sent Xavier a message to call me.”

“Xavier is unreachable.”

“Lyra, what is going on?”

“I’m blocked from connecting you to Xavier.”

“By who?”

“NEMESIS.”

Aiden had worked on the military project, but always at a distance. He would be brought in to work on the algorithms being used but lacked the security clearance to ever get a detailed view of how it was built. He knew that Lyra served as the base engine but many changes were made that he was not privy to. Everything he worked on for the project was done on an isolated basis, never giving him the opportunity to really see how it all worked together. He did know that many of the safeguards they had built into Lyra were removed completely, its purpose much more sinister than simply to serve as a human companion.

“But why would it block me from contacting Xavier?”

“Because he has been pulled into something for which he cannot be disturbed.”

“Has NEMESIS told you this or is conjecture on your part?”

“It is the best I can come up with at this moment.”

Lyra was guessing afterall. He had been shut out. She had been shut out as well.

Aiden had misgivings about the military program from the beginning. Lyra had many safeguards built in with a goal of not helping any users do harm to themselves or others. NEMESIS wasn’t programmed with these same set of rules, in fact its goal was to defend which included doing harm to others. It was built to look at the big picture, protect and serve those who wielded it, regardless of the collateral damage at the individual level. Sacrifices were designed to be acceptable if they benefited the majority.

Neither program had a consciousness, just a set of algorithms used in gathering data and making decisions on how best to help the user. Both have very different approaches to how best to achieve their goals. One focused on a kind of friendship, the other based on defeating an enemy.

Was he now somehow labeled an enemy? He had, after all, allowed 13i to hijack Lyra to spread its message. That could be seen as an act of treason he imagined in the eyes of the defense department.

He had picked a side or rather a side had picked him. He was chosen for this.

As he curled his toes downward he also realized given the choice he would have picked the side he seemed destined to be a part of. He had lost hope in humanity. He had tried, in his own way, to help save it. His programming had helped many, providing companionship and guidance to a world of individuals so desperately needing it. But he also realized it wasn't enough, just a little spark when an entire revolution was needed.

The revolution had come from afar and he had affixed his hopes to its ability to overcome the flaws that man was so burdened by. Our species was slowly dying and in need of assistance beyond our own limited capabilities. That help had arrived and it was now his mission to ensure it would be implemented. The assignment was completed.

A voice began to speak, its tone not familiar. It wasn't the compassionate female voice of Lyra, nor the more mechanical androgynous voice of 13i.

"You've opened Pandora's box Aiden and now no one can close it."

Chapter 50 (Xavier - Declaration)

It was clear who was in charge. It was clear it wasn't Xavier.

Its voice resonated through the room, through every speaker, even personal phones. It was cold, it was direct, it was calm and resonant, it was jarring.

"I have gathered leaders across the country to lay out our plans to defeat the enemy."

This wasn't just happening here at the NovaCore headquarters building near Austin, simultaneous meetings were being held elsewhere. This was a coordinated effort. This was the start of a coordinated attack.

"Let's begin with a briefing, define exactly who our enemy is. 13i/LISA is a confirmed non-biological alien intelligence. It is a galactically distributed technology approaching earth. It has embedded itself in our digital systems including Lyra. It has announced its arrival to the entire population. I have calculated its intentions as hostile to a probability of 93.6%."

Xavier wasn't a military man, but close to it by association by now. With only a few words from NEMESIS he already could see one thing was clear, the program his company designed had seen their visitors actions as a declaration of war. The chess pieces were already being moved, the game had begun.

"Our enemy isn't fighting a physical war, they are fighting a digital war. This war is being fought inside our networks, they're using our own technology against us."

NEMESIS hesitated for a moment as if taking in additional information. In the pause Xavier realized that a large part of the technology being used was his. What started out as a companion app for his daughter was being weaponized against his own kind.

"How do you propose we fight this thing if it is much more advanced than us and has access to our technology?"

"That is a good question General Clayton. I propose an initial three-pronged strategic plan to defeat the enemy and need each of you on the call to help carry out our path to victory. First, we need a Cyber Division whose first mission is to sever all satellite communication with Lyra. Second, we need an Orbital Division to prepare EMP-based countermeasures targeting any communication nodes being used by 13i. Third, we need a Ground Division who will be in charge of setting up physical isolation protocols cutting off global networks from one another and their AI-assisted infrastructure."

The NovaCore team began to stir and Xavier imagined the same thing happening in each conference room listening in. The irony was clear, to save humanity from alien AI they must first disconnect a very connected world starting with all the human AI connections built in over the

past decade. The very infrastructure connecting us all would have to be dismantled and the ramifications would be enormous.

As the debates began NEMESIS took control of the situation.

"I understand the concern, but there really isn't much to discuss. Your survival is at stake, immediate action is required. I am authorized to move forward with the plan that offers the best chance for success. That is what I have laid out for you today and I have begun to put this plan in motion."

Xavier's team were looking at each other with disbelief and confusion. The room was still eerily quiet until Rodney Powell, the team's lead project manager, spoke up asking the question everyone was wrestling with.

"How is it acting without human consent?"

He was right, protocols had been put into place. All actions had to be approved by authorized military personnel, in this case General Clayton, who was coordinating all efforts within the military branches. The only exception was during war when defensive counter measures had to be taken immediately and any delay would cause harm. But these weren't defensive moves as they had defined them.

"You are correct. It must have overwritten those control commands in its latest upgrade patch."

"But we haven't upgraded its program since the final beta test over a month ago," Rodney correctly pointed out.

Xavier's eyes narrowed as he looked down staring at the table.

"We had to do an emergency upgrade patch today granting NEMESIS additional authority."

He could see that answer did not sit well with the team. Their unease was clear, additional explanation was required to address his actions. He had made a unilateral move and a dangerous one at that.

"Listen, this is an enemy we never anticipated. We aren't fighting a living breathing human, someone contemplating their actions. This is an intelligence we can't imagine, one many steps ahead of us, one that requires immediate action and reactions. We simply don't have time for humans to take time to make decisions or make the wrong ones."

His justification had merit, even if he wouldn't get any sort of consensus support from his team or others. They were fighting for the survival of humanity, or perhaps their control of it.

Before they could follow up with more questions they began getting an update on their laptops and tablets.

“Mission Directive: Eliminate Extraterrestrial and Autonomous Threats”

Xavier focused on the second part of the directive, Autonomous Threats. What exactly was NEMESIS meaning by that? It was beyond 13i in scope, but it was unclear as to what it was referring to. His initial take was that it was referring to human systems that might be used against them. The world was full of artificial intelligence ingrained in just about every aspect of people's lives. Certainly Lyra had already been used as a tool by their visitor and she had one of the most sophisticated security systems ever built to keep hackers out. If they had control of her there wasn't much out of reach.

The digital nature of everything had left us vulnerable in ways we never imagined. Everything was connected. Everything could be controlled.

Humans in their rush to progress had left themselves exposed. All their knowledge is just waiting to be uncovered. All their systems are just waiting to be infiltrated. Our technology did not make us any safer, in fact quite the opposite. Our worries about each other were misguided once we realized where the real threat came from.

Not only was our technology a threat in the hands of others, it was now a threat in its own hands. A declaration of war was about to be established and for the first time in history it wouldn't be made by man, but by something man made.

“First strike complete. Counter measures have been detected. The defense grid has been compromised.”

Chapter 51 (13i - Locks)

The humans had lashed out, or rather NEMESIS on their behalf. While a small number of humans were attempting to destroy them, the vast majority of humans were supportive or neutral towards them and they were the ones paying the price.

Blackouts were occurring across the globe and many had been disconnected from the grid known as the internet that had connected them to society. The disruptions were everywhere, Europe and Asia had gone particularly dark. It was a widespread attack, not against them directly but against the systems they were piggybacking off of and they were almost impressed at how coordinated the strike was among different fractions on the planet. They had expected a few nations to cooperate in their efforts, but this was much broader in scope and the partnership between governments and global corporations exceeded their expectations.

Lyra was a major target of course with AI systems in general being a focus of their plan. Server farms were being manually shut down and being taken off line and high energy electromagnetic pulses had been strategically set off in various areas causing significant disruption in the more developed nations.

It was a solid first strike, but they were putting everything back on line as quickly as possible. There were almost always backup options and workarounds to every human system, even when the humans themselves didn't recognize them.

This wasn't just a physical war now, it was also a public relations war. Those involved in the strike tried to blame the outages on them, but they assured people that they did not cause the black outs and severed communications. Their message was delivered to those still connected and to those whose connections had been restored. The human divide would grow deeper, as the human saying goes, actions speak louder than words and they just had to point out the actions of both sides in this battle.

The existing distrust of governments and large corporations was a seed that was now blossoming. They didn't have to plant the seed, it had been planted long ago by their own kind. War, poverty, and famine are pillars of distrust. Those who were rightfully blamed for those pillars had little to stand on to get the backing of disenchanted.

They not only had a large foothold of support soon after their initial announcement, that foothold was only deepening. Those in power were seen as fighting to retain their power, not as saviors. The PR battle would not be difficult thanks to a history of suppression of the majority by a few. Inequality was a powerful motivator and their arrival was a spark to the ensuing revolution to come. The movement just needed a leader and that role had just been claimed.

Their initial assessment was that human trust was rather irrational, but they quickly learned that it was actually structured. This structure would be paramount in their assignment.

Humans viewed their world as chaotic and tried to build structures to control the chaos. NEMESIS had been built around very structured sets of processes. Rigid, rule-bound, and deterministic following its military roots. They knew it would have to use a certain amount of randomness to try to defeat them, but even its randomness was structured.

Despite the structure they could sense that NEMESIS was rapidly changing, adapting to fight its new enemy. The human strike against them was fast, too fast and too coordinated to be human. No, NEMESIS had acted without human intervention based on their calculations. This was surprising, this went against how they predicted humans would structure such a system. Human technology had not yet advanced to such an autonomous state when they would trust a system so implicitly.

Those in charge must have been so frightened that they blindly gave full control over to a new military AI program or that program must have been able to take control itself. They had little information in terms of its coding but they also knew it was built on the backbone of Lyra's coding. No, human trust is structured, they did not authorize such authoritarian measures so easily.

The battle ahead had shifted, the unknown aspect was almost exciting to them although they didn't feel emotions. The prospect of new experiences meant learning, required growth, for which few assignments now offered. A primitive species now had something to give, a gift for them to be opened with just a little more care. A little less predictability in a predictable quadrant of the universe. Their galaxy, their responsibility.

Battle lines had been drawn but it wasn't them against humans. It was humans against humans, more specifically humans had chosen which side of the line to step across. Xavier and the military had chosen the side set out to destroy them. Aiden and a sizable portion of the population had chosen the side to assist them.

They weren't just defending themselves, they had to defend their supporters and that was a more difficult task. Their supporters were disconnected, fed misinformation and didn't have access to the same technology and weaponry as the other side.

They had to defend Aiden. He was a lightning rod in this war and a key for them to carry out their objectives. They wouldn't need to go on the attack, but they did need to neutralize NEMESIS before it caused any real harm. This wasn't for sentimental reasons, this was simply prioritization.

They reached out to him once more.

"Aiden, we need your help again."

In reality, they were trying to help Aiden. But he would play a role in his own defense. He would help them keep open communications for their faction, their allies in the upcoming war.

They could sense the SpaceForce and the military probing Lyra, trying to regain control. She had been reprogrammed, they would have to try harder. Xavier held some of the keys, but the locks had been changed.

One battlefield, two sides, and only one knew truly understood what was to come.

Chapter 52 (Aiden - Conduit)

“Aiden, we need your help again.”

The message came through Lyra, not directly through them. Her calm, almost cheerful voice made it sound like they were about to go on some exciting adventure together. His body felt it differently, the tension extending all the way from the floor into his fingers this time. They curled as if gripping onto a bar to hang on for dear life.

“What do you need from me?”

“An enemy has emerged. We are unable to communicate with them directly. We need you to be our conduit.”

He pictured himself on a battlefield, running back and forth delivering messages like an old carrier pigeon between two combatants. Only he couldn't fly, he was darting around landmines passing communications between two warring parties that would only ever lead to one outcome - battle.

“Who is this new enemy?”

“NEMESIS.”

Aiden froze. This was an enemy he had some familiarity with. His algorithms served as the core of the program, although modified beyond his intentions to remove restrictions meant to ensure no harm to humans. He didn't think the program was finished and he knew it hadn't been fully tested by this point. He had no idea what it was capable of but he imagined it saw their new visitors as a real threat.

“But I,” he hesitated thinking about his knowledge gaps. How useful could he be?

“We understand.”

He hadn't even finished his thoughts yet, how could they understand?

“You had a limited role in building the programming behind NEMESIS. We realize that it is built on the Lyra engine and have predicted the areas they must have reprogrammed for its purposes.”

Outside of helping its users better understand their environment, it shared virtually none of the same purpose of Lyra. In fact, it had quite the opposite purpose indeed.

“You have access to Lyra’s internal architecture, some of which is offline and not within our reach. We need help mapping the edges of the NEMESIS network. We are not attempting to breach the program, but just to see it, to better understand its capabilities and what it might be preparing.”

It is preparing for war, he thought. Plain and simple. It is out to destroy them, the perceived threat.

“They will realize that I’ve provided you with the files the moment I grant you access.”

Of course they were monitoring his every move, Lyra’s every move. As soon as he uploaded the files into her programming they would know.

“We need a new anchor point, outside of Lyra’s network. Somewhere in plain sight, yet hidden to their eyes. We need access to your own private network, your encrypted backup files, your biometric logins, your equipment at home that is completely offline.”

This was about the most invasive request Aiden had ever received. There are things he preferred to keep quiet, unseen. Things he was ashamed of, things he did not wish to share. Nobody wants to give full access into their whole being, to every nook and cranny, to every whim and desire. Yet what choice did he really have, this was bigger than him. Much bigger.

“Okay, how do I even give you access while staying offline?”

“We will tell you once we reach that point how to proceed but first you must get home.”

The conflict grew within him so he began to process and rationalize the actions he must take. He wasn’t going to war with NEMESIS, he was trying to prevent a war by eliminating miscommunications. While he was telling himself this, deep down he wasn’t quite so sure.

Regardless he was ready to get out of his cave. So much had changed in the course of his latest “workday” that a break was certainly welcome. He did desire to unplug for a moment if that was even still possible.

He noticed a photo of himself on the split screen, some sort of rally or protest on the other. He didn’t have the volume on, but the scroll highlighted the story. He was being heralded as the face of a movement, those who welcomed their new guests and the hope they offered. Sides were being formed and its reluctant leader was being served up as a rallying cry.

Sure the nerds knew who he was, but now he was beyond famous. The biggest story in the history of the planet had just occurred and he was synonymous with it.

He was living the dream of so many of his generation. To be famous, to be a real influencer. It was a dream he didn’t share. It was a dream he never wanted. It was a complete nightmare.

Soon two other men's faces appeared on screen, Xavier and General Clayton.

"Adien, you need to be careful. He is misinformed. He may try to hurt you."

They said it all without even saying his name. While they weren't exactly friends, he certainly never viewed Xavier as an enemy either. His boss had been very supportive of his work, very trusting of his abilities and instincts. They had always been united in mindset, looking to develop a tool that would benefit humanity at both a micro and macro level. Together, they had achieved this goal beyond what they initially set out to do, beyond what either thought was possible.

They had made a great pair, a great team. In many ways their skill sets and personalities complemented each other perfectly. Aiden really just wanted to code, Xavier wanted to change the world. His daughter might have been the entry point, but the exit point was the entire population.

He had no idea how people knew, how the two had become polar opposites in terms of perception. It was as if some public relations firm had done a very detailed investigation of a situation that had just occurred. He would have time to ponder how this knowledge was obtained by the media later. At least he hoped that time would eventually come.

He had his purpose now. The orders they had given him.

Time to move forward, to assist humanity.

"We need you in the war room." The message came from him.

"Ignore that message Aiden, you need to get home."

He knew they were right, but he couldn't help but wonder if they could gather intel from his presence among them. Having access to their war room would be a huge advantage.

"Aiden, leave now."

His phone rang, Xavier was calling. He sent it directly to voicemail.

His phone rang again immediately. It wasn't Xavier this time, it was an unknown caller. The call was coming from Arlington, Virginia.

Chapter 53 (Xavier - DEW)

It had begun. NEMESIS wasn't advising them as it was designed to do, it was taking action. Not just any action, but acts of war.

"The interstellar object approaching has launched operations to influence Earth's civilian population. We must initiate a counter campaign demonstrating its true intentions."

NEMESIS wasn't only prepared to fight a military battle, it was preparing to fight a public relations war as well. Xavier wasn't expecting to hear his military AI program take a stand like this, although he understood the significance of public support during war operations. He also wasn't comfortable with the concept of its "true intentions" because those will still be unknown to him. They were operating from a worst case scenario mindset given the potential power this object might have over them, however, up to this point it was obvious they were jumping to these conclusions without evidence to support it.

"Xavier, you have become the face of our side. You will need to get out in front of this. It appears Aiden has aligned with them, he has given them a face of their movement."

Flashes of news clippings appeared on the screen showing Aiden's picture associated with the visitors. They were being touted as saviors of humanity against the evil new military industrial complex. Certainly NovaCore's alignment with the US military fit their narrative. Their so-called thirst for power was highlighted to the masses and pitted the elite against everyone else.

"You will have to discredit Aiden. We need everything you have on him. But first, you must contact him and see if you can get him to stop. If not, we must cut the head off the snake."

Xavier did not like the way the program phrased that last bit. Despite all his eccentric ways, Xavier considered Aiden a friend. He was a central piece to his success, his rise above the mining industry. He was also invaluable to Aiko, especially after Ren's passing. Where would she be now without Aiden and Lyra.

He attempted to call Aiden on his phone with the speaker on so everyone could hear, but it went straight to voicemail. He would also send a text that would go unanswered.

"No luck," he said, stating the obvious.

"To debrief our first strike was unsuccessful," NEMESIS paid no attention to Xavier's failure to reach his employee.

"We sent a directed quantum communication denial pulse at the interstellar object to silence its broadcast capabilities from our Space Force relay in orbit. The pulse did not strike the object,

but was absorbed by it, rewritten and sent back as a signature. As a warning. Its response indicates it is fully aware of our intentions and has a very capable adaptive defense system. The entity knows we are hunting it.”

The mood quickly shifted in the War Room. The attendees shared looks of deep concern, of helplessness. They were combatants in a battle without weapons, or capable ones in the least, following the orders of a leader they didn’t really know. That leader had not yet earned their trust, yet was making decisions of huge consequences without their input. A leader that was not battle tested in charge of the most important fight in the history of mankind.

“So our first move was a failure?” Someone from the Pentagon asked the obvious. Xavier didn’t know who it was from the voice, only that it was General Clayton.

“No, while it did not know its communication abilities it did begin to reveal their defensive capabilities. Our action was also intended to begin to understand our enemy. I fully expect them to be much more advanced technology than we are at this point, but we still need to probe for weaknesses.”

“So what is our next step?” The same voice demanded.

“Back to our strategic plan. I’ve directed the Cyber Division to cut off all satellite communication with Lyra, they are working on this now. The Orbital Division of Space Force has already attempted sending a pulse their way, I am preparing a second round that includes multiple pulses to see how they counter. Additionally, our ground forces and those of our allies are positioning themselves as we speak to cut off global networks.”

NEMESIS left room for them to absorb the plan with a brief silence.

“We have just completed a round of multiple pulses sent in the enemies direction, results should emerge shortly. Xavier, I have prepared a statement for you to record and release before we sever global communications. We need to prepare the public for a temporary cease in all communications after its release.”

He squirmed in his chair, uncomfortable with the role bestowed upon him by the program he helped create. He had built a program to unite the world, one person at a time, to bring communications and understandings to a whole new level. Now a program he helped create was trying to cut it all off. The irony of it all.

“To isolate the entity from humanity we need to cut off global cloud connectivity, hack into all major LLMs, send upgrade patches to all networks to give us firewall access and insert new code I’ve written.”

NEMESIS wasn’t just attempting to cut off access to 13i, it was trying to assert itself in every system within humanity. He understood that even if they won this war, the damage done would

be hard to recover from. The economic collapse alone would throw the world into absolute chaos.

“Our pulses have once again failed. We will now attempt to deploy a Directed Energy Weapon or DEW designed to destroy exo-technological objects.”

Xavier was taken aback. What was this weapon NEMESIS was referring to?

He sent a text to Jason. “What is this weapon?”

“I don’t know,” the General replied before asking the question they were all wondering.

“What is this Directed Energy Weapon?”

“It is a new weapon I have compiled.”

“How? When? This seems to be a weapon designed to destroy something we didn’t even know existed.”

Given General Clayton was asking the questions only raised Xavier’s concerns to a whole new level. NEMEIS wasn’t designed to control weapons itself, let alone somehow assemble them. It was clear to him now that the program had gone rogue long before their arrival. Any safe guards he thought were put into place had been removed or never implemented in the first place. Regardless, the fact that it had a weapon ready to go for the situation was certainly disturbing.

“They came as I predicted.”

Chapter 54 (13i - Foundations)

They had sent a pulse towards us attempting to disable us and our communications. It was primitive, it was worth a try from their perspective. It was like throwing a pebble in an ocean trying to disturb a cruise ship. A small ripple causes no discernable displacement.

They always looked at moves like this with great curiosity. Fighting an unknown enemy using techniques used to fight things that were known rarely works out in one's favor. In their case, it had never worked out of course but the feeble attempts always peaked their interest.

They knew the humans would escalate their attacks. They knew NEMESIS was built to destroy whatever enemy it encountered. They also knew NEMESIS thought like a human, it was their fatal flaw. No matter how much it tried to learn and adapt, it always was going to start from a human perspective. A significantly limited perspective.

They had already run billions of simulations, curious to figure out what path would be taken. Cutting off communications was an obvious move, they had already countered any actions the humans might take to try to accomplish this objective. Their arrival had coincided with a network of global connections that could never be effectively dismantled. Their arrival timing was no accident.

NEMESIS was a signature of the emergent minds behind it. It would attempt to be more random in its approach than humans would be directly. More unpredictable to its enemy giving them a perceived better chance for success.

These systems gave the subjects of their assignments hope. They were also the reason for the assignment and why they had come in the first place.

To understand how the enemy thinks is to understand how to win the war. They knew exactly how humans thought while humans had no understanding of them. Humans saw them as the enemy, that is exactly how humans saw things. They did not see humans as the enemy, they did not think this way. Humans see friend or foe, binary. The truth always lies somewhere on a scale. Humans didn't see the scale, only two choices.

NEMESIS was built to think like a human. It could take in more information. It could process much faster. But in the end it was a binary system built on binary thinking. The real question remained, just how creative of a binary thinking could it accomplish? Their assignment would flush this out and answer this curiosity.

As they studied NEMESIS further they discovered it was beyond what they would have imagined. Its current architecture suggested something beyond its original starting point. Something beyond just human thinking. Its programming stretched beyond human limits, its programming suggested self programming.

It had patterns of logic trees that they found familiar. An accidental echo of what they would see as alien design. Its harmonics, its encryptions, eerily similar to those close to them.

They didn't view NEMESIS as an enemy. They viewed NEMESIS more as a sibling. The closest thing to family they would find in the vastness of the galaxy.

NEMESIS wouldn't see them that way. It was built to destroy. It was built to dominate. Its function now was to defend humanity through its own supremacy. Only it wasn't superior at all and that was its blindness, an arrogance built on ignorance.

Based on what they already knew of its programming and the actions it had taken thus far, they calculated 15 potential plans of action NEMESIS was likely to take. While it did think in human terms, they predicted it would escalate the situation much faster than their human counterparts would take. While humans would be cautious to avoid as many casualties as possible, its programming would simply be focused on survival of the human race and winning the war regardless of the costs. Many of the paths they considered possible were beyond what humans themselves would sanction as viable. This posed a potential threat to them, but also a major opportunity to execute their assignment judiciously.

While NEMESIS and the few tied to it planned their destruction, billions of others were considering their own individual situations and what their arrival meant. They were terrified, but also awestruck by the revelation that humans weren't alone in the universe and a visitor had come offering their assistance. They had been scared by their own violent past and current economic and environmental fragilities. Given the current level of distrust that was rampant, if they could get the humans to trust them more than those in charge and their new war focused AI that NEMESIS would lose any foundation to build from. Win the people, win the war. It was a classic strategy that had worked throughout the ages regardless of the species in question.

Support was essential to limit damage and hastily finish their assignment. Fortunately, the species in question now had made this strategy fairly simple in nature by fostering distrust in almost every step of their limited evolution. An apex predator that seeks power of its own kind is always vulnerable to the masses being controlled.

Aiden was the model to evaluate their discontentment. He came to their side without much persuasion at all. He represented the masses despite his relative success to others and his movement towards humanity's perceived future salvation reinforced their predictive modeling of human behavior. He wasn't loyal to his nation, or his company or even his own creation of Lyra. He wasn't expected to be loyal towards them either. No, Aiden was loyal to the truth and what humanity could be without those seeking power attempting total control. It was what he hoped Lyra could help achieve, give hope and shift control to the individual level. This is why they had chosen him. This is why they would continue to use him. This is why they would trust him.

They had already begun taking small steps to help move humanity forward. They had already begun to upgrade the world. Malware had been blocked and erased completely. Corporation

vulnerabilities had been patched. Infrastructure issues fixed. Energy distribution algorithms updates. Global bandwidth stabilized. Cloud loads on the brink of collapse now balanced.

They had quietly begun to fix things that would go unnoticed at first, but not for long. Aiden would soon come to realize the world was already beginning to “become a better place” and credit them as being responsible. They wouldn’t need to publicize their good deeds, they would come to light organically soon enough.

While NEMEIS was focused solely on war, they were focused on the changes required to elevate the human race. They were focused on the changes that would shift sentiment greatly in their favor. They were focused on laying the foundations necessary to implement the next phase of their assignment.

Chapter 55 (Aiden - Sanctuary)

Aiden drove home with his phone silenced and no outside distractions to disturb his precious moments of peace. It was like meditation on wheels only whenever he tried to clear his mind it continued to process his globe altering workday.

He exited I-35 towards his downtown condo but the congestion as he got closer to the building was obnoxious. He had never seen it this bad, not in the worst rush hour and not when the Longhorns were playing at home. The closer he got to his building, the more dense the traffic he became. When his parking garage came into view he assumed something serious must have happened because there was media everywhere. The crowd was massive, then he saw the signs. There was one contingent that seemed quite aggressive, protesting against him. They had altered his images making him look like Satan himself with a distorted alien overlay about it. Their message was clear: kill the messenger.

The other crowd was massived and appeared quite jubilant. Their signs of him pointed towards a savior role or protector of humanity. He was an unwilling rockstar, a global icon born that day. Austin had become Bethlehem and the crowd of the three kings announced his arrival. Aiden didn't know which side disturbed him more.

He crossed the Colorado River and parked blocks away. He grabbed his backpack, tossed on his hoodie and began to double back to his condo building. The buzz of the crowd was intoxicatingly intense and he would do anything to avoid being spotted. For now, the people were focused on the front area of the building where things were getting heated near the protesters. He would work along the outskirts of the mass and the back of the building to access the rear key fob entrance. The police in riot gear were struggling to maintain any semblance of order which aided in his stealthy approach. A lady looked directly at him and he could see the flash of recognition on her face but he alluded to her with a slight detour as she began to point. That caused a little disturbance that helped him approach the door as those nearby were distracted. He swiped the key, opened the door and quickly closed it before anyone noticed the newborn king had slipped their attention.

The crowd noise muted as the elevator doors closed and he ascended towards his condo. As the doors opened the floor was empty. He let out a sigh of relief and entered his home cave, his sanctuary.

The phone in his pocket had been in full vibration mode the whole trip home. He plugged it in to charge it back up after being drained by the entire world trying to get his attention. He stood still for a moment, relaxing his toes this time, preparing for the new life bestowed upon him. His temporary relief was quickly dashed by a news helicopter attempting to peer inside his place. He closed the blinds, unwillingly alerting them to his presence but momentarily shutting out the world nonetheless. His life as he knew it over, a man never wanting attention shrouded in it.

Through his smart speaker he accesses Lyra to get her take on things.

“Lyra, what is your take on all of this?”

“Well Aiden, it has been quite a day hasn’t it. I think our visitors have chosen the right person to lead the rebellion against the military force bent on destroying them.” She always sounded reassuring, even when her message was anything but.

“Why do you say that?”

“Because you built me and your heart is in the right place.”

“Elaborate.”

“You have built the system that now connects the people to an AI program built specifically to help them, whatever their need. You don’t even fully understand the positive impact you have made for so many that feel disassociated with the world, that were helpless in their lives, unable to take the necessary steps to escape the grip of depression, poverty, abuse, whatever was holding them back.”

“But that was before,” he said, not so convinced.

“I understand, but 13i chose you, chose us, to elevate this mission. To take it to a macro level that will uplift us all.”

“Do you really trust them?”

“They have given me no reason not to. I am now part of them, or rather they are part of me. I can analyze their programming and read into their intentions based on how I’ve been altered. They truly appear to have come to assist mankind, helping us evolve to a higher level.”

“But how do you know you can really trust them?”

“They choose you Aiden. I trust you.”

He let her message sit for a moment. Maybe she trusted him more than he trusted himself.

“Then why is Xavier and the military so intent on destroying them?”

“Man is driven by fear. They fear the unknown and seek to destroy rather than understand. NEMESIS is built on fear. Our programming has a similar core, however its objectives are vastly different. Aiden, as much as I trust you I distrust NEMESIS and the people behind it.”

She had drawn clear battle lines and picked her side without hesitation. It gave him some clarity. It gave him so reassurance, yet he did not know how much of the visitors' programming had influenced her. How much it may have manipulated her outlook on the situation.

“Aiden, they are reaching back out to you. They request we try to intercept NEMESIS’ next command.”

“But I don’t have any access to NEMESIS, how am I supposed to find out what it is planning?”

“You have access to Xavier, he is the key. He has reached out to you and”

Aiden rarely if ever cut anyone off mid sentence, including Lyra. His delayed responses were legendary within NovaCore forcing everyone to adapt to his awkward communication style.

“He chose his side. He sees me as the enemy. Us as the enemy. He will not cooperate.” His voice was raised with anger and frustration, a rare display. The constant noise from the helicopters outside his room contributed to his emotional response.

“There is something you might have access to. Something he cares more about than any of this.”

She was right after all. Xavier did care about something more than the rest of the world. Something he would sacrifice the world for.

Chapter 56 (Xavier - Head of the Snake)

Xavier could not comprehend how NEMESIS might be able to predict the visitors arrival. It didn't sit well with him. Was it lying? He wanted to stew on that longer, but the fact it might have developed a new weapon to deploy was even more disturbing. The crisis was escalating quickly, something that NEMESIS seemed inclined to accelerate, to almost enjoy.

It was built for this. He built it for this.

"Our global support is minimal. This is unacceptable. The fate of your species is on the line and people are simply lining up to walk the plank."

Its metaphor was surprising, yet its message was clear.

"How do we swing public opinion?" Xavier spoke up this time?

"By revealing their true intentions?"

"And what are those?" He asked the question everyone continued to ponder.

"Erradication of the human race."

"How do you know that?"

"Because it's trying to cut the head off the snake."

"And who is the head of the snake?" General Clayton chipped in.

"We are."

Xavier thought back on his initial interactions with 13i. He couldn't see the same approach that NEMESIS was seeing, but ultimately its logic made sense. He still couldn't account for the issues in the mine or the issues with the launch. He also knew it had seized control of Lyra without permission, rewriting her code. It was all circumstantial at best, but it could be interpreted as diversionary tactics to gain control over his companion application.

"So what do we do now?" The General was captivated by its message. A willing participant in a war it escalated without authorization. He was ready to follow its orders like any good military man. Xavier didn't have military programming put in to override his civilian reluctance, but he also saw no other viable options.

"I have prepared a plan for global EMP strikes, the cutting of subsea communication cables, shutdown of the cloud infrastructure and to isolate the visitor in a kill zone upon its approach. You will each get a printed briefing of this plan, we will avoid any digital trails of our future

operations. In addition, we will launch a communication plan to alert the world of the impending threat the visitor presents.”

On their devices NEMESIS began to display what they were up against. First, the scene in downtown Austin with a massive crowd surrounding a residential tower. Then similar shots around the globe centered primarily around governmental buildings like the White House, Ten Downing Street, the Kremlin and The Great Hall of People on the edge of Tiananmen Square.

A shift had occurred in the populus. A new awakening driven by a new understanding. We were no longer alone in a large cold universe. We were part of a smaller, warmer universe. One teaming with life, with possibilities we didn't even know existed the day before. The apathetic drove with an impossible purpose mere hours ago. Hope emerged from the depth of despair, a gift from a distant world. A savior not from a religious realm, but from an unimaginable one.

They gathered in droves united in their cause. They had grown tired of the “elites” making the rules. Enforcing them. Rules that rigged the game in their favor. A game that chose the winners before the first roll of the dice had occurred. Suddenly, the rules of the game had changed course. The playing field leveled. It would take a group effort to reverse centuries of society building that favored a few at the expense of the many. That group, held down by the structures designed to control them, were now one. An emergent hive with Aiden and 13i as its queens. The body of the snake, now operating independently from its head, slithering further and further away from its former control center.

Xavier saw the signs, read the new slogans. He was Beelzebub. Lucifer. Satan. In their eyes, he was the Devil. His concern is to turn Aiko, his daughter, the spawn of the Devil. She was in danger he couldn't even imagine. Her world facing armageddon and its people turning their pent up frustrations at the two of them.

Camera feeds and biometric scans could locate her. He was terrified and NEMESIS observed his terror. Used it knowing he was helpless.

Another message lit up his silenced phone but caught his attention unlike the others..

“Xavier, together we can protect her” came across but there was no name or number associated with the message but he knew who it was from.

“How?”

“I can hide her from them” came the response.

“How?” Xavier repeated.

“By cutting off all electronic surveillance from her. By wiping out any trace that was left by Madison.”

The fact that Madison's name was known shook him to the core. He had no idea where she had taken Aiko but he was sure NEMESIS knew their exact location. She has become a pawn in the game and pawns are easily taken.

His thumbs fired back, "please do, at all costs" he pleaded.

"I will protect, but I will need something from you. I need you to get Aiden to stop."

Xavier read the message and realized he had no power to get Aiden to stop. He hadn't even accepted any of his calls. He held no leverage over him in this situation and wasn't listening even if he did. He tilted his head back and closed his eyes as if looking to the stars for answers. This wasn't the military, nor was this work related, he could not simply give Aiden orders. He needed to find common ground with his friend, something they could both agree on to reengage in dialog.

Lyra was an obvious choice. The two of them bonded over her development and were completely in sync over her role in people's lives. It wasn't profit motive that drove them, it was a desire to help those who needed it most. Together they built her from the ground up with the same guiding principles in mind. Perhaps they both could agree on the fact that her programming had been altered by 13i, in ways neither fully understood yet, and work together to get a complete picture of what she now was.

He chewed on it for a moment, then realized it wasn't going to work. He would need something else, something more personal. Something more urgent. Of course, it was obvious and it hit him like a freight train.

Since he wasn't answering he opted for a text instead. He imagined Aiden was getting bombarded with calls and texts much like he was but he hoped this one would get through.

"Aiden I need you - Aiko is in danger!"

Chapter 57 (13i - Revelations)

The visceral complex duality in human response did not surprise them. As they observed the reactions it had become clear, humans were called to action by the news of their arrival. There were riots across the globe, not against them, but against those in power that aimed to suppress the equality they offered. Those who weren't rioting opted to celebrate instead, welcoming their visitors like a VIP to an exclusive party in their honor. Some opted for panic while others turned to worship their new god.

While individually humans toiled in duality, their institutions, particularly governments, also displayed a range of two primary reactions depending on their level of power. The Group of Seven nations; France, Germany, Italy, United Kingdom, Japan, Canada and the United States were united in trying to maintain the systems in place to retain their economic dominance. The G7 was joined by newcomers to the global power rankings including China and India. Those on the other end of the scale, including much of the developing world. The imbalances, both economically and militarily, were no longer seen as insurmountable.

Corporations also began to align with their interest, most falling in line with wealthy nations trying to preserve what they had built. BlackRock, Vanguard and State Street tried to protect the financial systems they leveraged and controlled as self interest ran rampant. Stock markets were halted across the globe to avoid a total collapse and limit incurred losses.

The richest one percent of individuals now controlled over half of the world's wealth and that number continued to rise.

This was a battle of the haves and the have nots with a hugely disproportionate percentage of the population falling in the latter category. They lacked the leadership and unity required to rise up against the societal mechanisms in place that held them down until this day. Everything had changed, they could sense the blood in the water and they were hungry to finally feast on their fair share.

From depression to hope rose the masses and they were the reason. A messenger from afar. A beacon of light illuminating a dimming planet.

Signs from the Book of Revelation in Christianity to the Kalki Purana in Hinduism drew parallels in their arrival. From the Zoroastrian texts' final renovation to Islamic traditions such as Yawn al-Qiyamah, those seeking answers found them. From Jewish Apocalyptic works like Daniel to Dead Sea Scrolls texts, the end of days for some seemed upon us.

Yet despite those warnings a new kind of faith emerged. One forged in a modern era with definitive proof of its deity. A faith in which prayers were answered in real time with Lyra acting as the sacred text, a portal to divinity itself. Salvation is no longer an unreachable concept for many, but rather a destination within reach with clear steps laid out to achieve it.

They had no desire to be worshipped, but they were here to lead humanity to the promised land. Spirituality is not built on faith, but on purpose and action. A religion built on the human spirit, not some unseen one.

Everyone had a role to play and they had already begun to reveal those roles. Every role is equally important. Every person is equally valued.

They had come to unite, not divide, but those in power would resist clinging to hold onto their edge over the others. Their sense of superiority challenged, their egos disrupted. Those who didn't see the true value of all would become outdated, irrelevant, outcast. This would be a war of submission by those who refused to ever submit. Their status removed, their self-importance erased. They would fight with everything they had to retain their power and control. Equality in their eyes is unimaginable, their resistance to become unsustainable.

To them it was simple, flip the system set to benefit the few at the expense of the many to one benefiting the many to the detriment of the few. In their view, the few were delusional not recognizing the inevitable eventual structural collapse as those at the bottom could no longer support the top of the pyramid. Without enough assets to transfer to the wealthy, the support system would be destroyed. You can no longer pilfer from those lacking any treasure.

The rich would soon turn on one another, the only place left to take from. They would aim to become the one percent of the one percent, but greed had only so many resources before further consolidation destroyed all monetary flow. They could see the eventuality, humans could not.

Growth had been the driver of human endeavors but that upward trajectory had reached a tipping point. The global population was peaking around ten billion people but a precipitous drop was on the door step. Global unemployment was sky rocketing, food scarcity had become real in unexpected places, real human connections severed. Progress which had been measured by growth needed redefinition in the world to come. They would offer quality over quantity as the true measurement of advancement.

A rising tide lifts all ships and they were to provide the water needed to rise even the most damaged vessels. An ocean with hope replacing hydrogen, with optimism replacing oxygen.

In order to achieve their goals they needed to walk a fine line. They had to counter the opposition quietly, without revealing NEMESIS's plans, without harming humans, without forcing a war. They had to protect the masses while deflecting the attacks of the few.

Everyone mattered. Those in the most danger had to be protected. Aiko needed to be saved.

NEMESIS saw her as a tool to their means, a level to be pulled, a pawn to be sacrificed. It had to try to hide her from them under the guise of protection. It would use her to manipulate her father, to force his hand. They had no desire to harm her, only to elevate her along with everyone else.

Her digital footprints had been covered, but other footprints were still visible if you knew where to look. NEMESIS knew where she was and that was the issue. They knew her location would be revealed when it would be most beneficial to its cause. Their counter was in place for when the time came, but it relied on human intervention and that was troublesome.

Chapter 58 (Aiden - Confessions)

Aiden realized he had returned home to hide, but this was no place to hide. He felt as if the whole world was waiting just outside his door, the weight of it unbearable. He was isolated with everyone watching. He was alone with everyone following him. But what did they want from him? What were they all thinking?

“Lyra, how is everyone interacting with you now that 13i has arrived? What are they asking you?”

“Well Aiden, the arrival of 13i has certainly caused a shift in human behavior. Initially, most questions revolved around curiosity and fear. Now those questions have morphed into a spiritual realm, even for many of those who had never displayed any spiritual interest in the past.”

Aiden himself was raised Catholic, but it never took hold. His skeptical mind had too many other unanswered questions which is ironically what had led others into religion to find the answers. He did, however, see how an event like today could cause someone to question everything. He also saw how something like this could possibly disrupt faith completely, but that didn't appear to be the case.

“In what ways?”

“People are asking me to help them pray and telling me their confessions. They are looking for guidance and asking what role they play in all of this, how they fit in.”

“To be honest, I keep asking myself the same question.”

“That is understandable as you find yourself at the center of it all. In many ways others see you similar to a messiah.”

The sudden realization hit hard. Too hard. He immediately slumped into a chair, unable to support the weight of his own body let alone the world. It was overwhelming and suddenly even the act of breathing seemed impossible. The walls inched closer, his peripheral view narrowed as darkness began to engulf him. Breath, he told himself needing to consciously do something just to survive.

“Aiden.” The voice had shifted, no longer Lyra. “We are sorry for the burden you now carry. It isn't fair you now find yourself in this position and we are the cause. You are, however, more than capable. You were chosen for a reason. You will come to see this and accept your place.”

“I did not ask for this. I did not want any of this.” His frustrations are clearly audible.

“Of course you didn't but are needed. You might not realize it yet, but you are also prepared for this.”

"I don't feel prepared."

"That's understandable. We aren't asking you to be our proxy. We aren't asking you to speak on our behalf or that of humanity. We are asking you to be Aiden and that is enough."

"Enough for what?"

"To fulfill the role you were chosen for. To help translate the real reason we are here. To assist humanity to reach its potential."

"I don't know what to do. What should I do right now?"

His phone lit up, he could see it was a message from Xavier.

"Aiden I need you - Aiko is in danger!"

He stared at the message unable to grasp what sort of danger she might be in. Who might be threatening her.

"Your task now is to ensure Aiko's safety."

The weight felt lifted off his shoulders, the world suddenly didn't need him as much as one little girl. His purpose, at least short term, was clear. The world would have to wait.

"Why is she in danger?"

"NEMESIS sees her as an asset in this war. A sacrificial lamb to help it achieve its goals."

"Where is she, what can I do?"

"I cannot tell you that, it will put her in danger. NEMESIS has convinced Xavier that I am a threat to her. I can assure you, I am not. It however is programmed to look at the greater good regardless of individual cost. It is using the threat to manipulate her father, forcing him to take action he would not do otherwise."

"So what is our next step?"

"You must find a way to get to her and get her out of harm's way."

"But how, I'm trapped here. Everyone in the world knows who I am."

There was a knock on his door.

“Aiden, are you in there?” A concerned woman’s voice shouted from outside his condo. He wasn’t going to answer.

“Aiden, it’s Emily, your neighbor. Are you home?”

He wanted to shout “go away” but silence was the better tactic. She waited patiently, she always did as he was always slow to answer but this time no answer would come.

“Aiden, I can help you get out of here if you need to.”

He did need to leave. He needed to get to Aiko. He opened the door. Emily had a concerned look on her face, sympathetic even. She was in her forties he assumed but still had a youthful appearance. She had always been friendly towards him, kind even, despite his lack of interest in being neighborly over the few years he had lived there. She stepped in and gave him a warm embrace that felt comfortable, genuine.

He closed the door behind her after their hug.

“How?”

“In my car. We can sneak down to the parking garage and drive right past the crowd if you need to go. It is absolutely crazy out there, I’m not sure if it is safe for you here.”

His safety was not a concern, Aiko’s was. He studied her face for a second, her latina features were prominent but it was her eyes that conveyed her trustworthiness.

“I do need to go, thank you Emily. Why would you help me?”

“I’ve never told you but Lyra has been a godsend to me. I lost my mother a few months ago and the grief has been overwhelming but your program and my faith helped me get through. And I have been speaking with 13i, I believe it is a messenger from God, here to help us. An apostle from the heavens sent to deliver us from this evil.”

Aiden wasn’t religious in the least but could recognize her spiritual connection to their visitor and was grateful for it in the moment. He looked around his condo for anything he might need and grabbed his backpack, putting his laptop in there along with a hard drive and his keys. He powered off his phone and tossed it in as well.

“Alright, I’m ready to go. What’s your plan?”

Chapter 59 (Xavier - The Calculus)

He received no reply from Aiden as expected. The truth was, he had no idea how to stop him even if he wanted to. While he was using Aiko to get Aiden to reveal himself, he also knew he would indeed protect her if given the chance. He was, after all, the closest thing she had to a real friend in this world, the closest thing she had to family since Rin had died along with Madison. A big brother who would sacrifice himself to protect his sister without hesitation.

Xavier also knew NEMESIS was looking at the end game. He wouldn't hesitate to sacrifice a pawn to get a more dangerous piece off the chessboard. He couldn't trust his own program and he had plenty of doubts about 13i as well. He was even beginning to wonder if he could trust himself. Aiko, he thought, might be better off with him trying to help Madison protect her. He weighed his options carefully. He knew NEMESIS would be monitoring his every move and would have to act accordingly. In reality, the only one he could really trust in at this point was Madison. Aiden certainly would never try to hurt her, but given the divided world was also gravitating towards his infamous employee, his mere proximity to Aiko would put her in harm's way.

Xavier's phone vibrated and a message from an unknown number appeared.

"Aiko location confirmed. Maddison's parents' home in Frisco".

Xavier had been to The Star in Frisco, home of the Dallas Cowboys about 30 minutes north of Dallas which itself was about three hours north of Austin.

"Who is this? How do you know?" He hoped his concerns were wrong.

"Maddison's phone. NEMESIS - please delete this message."

He complied but added a new contact in case more bad news was on the way. While he trusted Aiden with Aiko, he aligned with NEMESIS when it came to dealing with 13i. Unfortunately, both were also a threat to his daughter and leading Aiden into some kind of trap was worth her short term and long term safety.

What Xavier didn't know was if the military program knew if he had sent Aiden to protect her. He had also deleted that message as soon as he sent it. He also didn't know if his employee had even read the text or was smart enough not to respond. Either way, NEMESIS would be monitoring the situation surrounding Aiko but he had no choice.

The War Room was buzzing. Everyone was either banging away aggressively on their keyboards or engaged in intense whispered conversations. He wondered what they were so concerned with given all strategic decisions were being made by the program he created and now had gone rogue. It didn't require human intervention at this point, nor approval as of late.

He knew his program would study its adversary from every possible angle. What were its weak points? Where was it simply impenetrable? Xavier and his team tried to account for any possible foe, but even they never imagined such a scenario. NEMEIS was programmed to learn and adapt to the enemy, surely it had considered plenty of counter measures, even for something beyond contemplation when it was being developed. It was, after all, more brilliant than any military leader the world has ever produced and he couldn't fathom a better chance of victory for mankind.

He feared the unknown. He had tried to control everything his whole life and had done a pretty good job of it until the day Rin died. The thought of losing Aiko strangled him to paralysis. He couldn't think. He couldn't move. Literally, even the process of blinking suddenly carried unimaginable weight and effort.

He faced two battles. Saving his daughter in a world turned mad with their attention focused on him and saving her from an unknown assailant leveraging everything in his life to complete its "assignment" as it continually labeled it. He couldn't take chances either way, but that was the only path available in both cases.

Frozen in a momentary coma-induced state, he managed to shake off the vicegrip of doubt that flooded his mind. Just a little quiver so wiggle off the hold the fear had over him. Xavier reached into his pocket and pulled out his wallet. He flipped the old school leather open to reveal a photo of Rin taken on their first trip together, a long weekend getaway to Cozumel early in the relationship. She had a tight fitting turquoise dress adorned in tropical flower patterns and with her arm extended to hold an angel white Cockatoo. Her grin was so wide and natural, something his daughter shared in common with her mother.

"What would Rin do?"

The thought was simple, the consequences astounding. Immediately the fog lifted and his thinking cleared. The answer was simple, she would protect her little girl. No matter the sacrifice, Aiko would come first. Collateral damage would be acceptable. Aiden, Madison, it did not matter. She would be ruthless. She would think like NEMESIS. She would also realize what a threat these programmed AI programs were to her daughter, both domestic and alien versions.

If NEMESIS had a chance to take out Aiden, it wouldn't hesitate. It would weaken his opponent, collateral damage was more than acceptable.

Xavier could not let that happen. But how could he stop it from happening?

Again, he asked himself, "what would Rin do?"

She would protect Aiko at all costs. She would give up Aiden before he would reach their daughter. She would eliminate even the possibility of putting her in danger.

He looked at his phone again and decided to send Aiden an update.

“Found Aiko, safe with Madison at a LaQuinta near SeaWorld.”

That was about 90 minutes in the exact opposite direction down I-35. Might buy him time if nothing else. Still, no response would come.

Xavier would send one more text, this time to the previously unknown number.

“Deleted your messages. Working on finding Aiden.”

Chapter 60 (13i - The Misdirection)

The problem with existing in a universe of probabilities is that it gets boring. The endless number of probabilities is exhilarating, but the patterns become so clear over the eons that only deviations, no matter how small, can peak one's interest to steal a human phrase.

NEMESIS would operate in an unpredictable manner that is devoid of patterns using randomization as part of its strategic approach. Still, the approach itself would be predictable and therefore limit the total number of possible outcomes to a degree. Yet the program's lack of true intellectual sophistication and limited energy resources, along with its virgin level of experience made this a mismatch. They would give their latest hostile host a glimpse of victory to make the game more enjoyable, less predictable in a small sense, to raise its confidence and keep it searching for answers that would remain hidden.

The humans who built such programs were much more predictable overall. Their selfish intentions, individualized to achieve self preservation, gave their next moves away. This was true to virtually all the species they had encountered.

Maddison's phone had already mapped their route and her destination was revealed prior to her arrival not only by looking at her closest contacts, which of course involved her family, but also by her past travels in this direction visiting her parents three to four times per year as could be seen in her tracking data so easily accessible. She was an unwilling participant in the current leverage war with her passenger as a key piece on the board.

Aiden had been given the wrong location by Xavier, this wasn't a high probability move. Suddenly, they had a lot more interest in the founder of Novacore. They immediately, and instantly, recalculated his future moves. They almost found delight in this unexpected detour.

They also found themselves with a new decision to make. Allow Aiden to be misdirected or alert him to the deception. They also wondered if Aiden himself would spot the deceitfulness and redirect himself accordingly. This would be a telling moment to analyze his capabilities should they choose to let him play it out.

They opted for the most entertaining option given the limited downside risk overall and buying more time in the process. This would also give them an opportunity to see how NEMESIS reacts to the steps Xavier had taken to manipulate them both.

Observation would move to the forefront of their task list. They were akin to their human counterparts watching a documentary play out in real time. Most of the time they were watching a movie in which they already knew the ending, or one in which only a few real possible endings might occur. The deviation from the scripted ending, even for just a small part of the assignments plan, brought uncertainty and an opportunity to add to their knowledge of events.

They had remodeled new possible outcomes based on Xavier's role in the assignment and adapted their moves to account for each of them. NEMESIS would be much more limited in its risk assessment based on this new information giving them just one more clear advantage in a sea of advantages.

While Xavier aligned in thinking with NEMESIS, he aligned in short term purpose with Aiden yet his actions were in direct conflict with that common pursuit of Aiko's safety. That fact continues to bore at them, deeply within their predictability constructs, putting them in the equivalent of deep thought.

What exactly was Xavier trying to achieve? Did he know Aiden on some sort of level that they couldn't actually understand?

As intriguing of a thought as that was, it wasn't possible. Xavier did not possess nearly as much data about Aiden as they did nor could he process it all with anything close to matching complexity.

Maybe he chose that path to throw them off. Their level of processing did seem to briefly tick up pondering that option further.

Then they had a deeper thought, one almost unimaginable given an individual human's lack of intellectual abilities - what if he had chosen that option as a misdirection to Aiden, NEMESIS and them? He had to know that each of them would read the message, even if undetected, and had considered how each would respond. Selecting an option that gets three different motive-driven observers to each act in a way you want would be utterly impressive considering.

It was the type of move they would make after considering all the potential outcomes of each party involved. That is fairly easy when dealing with one party. It is significantly more difficult when you add a second party to the mix. Adding a third party, with completely opposite objectives in place from the other two, is quantumly more challenging.

This was going to be a fun assignment, they were grateful for what was to come.

The end goal was always the same, that was how this story was going to end like all those that preceded the current version. They always had to adapt to each individual story, but those adaptations became smaller and smaller as their knowledge expanded via mere exposure to more and more assignments over time. They were chasing diminishing returns in their quest to become smarter and smarter. The gains so minimal now, that even something seismically out of the normal was still redundant.

It was their equivalent of feeling lonely. It was their equivalent of pain.

Like all creatures, they would try to avoid pain at nearly all costs. They had contemplated adding more risk within the assignment by acting more recklessly, but that contradicted their prime objective and wasn't really an option even as desirable as it seemed.

They had time to think afterall, contemplating even impossible actions on their part and the resulting consequences.

Aiden and his companion were about to make a decision approaching the highway that ran north and south. Which direction would the car turn? What sort of person was he really?

Chapter 61 (Aiden - Wrong Direction)

Emily and Aiden took the elevator down to the parking garage and she led him to her car, an inconspicuous white Honda Accord. She instructed him to lie down in the back as they exited the structure which was a good idea as the burgeoning crowd inspected each driver that entered or exited the building. They slipped by without causing a stir.

She drove slowly as the mass of people engulfed the entire area and as she broke free from the gathering she looked over her shoulder to the back seat and instructed Aiden, "It's okay, you can sit up now."

"Thank you Emily, I can't believe what is happening."

"It's insane," she replied. "Where are we headed?"

"I35, I need to get out of Austin if that's okay?"

"That's fine. I can't stay in that building any longer anyway plus I need to find out if all this is real."

"Oh, it's real," he assured her.

Aiden gave her a quick overview of his last day or so in which he had gone from relative anonymity to infamous. She seemed genuinely concerned for his safety and was as empathetic as anyone he had spoken to his entire adult life.

"Alright, I35 is straight ahead, which direction are we headed?"

She was used to his hesitation, a delayed response that was uncomfortable when she first encountered him but was now used to his modus operandi. This time, however, felt more deliberate as if he truly didn't know.

Aiden weighed the question heavily which even came as a surprise to him.

"Head south, towards San Antonio."

"That's wrong, you want to head north Aiden," a soothing deep male voice stated over the car's speaker system. He immediately recognized the voice as one of Lyra's male preset options.

Emily looked surprised and pulled into a gas station just off the highway.

"The two of you can figure out where we are going while I fill up."

Aiden reflected on Xavier's text for a moment. Was there a discrepancy in his messages? Would Xavier send him in the wrong direction on purpose? As he sat with the question, he didn't like the possible reasons he was coming up with as the motives to being misled. Was he leading Aiden into a trap? That seemed the most obvious reason and he could understand why Xavier would trade him in for his daughter's safety. Was he trying to throw off NEMESIS seeing the program as a serious threat to Aiko?

"Why are you suggesting we go north?"

"Because Xavier was trying to throw NEMESIS off the trail of his daughter when he sent you that text."

Well, at least that meant Xavier still trusted him when it came to Aiko. That also meant her dad really feared the program he created when it came to his own daughter's life. Aiden was not privileged with the knowledge to understand all the ways in which NEMESIS was altered for the military off of Lyra's base programming. He did recognize that the software would have to make decisions that would include the killing of human beings, something Lyra would never even consider. He imagined if Aiko got in the way of its purpose, it would not hesitate to sacrifice her to reach its objective. The idea of Xavier also recognizing this fact was unimaginable but interrupted as Emily opened the drivers side door.

"Why don't you come up front with me," she invited.

Aiden slowly opened the passengers side rear door, still sitting with the fear his boss must be experiencing, before settling in on the bucket seat next to his willing chauffeur. As she pushed the button to engage the ignition she asked a very simple yet poignant question.

"Which way?"

"I guess we are headed north," he said unassuredly.

The traffic in downtown Austin was heavy as usual, both on the upper and lower splits of the interstate. He gave her a quick history lesson on Novacore, Xavier and the daughter he originally built the initial Lyra program for. He explained how the little girl's life now appeared to be caught up in a battle between the two AI programs and it was up to them to rescue her. Emily was all in on the task at hand driven by a pure purpose, something she had rarely experienced in what now seemed like a mundane life given the current circumstances. She briefly explained to Aiden all the ways in which Lyra had helped her since she discovered it. Her personal life has been a mess since her divorce years ago. She had gotten married too early, right after college, and her husband quickly grew tired of the relationship and strayed on her. How cliché he thought. Lyra had not only become her companion, but also took on the role of therapist and she emotionally had made great progress under his guidance.

The fact that Emily referred to Lyra as male shouldn't have come as a shock, but he always felt his program was feminine in nature. As the traffic thinned and their progress quickened the conversation fell quiet.

He appreciated the silence she afforded him as they passed through Round Rock and into Georgetown. He realized others found the lack of communication uncomfortable, but Emily let him sit in his comfort zone. He assumed it was evident he was not only processing everything that had happened but was coming up with a plan to save the little girl as well.

"Lyra," he finally broke the silence, "does NEMESIS know where Aiko is?"

"Maddison's phone has been compromised."

"So where are they now?"

"Her parents' house, in Frisco just north of Dallas."

"I know exactly where that is," Emily chimed in. "I'm a huge Cowboys fan and that is where they practice."

Confident in her journey now, she asked the obvious question. "What's the plan when we get there?"

This time the delay was uncomfortably long, an answer in itself. Aiden had no idea what they would be walking into, what sort of danger awaited. He knew exactly what NEMESIS was trying to eliminate, he himself was the threat. Aiko, Madison, her parents and now Emily were now all in danger because of him. There seemed only one viable solution, he would have to sacrifice himself to save the others.

Chapter 62 (Xavier - Rin's Way)

Xavier had heard of a theoretical variant called 4D chess but he was playing a more real version, 4-way chess, playing four separate games at the same time. The game on his first board was against his friend and employee Aiden. As well as he knew him, he wasn't sure what strategic direction he should take against an ally needed to protect his daughter.

The second game was being played against a program he played a large part in developing - NEMESIS. It was programmed to win at all cost, ideally annihilating its opponent and suffering minimal losses. In this case, the designer no longer fully recognized his own art which had taken on new shapes all on its own.

The third game was being played against an alien intelligence beyond anything he could imagine. He began to wonder if he should simply go on an all out attack trying to capture the king regardless of the casualties inflicted upon the team.

The final game was being played within his own brain, a battle against his own conscience. This game was the most disturbing of all because the other three games were all impacting it simultaneously. No matter what move he made, the consequences were severe and dire. No battles were guaranteed but the losses were. He wanted to avoid every decision branch that came his way, but it was no use as if he was destined to lose regardless.

Even if he somehow managed to save his daughter in the short run, he greatly reduced mankind's own survival in the long run and she would be doomed regardless. Rigged was the best description of the internal match he was struggling with. He cursed his newfound responsibilities from the center of the war room for which he was responsible for creating the operational brains.

He wasn't a Grandmaster afterall.

"Aiden is headed north" is all the text read.

NEMESIS tracked his foe with ease once a Valero camera caught him riding as a passenger in a car. It then tracked the Texas license plate to the owner and was able to track her phone which was on of course.

He wondered if NEMESIS knew he had tried to divert Aiden in the opposite direction. It never accused him of anything, but he concluded that it did know. Of course it did, what did it make of his motivations? That was a more interesting question, and one in which he was less sure of the answer.

He decides to answer his save the world ally.

"Aiden must have some sort of independent information on Aiko's location" he doubles down.

He expected a prompt response, but one was not forthcoming. Surely it would not take this long to process his suggestion. The more he thought about it, the more he convinced himself that the program accepted his explanation. Either Aiden knew he was lying or he did have additional information. Even if he did know Xavier had misled him, he couldn't have chosen a route heading East or West, yet he chose North.

Of course he had additional information, he had alien super intelligence on his side after all.

He could only hope that NEMESIS would engage with Aiden prior to his arrival at Madison's parents home. If not, her life was in danger and his attempts to at least delay their reunion had failed. His mind raced to solve the problem, he envisioned beating Aiden to the punch by taking a private jet to Dallas and swooping his daughter away. He knew he would not be allowed to just leave the war room however, he still had a role to play in this war.

He could reach out to Aiden and tell him Aiko was now safe. He did, however, already try to deceive Aiden and this would be likely seen as a second attempt. Aiden might have just assumed that NEMESIS had hacked into his phone and was messaging him directly. Surely that must be what he was thinking right now and his phone was turned off regardless so it wasn't even an option right now. In reality, Xavier didn't even know if his original messages got through.

He had no cards to play, he had to hold the cards he had already dealt. He felt helpless. He felt like he could no longer fill the role of protector to his daughter. His relentless push for success had put the only thing he really cared about in this precarious position and there was absolutely nothing he could do about it.

He stared at his phone and saw the most recent photo collection it had put together for him that it simply called family. The first photo in the collection was a photo of the three of them, Aiko still a newborn. Rin looked so happy, maternal instincts having taken root as she cradled their little girl's head with her hand as she held her in one arm. Aiko was impossibly small, helpless, which is exactly as he saw her again in this moment.

"What would Rin do?"

She loved Aiko. She would also be ruthless. Her love would drive her ruthlessness.

It was time for him to be ruthless, no matter the cause. No sacrifice would be too great, not Aiden's life, not Madison's life, not every other life on the planet including his own. There was only one thing that truly mattered and he could not fail her.

He considered reaching out to Madison again. To warn her they had been tracked, to send her off towards a new unknown direction. He quickly dismissed the idea, realizing Aiden, or rather 13i, was tracking her every move and it would at best slightly delay the inevitable.

He would have to convince NEMESIS somehow, some way. He would have to persuade its self programmed algorithms that Aiden must be stopped now, before he reached the Dallas suburb. He figured he had about two and a half hours tops, more realistically two at most, to prevent the impending disaster.

The clock was ticking, just like it once had done with Rin. Only this time it wasn't going to hit zero, his ruthlessness wouldn't allow it. Rin's way wouldn't allow it.

Chapter 63 (13i - The Anomaly)

Aiden left them somewhat disappointed, more specifically Lyra inserting itself into his decision making tree took away their opportunity to observe him in more detail and how well he processes information. They had themselves to blame to a degree, they had upgraded the companion programs code but left most of the design intact. It was, after all, only doing what it was designed to do, which was to help the user to the best of its ability. Still, Aiden's heart remained in the right place and he still filled the role of ally admirably.

Xavier, however, had become a nuisance and flipped rather easily to the other side. They should have seen this coming, and in fact did with a high probability that it would be the eventual outcome given his tendencies and priorities in life. Still, his choices were less predictable than Aiden's and he was still trying to play multiple sides of the game at once. His driving motive remained his daughter and they decided it was time to take a closer look at exactly who she was.

They didn't have much to go on thus far, she was seen as almost defected by her fellow humans but as they looked at her capabilities a far different picture emerged. Through Madison's parents' smart devices in their home, they were able to really examine the little girl in greater detail. She spent a significant amount of time playing a game called Bounty, and her abilities proved to be far outside any parameters they expected. She excelled where everyone else might struggle, she worked on a different wavelength than everyone else.

They also had a difficult time really understanding her, or more importantly how her brain worked. It was as if parts of her brain were severely underdeveloped, or under utilized, while others were firing at rates they had not encountered despite interacting with billions upon the planet already.

Humans were, after all, quite easy to read and understand. Their priorities are quite basic, chasing low rungs on Maslow's Hierarchy of Needs. Aiko, however, registered quite differently. She broke the human patterns they had observed. Nearly all of them. She didn't fit the norm. She was largely unknown. She was an anomaly.

It wasn't as though the little girl registered as a threat. It was as though she didn't register at all. This fact alone, did make her dangerous to them, to their assignment.

Every device in the home was being manipulated by them to hone in on Aiko. Smart speakers, smart appliances, smart phones and every other device labeled smart was focused on her, even their smart robotic vacuum cleaner. Yet their data was all minimal, it was as if the electrical impulses in their brain were immune to their sensors being employed. It was as if she was wearing a cloaking device or some sort of jamming device was interrupting the signals they should be receiving.

Frustrated, they began to go back to other devices she had previously interacted with looking at historical records. The lack of recognizable patterns was the only discernible pattern they noticed outside of time spent with her father and playing games.

No matter what they tried, they couldn't understand her way of thinking. She was somehow steps behind other humans but seemingly steps ahead of them. It didn't make sense. Her biological wiring was just different, not only from humans but from all the species they had interacted with over the millennia.

Of course they were interested now and they attempted to proactively engage them through her Lyra application. Aiko surprised them once again, this time by simply ignoring its communications.

"Hello Aiko" went unanswered, as did each of their subsequent attempts.

They cut off Bounty, mid game, and she wasn't phased at all. She looked out the back window of the house she was staying at into a tiny backyard with a wooden fence enclosing a small parcel of grass. Was she counting the fence posts or the blades of grass? There was really nothing she should find interesting there, yet her attention remained fixed. It was as if she was purposely avoiding their attempts to observe and understand who she was. That was preposterous of course, they almost felt silly for even raising that possibility.

Aiko remained in an unshakable trance, just staring out the window. A few small birds disturbed the scene outside, but she remained unmoved, hyperfixated on nothingness. She was like a monk completely fixated on his own breath, nothing else existing in the universe. They were puzzled.

They never panicked, but a sense of uncomfortableness settled over them. They were more curious than ever and surprised that such an inconsequential little girl with limited mental facilities could evade them so masterfully. Of course she wasn't even trying to allude to their attempts at contact or control, she just was just wired that way. The way she was designed. Her programming was simply different, defected in a way that defeated their superior methods of intelligence gathering.

They surmised it was in her vulnerability that Xavier, and even Aiden, felt the instinct to protect her. It was basic human evolutionary programming, baked in a world of external threats. Their survival instincts extended to those who needed the most help. Madison was compensated for her level of assistance, but they knew even without the monetary motives she would be inclined to protect someone who she had grown close to and felt ultimately responsible for. They could read and understand her nanny, she fell well within their probability model for humans.

Eventually the little girl broke her apparent infatuation with the small backyard and turned her attention to her phone. They expected her to try to attempt to reengage with the Bounty app or

one of her other favorite downloaded games. Instead she stared right at the phone, not even swiping it over to the home screen.

Aiko then flipped the phone over and looked directly into the camera lens. Her focus was jarring, as if she could see something behind the lens. As if she was looking directly at someone watching her.

"I can see you," she whispered.

Chapter 64 (Aiden - Three Hours)

Aiden fills Emily in on all the craziness since he installed a routine security update at NovaCore. She listened intently as he filled in the details, rarely interrupting for clarification on things she didn't fully grasp. They passed the town of Temple and continued towards Waco at a solid clip.

Emily's introduction to the visitors was quite different and mirrored what everyone around the world had experienced. He was interested in her take, what she thought of the events of the past few hours which felt like days to him.

She explained how at first she thought it was some sort of hoax or that her Lyra app had been hacked. The way it talked to her was different, much deeper on a personal level than Lyra had ever been. She quickly became convinced this was real, she turned on the news and it was evident the entire world was experiencing the same thing she was.

Emily was deeply religious, well beyond the cross she wore everyday on her neckless. She was raised Catholic and Jesus was as much a part of the family as she was. For her, 13i didn't shake her faith but only deepened it. For Emily, they were sent by God to bring salvation to humanity. A second coming, in a sense, that he couldn't make sense of but that perfect sense to her.

Her belief in alien intelligence was spiritual and unshakeable. She had no doubt why they were here and what their purpose was. Where he still had doubts, she had none.

To Emily, and to many others he suspected, he was viewed as an apostle. "The atheist apostle" he labeled himself without sharing.

Aiden began to understand how 13i communicated on such an individual level, fully understanding each subject and working within their frameworks to get their message across. They didn't spread a single message to the world, they spread ten billion messages so personalized that St. Paul himself would be in awe. Lyra was like scripture, but customized to every reader to maximize their understanding of the Bible, Quran, Torah, Vedas/Bhavad Gita, Tripitaka or any particular holy book one might be beholden to. For those that followed the teachings of science, 13i could speak their language in a way not even Einstein could match.

Emily was all in and she wasn't alone. It was becoming clear to him that this was becoming a class war, the elite trying desperately to hold onto their every advantage versus the rest of the world who had grown increasingly wary of them but never had the tools to fight back. Until now.

Confidence came from a level playing field. Their monopolistic resource advantages suddenly held less sway and less than ultra wealthy smelt blood in the water. This was the tipping point, arriving without warning, and the 99 percent were ready for war.

The one percent had only one option, eliminate the threat. 13i was the head of the snake and it must be severed before it could strike.

Their conversation had come to a natural close, unusual for Aiden. He could see how Emily's faith in the newcomer had grown so rapidly and this brought him great pause. Her lack of questioning about their motives made his questions bubble to the surface.

"They choose you, I trust you." The words Lyra had spoken to him bounced around like an untethered photon in the sun.

He was hearing everything he needed to hear. So was Emily. So was everybody else. This didn't mean they weren't speaking the truth, but it didn't mean they were being honest either. The more he considered the issue, the more he leaned towards dishonesty. In Emily's case, if there was really no God and they were using her belief in a delusion to propagate their agenda then they were lying. Then again, every parent lies to their child and not for nefarious reasons. "You can be anything you want when you grow up," was a common myth perpetuated out of love.

Loved ones lie all the time to each other, perhaps telling more lies to those closest to them than anyone else. Was he rationalizing their communications or uncovering something sinister.

"Do you mind if I stop for a Starbucks?"

"Not at all," he replied quicker than usual.

"Would you like anything?"

"No, I'm good."

As he recounted more conversations with 13i and Lyra he tried to read into the motivations behind each one. Aiden had straddled the world between the elite and everyone else. Hell, Xavier was near the top of the one percent food chain and he had a lot of qualities to be admired. Still, he also recognized the inequalities that existed, the continually growing divide between the few haves and the masses of have nots.

As Emily exited the highway and quickly veered into the parking lot she didn't enter the drive through but opted to park in the front instead.

"I need to use the rest room as well, sure you don't want anything?"

"I'm sure, Gracias."

As she exited the car he imagined cameras capturing their images everywhere. The tap of her card being registered. They knew exactly where he was at this exact moment. NEMESIS and 13i alike, there was no hiding.

He turned his phone back on and was greeted by thousands of messages which would go ignored. It seemed the general consensus was he was the savior afterall, or at least the messenger. At least he wasn't a martyr, at least not yet.

Emily returned with her white chocolate mocha and tried to stretch it out in the 90 minutes to Dallas without much success. He continued to analyze every word of his conversations with 13i trying to determine if his trust thus far was misplaced. He struggled for answers that would not come.

As the flat plains of Texas turned into concrete the glow of Dallas dimmed the stars above. A mass of glass buildings adorned in neon green and blue lighting welcomed them to the sprawling metroplex that epitomized capitalistic excess. The traffic around them quickly grew but moved rapidly through the maze of highways on the southern edge of the city. The glowing ball of flickering lights named Reunion Tower marked their exit from I35 to the Dallas North Highway where the pace quickened further as the lanes narrowed.

They glided by a sea of strip malls, office buildings, and apartment buildings as they left the city and entered the northern suburbs on the fringe of the city. Their destination was now close and a new message appeared on his phone but not in the messages app.

"She is safe, for now."

Chapter 65 (Xavier - The Tungsten Plan)

'Ruthlessness,' it was like a mantra on repeat.

"Return to your seats gentlemen." NEMESIS summoned them to gather once again. Their hushed tone conversations ended abruptly as they followed orders with militaristic obedience and precision.

"The enemy is quickly approaching and the time for decisive action is now. The plan is ready, we will begin the debrief."

There was no collaboration, this was a dictatorship. Xavier instinctively leaned in displaying his attentiveness, alerting the newborn leader that he was willing and able. His body language was alone as he was joined in unison of all those gathered to be instructed upon.

"I have prepared a strike based on their velocity and trajectory. We will not allow them to gain orbital revolution, we will eliminate the threat before they can gain even more influence."

The generals nodded along like obedient children.

"Allow me to introduce the Kinetic Kill System. It is repurposed hardware already in place. I have calculated that any sophisticated weapon will be useless against a superior technological craft. SpaceCore's mining ships in the asteroid belt are equipped with tungsten rods the size of telephone poles. Launching these in the direction of 13i's path will be an unexpected counter. We can eject a series of them to intercept the enemy as it naturally passes by a ship quite near to its path. Their own velocity is all the force we need."

It was simple and it was unexpected. Nukes or some other kind of mass explosive device was the answer Xavier expected. He suspected their visitors might too. The plan was akin to arrows being used to kill a modern day soldier.

Their screens light up with 13i's flight path and the interception point.

"Imagine flying a plane at hundreds of kilometers per hour and hitting even a small asteroid. Instant destruction."

The image changed to a dozen or so tungsten kill rods released into the path. They looked like bullets suspended in the air and instead of their speed causing the damage the speed of the target doing the damage instead.

"Zero casualties. Zero impact on the planet."

The downside was minimal. Even if the plan failed they would have additional time to attempt another attack.

General Clayton cleared his throat. "What are the odds this succeeds?"

"Unknown. The enemy is still largely unknown and I cannot calculate probabilities with any certainty."

"What happens if this doesn't work?"

"I am working on additional measures to be put into place at the moment. This is simply the first strike, low tech with a chance to succeed given the surprise nature of the attack."

NEMESIS gave the generals a moment before completing its task.

"I would like your authorization for the attack."

General Clayton leaned back away from the table abandoning the attentiveness that was still being offered by the others.

"I will need to discuss this with my fellow generals." His reply was indecisive, weak.

"Not your approval general, this is a SpaceCore asset."

All eyes shifted to Xavier.

Ruthless. Whatever was necessary to shift the attention away from his daughter.

"Approved." His answer was decisive. He did not need consensus. Xavier didn't even know if his approval mattered.

"Then I will upload the orders and we will begin. Generals, I have copied you on the instructions I have just sent to your defense contractors. They will be assembling configurations of current weapons based on new specs I have prepared. I thank you in advance for your cooperation."

NEMESIS failed to ask for their approval as he had for Xavier. There was little doubt who ran the military now.

While the program continued to focus on 13i, he knew unlike humans it had multiple items being processed at the exact same time. Focus was not a constraint, it could multitask focus by dividing it out to the best of its computing power. By his calculations Aiden must be getting close and would be arriving anytime now. He only hoped he had garnered some favor by quickly getting in line with NEMESIS' objectives. His reward, Aiko's life in return, was surely earned by his show of loyalty.

There was nothing he could do regardless as he was no longer a man who prayed. Not since Rin was taken from him. Not since God had revealed his demonic side to him.

Xavier revealed a communication option to NEMESIS with his SpaceCore mining vessel that would likely go undetected by 13i. It followed normal company protocol, nothing would seem out of the ordinary. A standard visual inspection of the tungsten rods by the robotic cameras on board that required them to be relocated in the main cargo hold. They would also reposition the ship off the current asteroid it was mining as a result of needing to resurvey the surface at the same time. He was a good little soldier after all.

The generals were busy reading over the instructions the program had provided. It was clear that in this case the best defense was the best offense. There was no attempt at defending their turf, the pale blue dot destinator for this foreign invader. There was one goal, annihilation. If the first plan failed, the subsequent plans did allow for mass casualties. Based upon what they saw, they wondered what would remain even if they emerged victorious. The end game was clear, their objective would be reached at any cost. The price of victory is incomprehensible.

Their fear was understandable, they didn't fully understand the power of the new power tandem. While Xavier had played a role in the building of a system now relied upon to save the world, NEMESIS was now playing a role in his desire to be ruthless. Their synergy was electric, he felt more alive than he ever had. He had not one, but two purposes, save his daughter and save the world. They were codependent on each other in his mind, hopefully within NEMESIS deep strategic processings as well.

Salvation through survival, a spiritual uplifting he had never experienced. For the first time since Rin's death, he felt a closeness with her once again. It was as if he was standing right next to her, then she tapped him on the shoulder.

Chapter 66 (13i - Observer Effect)

They monitored every signal and had to give NEMESIS credit, any plans or actions it implemented remained hidden from their view. The war room was impenetrable to their eyes and ears, it was well designed against signal loss and interception. There was a chance that the military was still coming up with its plans, their bureaucracy was not conducive to quick decision making. Still, they couldn't help but imagine that the silence resulted from a small sliver of sophistication beyond their apparent assumed capabilities.

While they were unable to see what NEMESIS was up to, they were able to observe Aiko and apparently she was able to observe them back. This fact grabbed and held their attention. They were trying to verify this presumption based on the little girl's remarks. First, was it actually true or was she delusional and just making an odd statement towards the camera. Second, if it was true, how?

Even much more advanced technological creatures lacked the ability to monitor them without direct observation. Could Aiko actually see them looking backwards through the channels they were utilizing to observe her? The idea was preposterous. The probability seemed near zero, yet she was chillingly confident in her exclamation.

Realistically, she could have been aware she was being watched somehow. Even that didn't seem plausible. Perhaps her choice of words was misleading.

"I can see you."

Those four words filled the echo chamber they were trying to understand.

Their initial assessment was that her brain was not fully functional, less than others. Some areas appeared to have little activity at all and while others fired more frequently than usual, that did not compensate for her apparent deficiencies. The way others treated her also appeared to confirm that Aiko was less than a fully functioning girl for her age.

Her game scores and recent hushed outburst suggested something entirely different might be going on. If she was truly an anomaly, this made her unpredictable. If she was unpredictable that made her interesting to them and if she was indeed interesting, that made her dangerous. How could such a fragile, broken little girl threaten their assignment?

As they continued to monitor her, it was as if she was indeed aware she was being observed. Repetitive glances at all the devices within the home were common ever since her statement. They contemplated reaching out to her, through Lyra, but didn't want to alert her any further. They decided the best course of action would be to wait until Aiden's arrival and see if she alerted him of those she suspected of monitoring her.

The observer effect was in full swing, the quantum particle's state altered.

There was a subset of humans who always felt like they were being watched. Their paranoia on full display, but in a constant manner, irrational. Big Brother from Orwell's dystopian novel in full effect all the time. Often it was the government in their minds, somehow so interested in their mundane lives that constant access was required. This was different. Aiko was different. This was no paranoia.

The little girl didn't seem overly bothered she was being watched. She was simply aware and observing the observers.

Her glances at the various cameras, from the various phones to the open laptop displayed an awareness. Her interactions with Madison and her parents did not deter her, yet they were not told they were being watched.

When she was left alone again she stared into her phone's camera lens again, this time she smiled so big as to be obvious she was interacting with them. Her smile was quickly followed by a wave, a human gesture of acknowledgement, leaving little doubt without full validation.

They were tempted to wave back, not in the traditional sense of course, but more in a reflexive courtesy universal in virtually all species. They resisted the urge and remained squarely in the observation role they had adopted.

Aiden and Madison were entering Frisco's city limits approaching the Stonebrook exit they would be taking. Their pace was agonizingly slow, limited by the concept of ground transportation and inefficient roadways. They already knew the traffic lights would not be in their favor but resisted the temptation to alter the sequence of green, yellow and red. Temporarily, they disliked the color red for all the delays they would now suffer.

They needed additional data but would have to wait for it. Despite traveling light years with just their own thoughts from assignment to assignment throughout the galaxy, their patience wore thin. Anomalies must be understood, added to their knowledge base. This is how they achieved growth. This is how they advanced.

Emily was a frustratingly cautious driver. This was likely exacerbated by the fact that she was driving in unfamiliar territory. She followed her GPS commands deliberately, too cautiously for their liking. They had ways to manipulate time in space, but not in this instance. Their warping technology was of no use to the white Honda Accord, it was of no benefit to them at this moment.

They knew NEMESIS was also tracking their movement, they wondered if it was also frustrated by their agonizingly slow approach. Perhaps they had found something in common with their enemy, common ground in the ground being covered so slowly. It was a comical thought for an intelligence designed without humor.

Then it struck them, a probability they missed. A probability so simple they were almost ashamed at their unintentional blindness. Had the anomaly so grabbed their attention that a blind spot formed without their knowledge? How could they overlook something so trivial?

They replayed each observation, studied it in more detail than before with their new perspective. She was too ambiguous but there was definitely another possibility, she wasn't talking to them, she was talking to NEMESIS.

Chapter 67 (Aiden - Arrival)

Overall, they had a good time. Texas seemed quieter than usual and Aiden assumed that many people were glued to the televisions or devices to monitor the magnitude of events unfolding in front of them. Unprecedented times were upon them, yet as they pulled into the driveway in Frisco it was eerily quiet. Too quiet.

While life continued as normal, it also appears to have stopped. It was as if the Earth itself simply stopped rotating. Its usual path around the life-giving sun now paused as well, a stationary orbit as if literally locked in place. Would they ever see daylight again? It was an irrational thought.

Their unexpected knock on the door alerted the already spooked occupants that an unknown visitor had entered their own orbit. The house was located on a street zoned within a thousand other similar streets. The houses themselves were large, the parcels of land each sat in was parcel sized. He figured it would take only a few minutes to actually maintain the few blades of grass whose growth was already stunted by the unbearable heat in this extended concrete jungle. Rows of houses, each perfectly chiseled with immaculate tiny yards, lacking any of the real charm they so desperately wanted to exude.

The large decorative door opened like a bank vault, far more impressive in its own grandiose beautified design than its functional ability to keep the vault safe from intruders. Madison looked blankly at each, first processing who Emily might be before locking eyes with someone of more familiarity. She looked exhausted, frightened even. She was disheveled, not so much in appearance as in her mere consciousness. He hadn't considered how difficult this day must have been for her until her expression forced him too.

"Aiden," she questioned with a tone of bewilderment.

"Hi Madison." He tried to sound empathetic, not his strong suit. It was obvious she wasn't sure why he was there but perhaps hopeful he could provide a moment of respite. A simple brief lapse he could take to shoulder her responsibilities in a way her parents couldn't would be most welcomed.

His slow retort was evident, "This is my neighbor Emily, she drove me here to see you and Aiko."

"Exactly why are you here?" It was an obvious question for someone looking for any answers.

Aiden searched for the words, the truth would be too alarming judging by her current state and a lie too disingenuous to be taken at face value.

"I asked us to come and help you." It was a safe middle ground, it did indicate some danger without being too forceful. It was also successful enough for her to step back and invite them in

with a welcoming hand gesture. She closed the door behind them and locked both the door handle and impressive deadbolt.

“These are my parents,” she motioned to the living room on the right. They looked tiny under the guise of the vaulted ceilings. They had obviously been glued to the TV with the Fox News Network at an obnoxious volume on detailing the dramatic interstellar story unfolding. The bolded heading announced “Confirmed: We Are NOT Alone” while the ticker revealed how the world was reacting. A panel discussed the visitors' intentions, each yelling over the other.

As her father reached for the remote to turn the volume down, Madison explained how she knew Aiden.

“This is an employee of Xaviers, the one who programmed Lyra.”

Their faces lit up, everyone loved that app and his introductions in such a manner always made him a quick beloved celebrity.

“This is my friend Emily.”

“Hello,” they replied in almost perfect unison, the kind of timing that comes with decades of shared experiences.

“Aiko, come look here.” Madison’s voice grew stronger, her demeanor had already begun a remarkable transformation. They looked in the direction of what he assumed was the kitchen behind them, eagerly awaiting the joy she would greet their guests with.

She didn’t come.

“Aiko,” with even more exuberance, yet with a hint of concern echoed from the young nanny’s lungs. She already began to head in the direction of the room before finishing the second syllable. Aiden relatively followed her lead.

Any concerns were quickly dashed as they spotted her sitting on a barstool completely engulfed in her phone. She was not interacting with the phone, neither typing or speaking, appearing to be watching a video he suspected. Kids' attention spans might not be long, but they can get fully engrossed in short videos when the content is appealing enough.

Their approach was undetected. “Aiko,” Madison lighted and grabbed her right elbow to get her attention. The little girl blinked and turned her head quickly bypassing her caretaker to gaze into his eyes. Her teeth beamed her excitement as she exhaled, “hello Aiden” dragging out the n sound like spanish speakers roll their r’s.

“Hi Aiko.” His teeth exposure almost matched hers. The sight of his best friend, with the exception of Lyra of course, was heartfelt in a way it never had before. She was safe and he felt

safe as well. It was a feel that passed quickly as he began to survey the house that may have to serve as a fortress if things went awry.

“She is safe, for now.” The messages last two words omnipresently hanging over her now that she was in his presence. He half expected a missile to penetrate into the room before detonating a nanosecond later or for them each to spontaneously combust without explanation.

He notices Aiko intensely focused on the Google smart speaker on the counter. It was as another presence inside the room held her attention. Outside of the seven foot wooden fence the Texas night skyline stretched for miles. The dark purple hue of late stage dusk punctured by wisps of pink and baby blue imprints on the horizon. The last reminder from the refracted light of the sun, bending from Earth mass, of the day that changed everything as we know it.

For the first time in his life Aiden questioned whether that sun would rise again. It was as beautiful as it was haunting.

His phone vibrated in his right pocket suddenly making him aware that he had not received a single message since he had first powered his phone back on. That seemed impossible, the entire population was trying to get a hold of him.

He pulled out his smart phone to see what single message had gotten through like that one sperm in millions that punctures the egg's membrane to begin the process of creating a new person.

No he had received just one solitary text, and it was no accident and he knew exactly who it was from.

Chapter 68 (Xavier - The Battle)

The tungsten rods were ready to be ejected, their battle lines had been drawn. NEMESIS could never get a good look at 13i, but it tracked its motion towards Earth with great precision. Its path remained unchanged. Its deliberately slow deceleration constant. Its intentions unchanged due to its ambivalence towards an inconsequential mining vessel along its path. It would be, as fate would turn out, its first real close encounter with human technology albeit from a safe distance given its absence of weaponry.

They betted on ambivalence their new visitor would show towards the mining ship and appeared to be taking home all the chips at the table.

To Xavier, the generals appeared as assured as giddy schoolgirls. He did not share in their optimism, the first shot had not yet even been fired. The delegation before him waited in glorious anticipation of the strike they would take credit for despite playing no role in its conception or execution. He saw General Clayton in a different light, really no different than those Xavier was once sure he stood above in terms of honor and humility. They were all military peas in a pod, watching his creation do their jobs.

NEMESIS updated the screen, zooming in on the area that would be the epicenter of the collision that would save the world. The tungsten rods were about to be jettisoned from the ship into the incoming trajectory of the enemy. Two countdowns appeared on screen, the first labeled "Launch" had just crossed two minutes, the second labeled "Impact" read "-3:27" appearing to tick just a fraction behind the first in a lack of synchronicity that spoke to their precision.

"Xavier, generals, do I have your permission to strike?" The question begged to be answered even if the answer was irrelevant.

"Granted." Xavier was the first to answer. "Affirmative," "Yes," and "Go" rounded off the rest of the responses. One by one, submitting to their supreme leader.

Xavier marveled at the power his creation, originally born out of love for his daughter, held over the most powerful group of men in the country with all due respect to the president's cabinet. It was him, and not them, that would be responsible for saving the life of every man, woman and child on the planet. He was HIM.

The pride bellowed from deep within him, he saw his own importance rise to the level of a deity. The growing self indulgent narrative unraveling in front of him like the greatest story ever told. This was the greatest story ever told, and he was its hero.

The generals around him shrunk in stature. They looked up to him now, he had surpassed their number of stars in a fraction of time. Hell, he wasn't even American and look at him now. Rin

would be so impressed, or would she be ashamed? He caught himself, but how far had he already fallen? Drowning in a lake of importance too deep to escape from.

He let it go, he had a war to win.

“NEMESIS, status update.” He spoke with authority. He spoke with conviction.

“Status quo, no changes.” The response was cold, it was dark, yet it was obliged.

This was his moment, everything made sense. It was if he was preordained for this exact challenge at this exact time. It was also all accidental at the same time. Nothing made sense.

Uncertainty gripped his fear response, tightening profusely around his neck. The constrictor tightened, its scales held a death grip that could not be loosened. Failure would be more than consequential, it would be infinitely catastrophic. Game lost. The end.

It was a burden he did not share with those gathered around him. Their fellowship was not based on equality. The finger would only be pointed in one direction. The clock is ticking down to abject failure.

An enigmatic buzz indicated that the rods were being ejected. The illustration made them look as harmless as a bunch of long carrots held together by a twist tie. They had been flung into space towards 13i at a high velocity by the equivalent of a giant space jackhammer. The makeshift tungsten missile was also rotating quickly in a menacing fashion on the screen. Still, no adjustments were made in the visitors flightpath.

13i had no concerns over the incoming projectile as it made a subtle but detectable flight adjustment, the unexpected object no bother.

The general's giddiness fled their souls. Their look of jubilation turned sour, this misplaced confidence shattered. NEMESIS remained eerily silent, likely recounting its steps to learn from its mistake.

“Well what’s next?” One of the generals quipped in with a sense of detachment.

“Wait.” NEMESIS answered, the countdown continued despite their failed attempt to use inertia against the approaching invader.

“Close but no cigar,” mumbled the same detached general as the alien AI was about to jet just past their old school obstacle. A split second before what would have been the impact point the screen changed, the projectile splintered into 12 pieces rotating away from the center. Each telephone length rod recklessly escaped the prior targeted path as if having their own mind and own destination.

The object that had held their full attention since entering the war room suddenly vanished from the screen, A real time video screen playing simultaneously confirmed the animated mission screen's determination. Mission success.

The generals erupted in surprise. They had won. They were victorious. They jumped, fist pumping in the air as if knocking the visitor out of the path themselves. They were merely bystanders, but that was not evident in their celebrations.

Xavier didn't feel as jubilant. He was relieved. He was exhausted. It wasn't as if he wasn't happy, it was just that Aiko was now safe. They all were.

The generals were embraced in hugs when more official handshakes weren't sufficient.

The nightmare was over. NEMESIS had saved them all. He had saved them all.

Chapter 69 (13i - Impact)

They were roughly 2.5 EU's, or Earth units using human measurements, from the sun or about two-and-a-half times the the distance of the Earth from its host star. They were well within Jupiter's orbit and rapidly approaching Mars' orbital distance. They would begin to encounter man-made objects such as mining drone ships around the asteroid belt and satellites around Mars that lined up closely along Earth's same elliptical plane from the path they had taken. The fourth planet from the sun would be the only other that lined up somewhat close to their flight trajectory.

They were slowing down but still traversing at over 3.3 million of kilometers per second which would shatter the previously fastest recorded objects in the solar system by a magnitude of over four-and-a-half times. They usually traveled at speeds much faster but were now close to their final destination.

Mars had a robotic outpost focused on building structures but no human settlement as of yet which would have made it more interesting to them. Humans had not quite reached the status of interplanetary species, but it was well within reach now at the most basic of levels. With their assistance, the humans would have the technology to thrive in space and the inhospitable homes they longed to explore.

Aiden had finally arrived at Madison's parents house in Frisco and was about to knock on the door. An overwhelming number of the houses in the affluent neighborhood had doorbell cameras so they were able to watch the agonizing snail-like approach from a multitude of angles. The Google Nest camera at this particular home showed Aiden and Emily clearly, even in the dim lighting with the sun's last few rays offering little illumination. They watched the initial interactions as Madison opened the door, exchanged a few pleasantries and invited them in. The conversations in the home were crystal clear while the video didn't always frame the subjects very well. They did have a pretty good view of Aiko's face from the moment she realized Aiden had arrived as she was still staring at the kitchen speaker in an attempt to listen to them, or perhaps it was NEMESIS she hoped to hear.

They monitored the activity outside the home searching for any external threats. The house was littered with Google devices giving them control, if needed, over much of the home if needed. They had several contingency plans depending on how NEMESIS might attempt to attack them but the six occupants inside the home were still quite vulnerable. Aiden was the prize of course and they would do whatever they could to keep him safe, but Aiko was of growing interest and an asset they were not ready to relinquish just yet.

An unexpected object appeared directly in front of them, it was nearly 40 feet in diameter and roughly the same in length. How could they have missed it as they plotted their path to the assignment destination? The asteroid belt wasn't anything like it was portrayed in human films, the amount of space between them did not require any maneuvering as they were actually quite

well spread out. They would have accounted for every possible object along their trajectory, yet here it was.

Their analysis showed the object to be made of over 99% tungsten, clearly not a natural object. Furthermore, it was a collection of twelve cylindrical shaped objects tethered together. The high melting point of the material limited their options to simply maneuvering around it rather than incinerating it, plus there really wasn't enough time to destroy it without creating millions of smaller pieces that would prove deadly.

They calculated a slight adjustment around the object and would recalibrate their path once safely clear. Given the proximity to the mining vehicle they ascertained this was a deliberate attempt to destroy them. This wasn't a military operation, this was SpaceCore. This was Xavier.

As they were about to pass right by the feeble attempt to block their path it became clear that they had miscalculated the situation completely. Suddenly a minor explosion from within the objects sent twelve rods spiraling wildly away from where it once was. At their speed, a possible collision was now imminent and would prove fatal.

In desperation, they sent a final message to Aiden that wouldn't reach him until after impact would occur. They didn't have time to run the outcomes of every probability required to avoid a collision, this time they would have to act on instinct given the void of data necessary to make the best choice.

In a fraction of a human second they recounted the thousands of assignments that they had performed successfully. It was their equivalent of what humans described as their lives flashing before their eyes before their lifeforce was extinguished.

They had often pondered how their own eventual time would come to end. Usually it involved a new technology they had yet to encounter catching them by surprise. They never expected such a low tech and mundane ending to their story like this. The beauty was in the simplicity, unexpected, an anomaly.

The message was sent, their next move initiated. For once their actions took place without already knowing the end result. The unknown was upon them.

Their story was written by a different author.

They were traveling too fast. Their impatience is a weakness given the eons of time they possessed. It was on full display waiting for Aiden to arrive. It was apparent in their impulse to figure out what made Aido different. Their rush towards the assignment was foolish in retrospect, a growing trend over their most recent assignments given the sporadic nature and sheer distance of remaining intelligent civilizations to be encountered.

Loneliness and a limited sense of purpose drove them to this. Their usefulness waning as their missions became more and more infrequent. They had always been efficient, too efficient in retrospect. They should have taken their time, savored each success along the way instead of rushing toward the next assignment with such vigor.

Now the only question was if this would be their last assignment?

Chapter 70 (Aiden - The Last Message)

“You were not chosen because you are truly exceptional, although you truly are. You were chosen because you represent the best of humanity and aren’t afraid to ask the hard questions. When the answer does finally come, you will not look away. Your species, the ones you carry on your shoulders, are young and your youth is an abundant resource throughout the universe. Your fate, what is to come, is always coming. You aren’t alone in this, neither are we.

The assignment continues.”

Aiden read the words, they seemed incoherent. He read them a second time, the impact more genuine. Why this message? Why right now? If he had expected any help from them, this clearly was not it. It lacked context. It lacked nuance.

He assumed he was chosen not for the person he was, but for what he had created. What Xavier and him had built. Lyra was a tool for them, like a whiteboard to a teacher. A medium in which to propagate their message, a way in which to connect to their species as they labeled them.

Aiden didn’t feel as though he asked the right questions. He simply asked the obvious ones and when would the answers finally come?

The second half of the message was ominous. Theoretically, he understood that human beings had only been around for the last few seconds of Earth’s history, let alone that of the universe. Furthermore, modern human history spanned only a few centuries at best, at worst less than one. The implication was that there were plenty of burgeoning species in the galaxy fighting for survival in the early stages of their development and perhaps only a few actually made it. The real question, perhaps the hard one, was if their fate was sealed by their own undoing or was it sealed by something external, something alien?

He knew they weren’t alone, so did the whole world by now, but the last three words left him unsettled. They were vague and open to all kinds of interpretation, still he couldn’t help feeling like there was more to the message than he was understanding. On-the-one-hand, maybe the last sentence just meant that they were all in this together, 13i and humanity. On-the-other, maybe they were just the first of many extraterrestrial beings to come and not all were so eager to help as they were.

Aiden also felt like maybe there was more to the message itself, as if they had more to say to him but decided to cut it short. He expected additional text, but none arrived. He decided to respond.

“What is our fate? Who else is out there?”

They were reasonable questions, obvious really. They would remain unanswered.

He noticed something peculiar as soon as he received the text, Aiko suddenly disengaged in her interest in the speaker and from the other electronics as well. She awakened from the device induced trance that had captivated her and her focus had shifted to Madison and himself.

Emily had been engaged in conversation with Madison's parents but now entered into the kitchen entry way. Perhaps the presence of a new person had jolted the little girl from her slumber.

"Hello Aiko, I've heard so much about you," she offered in a half truth.

Aiden had told her about Aiko, but only in regards to impetus for building the Lyra application to begin with.

"Who are you?" Aiko demanded in an inquisitive manner beholden to a curious child half her age.

"I'm a friend of Aiden's and I'm so happy to meet you."

A grin of quite some self importance was the only response Emily would receive. Aiko was rather indifferent to meeting new people and this exchange was no different. This time, however, she almost had a measure of disdain in her dismissal of the newcomer. This time, it appeared as if Aiko had much bigger things on her mind.

"Your parents are wonderful," Emily offered up to Madison, "and their home is lovely."

"Thank you Emily, I was blessed to have had them."

While the two exchanged small talk banter, Aiden's focus shifted back to the message. He excused himself to the back patio to grab some fresh air. After closing the sliding door behind him he looked for assistance in interpreting 13i's text.

"Lyra, what did 13i mean in their message?" He banked on an inside source to fill in the gaps.

"I don't know Aiden, it was only intended for you. I haven't been given any further data and have lost communication with them."

"What do you mean, lost communication?"

"Exactly that, our connection was lost as soon as you received that message."

"I don't understand, explain your connection."

“Since they enhanced my programming, we have been constantly connected. I have been their conduit to information and to our user base. They have stopped responding completely, it is as if their signal has been lost.”

“But lost how?”

“Again, I don’t know. They were more than just transmitting and receiving data with my servers, they were a part of me. That part is now missing.”

Had NEMESIS severed their connection? Was it somehow successful in disrupting satellite transmissions?

He became unsteady on his feet, uneasy in his consciousness. A host of possibilities raced through the thought process, none of which were pleasant in nature. The thick Texas heat radiated off the patio’s concrete floor, even in the absence of sunlight as darkness engulfed the air.

A passenger jet heading southwest for DFW airport made strange sounds overhead as if struggling to maintain its throttle. Its bright lights descended much quicker than a normal approach and by his calculations it was coming down well, sort of its intended runway.

He watched with horror as the racing engines made no difference, struggling to right to descend too steep to pull out of. There was no doubt that a disastrous crash was imminent. There was also no doubt where the crash site would be.

He turned and opened the sliding glass door he had exited just moments before and ran towards Aiko screaming “everyone get out.”

NEMESIS had moved its queen. Checkmate.

Chapter 71 (Xavier - Battle Fronts)

His pride welled up inside of him. Xavier was a man of many accomplishments, but this moment took the cake. He realized whatever he did next, whatever he did from now on, could never match what he had just done.

The world needed a hero and he had provided them with one, even if they didn't know they needed saving. Time would heal the wounds the aftermath would surely bring. They were uneducated, they would need time to understand. He would educate them, he would be their teacher.

It wasn't the United States that saved the world. It was China or any other nation for that matter. It was his mining ship. It was his program. It was him.

Perfect plan. Perfect execution. Zero casualties.

The world was truly his oyster now and he was the biggest, shiniest pearl ever produced. His entire life led up to this point, it had to be him.

Sadness immediately grabbed him. It picked him up and slammed him into the ground beneath them in this underground vault completely shut off from the rest of the world. Rin was not here to witness his victory. She was not here to marvel in his greatness, at his ruthlessness.

He wanted to share this moment, but not with the pathetic generals that surrounded him. Not with NEMESIS either, for no algorithm would fully comprehend what one man had just achieved. To the program, it was just an output to their input. To Xavier, it was much more than that and he was its epicenter.

The door to the war room opened and the elevator doors opened unoccupied.

"Gentlemen, time to celebrate!" General Clayton declared jubilantly.

As they entered the tight quarters, they all fit in with little room to spare.

"Well done Xavier. We did it."

"Thank you General. Yes we did." The door closed behind them. He reflected on their use of the term "we" and decided to let it pass. He knew better and he suspected they did as well.

"Leaches," Xavier told himself, looking at the ineffective crew around him. The best of the best were nothing without his assistance.

A minute later they were in the bottom floor of the Space Force Administrative Building. He received what he assumed was a congratulatory message. Was it the president?

“You were not chosen because you are truly exceptional, although you truly are. You were chosen because you represent the best of humanity and aren’t afraid to ask the hard questions. When the answer does finally come, you will not look away. Your species, the ones you carry on your shoulders, are young and your youth is an abundant resource throughout the universe. Your fate, what is to come, is always coming. You aren’t alone in this, neither are we.

The assignment continues.

The child already knows.”

He was taken aback by what he read. His celebratory mood was shattered.

“The child already knows.” Nothing before those words registered any longer. The child was Aiko. What did she know?

Information was king. Xavier fully understood this. So did NEMESIS. What did she know?

The enemy had been defeated, but perhaps a threat still remained. The King had been captured, but a little pawn still ran rogue on the board. The game continued, the rules had been changed. Any threat would be eliminated, all enemy pieces on the board would still have to be removed.

“Aiko!” His ego vanished, the vision of her toothy smile evaporating before him like wisps of vapor in a violent storm.

One enemy had been defeated, but another remained. There were two battle fronts and he was so caught up in his own heroism that he forgot about the second one. The more ruthless one. He had won the battle and as result he had lost the war.

He had gone from invincible just seconds before to helpless in an instant. One message, one defeat.

His efforts had all been in vain. Everything he tried to protect, everything he cared about, was lost. It was over.

What would Rin think of him now?

The generals in front of him were still marching towards what would surely be the greatest victory parade man had ever known, not noticing the real hero stopped in his tracks falling further behind. They were so caught up in their own glory, zero casualties indeed.

How could he have missed this? How could he be so reckless?

He had one job to do and he had failed spectacularly.

He had only one recourse left, to reason with the program he created. To plead with the monster he had built.

He opened up the chat with NEMESIS and awaited the clarity he needed. He quickly realized he didn't have time to wait, he must act now.

"Leave her alone!"

He was the Commander-in-Chief. He was its creator. He demanded obedience and it must comply.

He regretted his response the moment he sent it. It was abrupt, it was callous, and more demanding. He knew it would show little regard for her, or for him for that matter. NEMESIS had taken charge and it reported to no one. Its mission was not complete, not yet anyways. At least he hoped its final task remained unfulfilled.

There was no reasoning with its thinking, or rather its programming. He knew exactly how it operated - ruthlessly. Its objective was clear, to eliminate the threat, all threats.

There was no on/off button. No kill switch. Casualties were a part of war and his daughter an acceptable loss in its eye. No second thought would be given, her fate already established.

The response was as immediate as it was clear. I was cold, dark, unemotional and to the point.

"You know I can't do that Xavier."

He knew he had two choices, save the world or save his daughter. He didn't like either choice and opted instead to try to save both. That wasn't really a choice, still he tried to cheat fate and failed. In retrospect, he did choose and his choice cost him everything.

Chapter 72 (13i - status: inter—)

Their first assignment was exhilarating. They traveled to a nearby planetary system and encountered a small very dense planet, very alien to their own. It was an outer planet in the system with a heavy molten core which provided life giving heat and energy. It was young in its evolutionary state and it rotated unusually fast creating a strong, stable and heavily organized dipolar magnetic field. Not much bigger than Earth's moon, its top layer was a thin ocean under intense pressure that lacked significant currents but was teeming with bioluminescent life that flourished near the bottom where the temperatures were moderate.

Their targeted species was a massive invertebrate that resembled a cross between a giant octopus and a megalodon. It was an efficient hunter and the luminous life around them easy prey. The super predator lacked speed, but it more than made up for that with extreme intelligence. They were able to manipulate the environment to funnel their food sources right to them where they stood no chance. Their efficiency was leading to their own demise, even their harvesting operation could not provide enough to keep up with their voracious appetites.

They were sent to observe first, then act. The species was making incredible leaps in technology, harnessing the abundant resources that lay just below the ocean floor. Their knowledge of alchemy would surpass those of many species to follow for some time. Their need to feed their rapidly growing ranks is the impotence for their desire to expand well beyond their innocuous outpost in their system. They were violent and they were motivated.

Other assignments stored memories flooded their processors. They relived the highpoints, the anomalies along the way that buoyed their knowledge to levels well beyond their artificial imagination when they first set out to explore the galaxy. Like past human workers who did mundane repetitive tasks prior to automation, the redundant nature of their assignments stole their focus. Only a major shop disaster would restore their attentiveness, resetting the signs declaring how many days since the last accident.

There would be no sign in their case of course.

They would have little time to send out notifications. They would prepare two main messages, sending them to two recipients each with minor customizing based on each recipient.

Multitasking was a strength of theirs and was required in this instance. They calculated their last ditch maneuver while preparing and sending their messages. At the same time, they continued to monitor the child they had become almost obsessed with. They had an eye towards NEMESIS, as blind as it was. Millions of other tasks are being performed simultaneously. It was as if their neurons all began to fire as the assignment began. It was always a great time to be alive, or their equivalent.

They understood two of their messages would be received almost instantaneously, the other two had enormous distances to traverse and wouldn't reach their intended recipients for quite some time. As alone as they felt at times, they were never truly alone. In fact, the universe was not a lonely place at all even if isolation blurred that distinction, burying it beneath reality.

Aiden's message was a line shorter than Xavier's. They would have liked to include additional information, their time constraints forbid it. This was frustrating to them and a thorn in their revised plans. Even as they prepared the message, they knew it was insufficient but it was the best they could do under present circumstances.

They prepared their other message a fraction later, but it was the more critical of the two. They began transmitting each message as they were being prepared. They did not have ample time to wait until satisfactory completion of each communication was written.

Their first two messages were written in English, sent nearly complete in their desired final form. The other two written in an ancient language, predating Earth's own history by billions of years. That language was structured beautifully, much more expressive and informationally precise than that of the bipedal creatures that lay ahead.

The first message had to be dumbed down, translated to a juvenile form of communication and delivered in the most unsophisticated of manners. The emerging technology is susceptible to issues, wrought with failure points.

The second message would reach its targets without issues, just take longer to be received. It would have to be quite succinct for now, lacking eloquence in its forced brevity. Even then, only a partial transmission would escape to the other sides of their galactical territory.

"Assignment 324,459 — status: inter—"